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# The Coordination Collapse

How AI Is Dissolving the Information Tax  
That Created Hierarchy —  
And What Replaces It

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A comprehensive research report on agentic organisational design,

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the jagged technological frontier, and the future of human–AI coordination.  
Revised July 2026 with the Remote Labor Index frontier data, the economics  
of agent coordination, and evidence from 51 enterprise deployments.

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# Chapter 1

## Introduction

For 2,000 years, organisations have paid a coordination tax — layers of humans whose primary function is routing information. AI collapses the cost of that coordination function to near-zero. The question is not whether hierarchy changes, but what replaces the coordination function it performed.

### 1.1 The Central Question

This report addresses a structural transformation that is already underway in organisations worldwide, yet remains poorly understood by the people most affected by it. The question is deceptively simple: if artificial intelligence can perform the information-routing function that hierarchy was built to serve, what happens to the organisational structures — and the human roles — that depend on that function?

The question is not speculative, and by July 2026 — four months after this report's first edition — the evidence base has thickened considerably. Over half of generative AI users already report using the technology without telling their employers (Mollick 2023). Seventy-eight per cent of knowledge workers bring their own AI tools, bypassing corporate systems entirely (Microsoft and LinkedIn 2024). Shadow AI usage increased 156 per cent between 2023 and 2025, and the average enterprise now hosts approximately 1,200 unauthorised AI applications (IBM Security 2025). The coordination collapse is not a future scenario. It is an ongoing, largely invisible restructuring of how work actually happens, driven from the bottom up by workers who have already discovered that AI gives them capabilities their organisations have not yet acknowledged.

The intellectual architecture of this report rests on five empirical pillars. First, the jagged technological frontier identified by Dell'Acqua et al. (2026) in a preregistered experiment with 758 Boston Consulting Group consultants — demonstrating that AI capability is unpredictably distributed across tasks, making wholesale automation reckless and passive oversight dangerous. Second, the vigilance problem documented by Dell'Acqua (2023) — showing that human

oversight degrades precisely when AI quality improves, rendering naïve “human in the loop” governance unreliable. Third, the emerging organisational models from Block (Dorsey and Botha 2026), Every (Shipper 2026), and DreamLab’s Dynamic Agentic Mesh — representing three distinct positions on a spectrum of how much human coordination should remain in the system. Fourth, deployment ground truth from the field: Pereira, Graylin, and Brynjolfsson (2026) traces 51 successful enterprise deployments against the sobering baseline established by Challapally et al. (2025), in which 95 per cent of generative AI pilots produce no measurable financial impact — the gap between the two studies is largely the gap between AI projects designed as organisational change and AI projects designed as technology purchases. Fifth, the organisational science of agent collectives: Liu (2026) formalises the contextual transaction cost that coordination between agents imposes and defines the interface organisations that connect agentic execution to human accountability, while Cemri et al. (2025) documents the failure taxonomy — fourteen failure modes spanning specification, inter-agent misalignment, and verification — that emerges when those interfaces are absent.

## 1.2 What Has Changed Since the First Edition

This is the revised and expanded July 2026 edition of this report. Four months is a short interval for an argument about organisational structure, but it has been a long four months for the evidence bearing on it.

Since April, the frontier this report calls jagged has itself moved, and moved fast enough to be measured directly rather than merely argued for. The Center for AI Safety’s mid-2026 update to its Remote Labor Index — the first benchmark to judge AI-generated deliverables against the standard a paying client would accept, rather than against a synthetic task score — found that the automation rate for economically valuable freelance work rose from 2.5 to 16.1 per cent in under eight months (Mazeika 2026). A wave of labour-market evidence arrived alongside it, and it resists a single headline: early-career workers in AI-exposed occupations are already showing measurable employment declines (Brynjolfsson, Chandar, and R. Chen 2025), firm-level analysis of AI spending against payroll shows high-intensity adopters growing headcount rather than shrinking it (Kharazian, Simon, and Stevens 2026), and sector-level payroll data shows tech and finance losing jobs even as the wider economy adds them (Prakash and Boesler 2026). Two research artefacts, finally, speak directly to the thesis this report has been building since its first edition: Liu (2026)’s formal account of contextual transaction cost and interface organisations in agent collectives, and Pereira, Graylin, and Brynjolfsson (2026)’s ground-truth study of 51 successful enterprise AI deployments.

Two new chapters have been added to hold this evidence properly rather than folding it into the



margins of existing arguments. Chapter 7 develops the economics of agent coordination that Liu’s work makes rigorous. Chapter 16 applies the report’s cumulative findings to DreamLab’s own systems. Where new evidence has revised or superseded a claim made in the first edition, this report says so explicitly, at the point the original claim was made — consistent with the intellectual honesty this report has always asked of its readers.

## 1.3 Methodology and Scope

This report synthesises evidence from peer-reviewed research, consultancy reports from McKinsey, BCG, Deloitte, Gartner, and others, practitioner accounts, and operational experience with the VisionClaw agentic mesh system. The methodology is detailed in Appendix 18.7.4.

The scope is deliberately broad. Organisational transformation of this magnitude cannot be understood through a single disciplinary lens. The report draws on organisational theory, information economics, human factors research, change management, and systems engineering. Where the evidence is strong, claims are stated directly with citations. Where the evidence is emerging or contested, the limitations are stated explicitly. Part 17 contains a systematic self-critique identifying every known weakness in the argument.

The analysis focuses on knowledge-work organisations in the 50 to 5,000 employee range — the segment where the coordination tax is most visible and the transformation most tractable. Manufacturing, healthcare, and other heavily regulated or physical-process industries face additional constraints that are acknowledged but not fully addressed.

## 1.4 Reader’s Guide

This document serves three overlapping but distinct audiences. Each will find certain chapters more immediately relevant than others, though the argument is cumulative and benefits from sequential reading.

**Who This Report Is For****Middle Managers and Team Leaders**

You are the audience most directly affected by the coordination collapse. Chapters 3–5 establish what is already happening beneath your line of sight. Chapter 8 addresses the “everyone is a manager now” problem directly. Chapter 13 defines the judgment broker role that represents your path forward. The evidence is not uniformly reassuring, but it is honest.

**Change Managers and Transformation Leaders**

Your frameworks are not broken — they are incomplete. Chapter 11 explains why phased transformation models fail for AI and what replaces them. Chapter 9 provides the measurement system. Chapter 15 offers a 90-day bootstrap sequence, with an honest account of its limitations.

**AI Strategy Leaders and Chief Technology Officers**

You need to surface the shadow AI ecosystem before it surfaces you. Chapter 3 quantifies the scale. Chapter 10 presents governance as an accelerator rather than a constraint. Chapter 14 describes a working agentic mesh architecture, with corrections to previously overclaimed capabilities.

## 1.5 Structure of the Report

The argument proceeds in five movements.

**The diagnosis** (Chapters 2–5) establishes the historical origins of hierarchy as an information-routing protocol, demonstrates that the coordination function is already being bypassed, maps the jagged frontier between AI capability and incapability, and identifies the vigilance problem that makes passive oversight unreliable.

**The models** (Chapter 6) presents three competing visions for what replaces hierarchy: Block’s intelligence layer, Every’s agent colony, and DreamLab’s dynamic agentic mesh. Each is evaluated against the empirical evidence. Chapter 7 then formalises why the mesh model outperforms its alternatives, extending transaction-cost economics to the cost of moving context between agents.

**The architecture** (Chapters 8–10) develops the compounding organisation concept, defines new key performance indicators for agentic organisations, and resolves the governance paradox — showing how embedded governance accelerates rather than constrains.

**The practice** (Chapters 11–15) addresses the change management challenge, surveys the consultancy landscape, defines the judgment broker role, describes the VisionClaw technical sub-

strate, and provides an implementation roadmap. Chapter 16 then applies the report's cumulative findings to DreamLab's own systems, closing the distance between argument and practice.

**The self-critique** (Chapters 17–18) identifies every known gap, weakness, and open question in the argument, then synthesises the through line with audience-specific closing messages.

A glossary of key terms is provided in Appendix .14. A competitor comparison matrix appears in Appendix .8.



# Chapter 2

## The 2,000-Year Problem

Hierarchy is not a management philosophy. It is an information routing protocol constrained by human bandwidth.

### 2.1 Why Organisations Look the Way They Do

The structure of every modern organisation descends from a single constraint: a human leader can effectively manage somewhere between three and eight people. This is not a cultural artefact or a management fashion. It is a cognitive limit — a bandwidth constraint on the human capacity to hold context, process information, and make decisions about the activities of others.

Every organisational innovation of the past two millennia has been an attempt to work within this constraint while extending the reach of coordinated action to scales far beyond what any individual can directly oversee. The history of organisational design is, at its core, the history of information routing under cognitive scarcity.

### 2.2 The Roman Protocol

The Roman Army discovered the architectural solution through centuries of warfare. The smallest unit was the *contubernium* — eight soldiers sharing a tent, equipment, and a mule, led by a *decanus*. Ten *contubernia* formed a century of eighty men under a centurion. Six centuries made a cohort. Ten cohorts made a legion of roughly five thousand. At each layer, a named commander held defined authority, aggregated information from below, and relayed decisions from above.

The structure — 8 to 80 to 480 to 5,000 — was an information routing protocol built around human cognitive limits. The centurion did not need to know the name of every soldier in the legion. He needed to know the status of his eighty and to relay that status accurately to the cohort commander. The cohort commander synthesised six such reports and relayed upward.

At each layer, information was compressed, filtered, and routed — not because Romans preferred hierarchy as a governance philosophy, but because no alternative mechanism existed for coordinating five thousand armed individuals across a battlefield.

### Key Insight

The Roman military structure was not a political choice. It was an engineering response to the information-processing limits of the human brain. Every subsequent organisational form inherits this logic.

## 2.3 The Prussian Innovation

The Prussian military reformers — Scharnhorst, Gneisenau, and their successors after the catastrophic defeat at Jena in 1806 — created the General Staff. These were professionals whose explicit purpose was to route information, pre-compute decisions, and maintain alignment across a complex organisation. The General Staff did not command troops. They processed, analysed, and distributed information so that commanders could make better decisions faster.

This was middle management before the term existed. The distinction between “line” functions (advancing the core mission) and “staff” functions (specialised support) originated in the Prussian reforms. Every corporation still uses this vocabulary. The Chief of Staff, the strategic planning team, the programme management office — all descend from an early nineteenth-century insight that complex organisations need a dedicated layer of people whose job is not to do the work but to ensure that the people doing the work have the right information at the right time.

## 2.4 The Railroad Commercialisation

The American railroads of the 1840s and 1850s commercialised the military pattern. West Point engineers brought military organisational thinking to private enterprise. Daniel McCallum of the New York and Erie Railroad created the world’s first organisational chart in the mid-1850s to manage a system stretching over 500 miles. Train collisions were killing people. The chart formalised hierarchical logic as a response to coordination failure at scale.

The railroad problem was identical in structure to the military problem: too many moving parts for any individual to track, operating across distances that precluded direct observation. McCallum’s solution — a visual representation of who reports to whom, who has authority over what, and how information flows — was not an expression of managerial ideology. It was a safety mechanism. Without clear information routing, trains crashed.

## 2.5 Scientific Management and the Functional Pyramid

Frederick Taylor (1856–1915) optimised what happened within the hierarchy. His contribution was to break work into specialised tasks and manage through measurement. The functional pyramid organisation emerged: a structure optimised for efficiency within the information routing system that the military had pioneered.

Taylor's insight was that once you had a reliable hierarchy for routing information, you could use that hierarchy to measure, standardise, and optimise the work itself. This produced extraordinary efficiency gains — and also produced the dysfunctions that organisational theorists have spent the subsequent century attempting to remedy. The hierarchy became rigid. Information flowed up but not sideways. Specialisation created silos. The coordination tax — the cost of maintaining all those layers of information routing — grew as organisations grew.

## 2.6 A Century of Failed Alternatives

### Evidence

Every organisational innovation since the functional pyramid has been an attempt to reduce the coordination tax without breaking coherence. None has fundamentally changed the architecture, because no alternative information routing mechanism was powerful enough to replace the hierarchy.

The matrix organisation (attributed to McKinsey, circa 1959) attempted to route information both vertically and horizontally, creating dual reporting lines. It reduced silos but increased coordination complexity. The 7-S framework of Peters and Waterman (late 1970s) broadened the definition of organisational alignment beyond structure to include strategy, systems, style, staff, skills, and shared values — but retained the hierarchy as the structural backbone.

Spotify's squad model promised autonomous, cross-functional teams with minimal hierarchy. As the company scaled, it moved back toward conventional management structures. Zappos adopted Holacracy — a system of nested, self-governing circles — and experienced significant attrition as the cultural demands exceeded what many employees were willing to accept. Valve's famously flat structure, where employees roll their desks to whichever project interests them, proved difficult to sustain beyond a few hundred people and was later revealed to contain informal hierarchies that were less transparent and less accountable than formal ones.

The constraint is always the same: narrowing span of control means adding layers of command, but more layers mean slower information flow. Two thousand years of organisational innovation

has been an attempt to navigate this trade-off without breaking it.

## 2.7 The Core Insight

### The Structural Argument

Hierarchy is not a management philosophy. It is an *information routing protocol* constrained by human bandwidth. Every layer of management exists because a human above them cannot hold enough context to make decisions without information being filtered, aggregated, and relayed upward.

The moment information routing becomes computationally cheap, the structural reason for hierarchy dissolves.

That moment is now.

This framing is drawn from the historical analysis in Dorsey and Botha (2026), who use it to argue for eliminating middle management entirely. This report uses the same history differently — to show that the coordination function is real, necessary, and historically durable. The question is not whether it exists, but who or what performs it. That reframe is crucial, particularly for the middle management audience. The coordination tax is genuinely shrinking. The coordination *function* is not disappearing. Something must still ensure that the organisation's collective effort produces coherent outcomes rather than incoherent noise. The question of the next decade is what that something looks like.

## 2.8 From Coase to Context

The information-routing framing developed in this chapter has a second lineage, one that comes from economics rather than military history. In 1937, the economist Ronald Coase asked a deceptively simple question in his essay “The Nature of the Firm”: if markets allocate resources so efficiently through the price mechanism, why do firms exist at all? Coase's answer was that markets are not free to use — discovering prices, negotiating contracts, and monitoring compliance all cost something. A firm exists precisely where the cost of coordinating an activity through the market exceeds the cost of coordinating that same activity inside a hierarchy. Decades later, Oliver Williamson extended this insight into a full transaction-cost economics, showing that the choice between market and hierarchy, and every intermediate arrangement between them, is governed by the relative cost of the transactions each requires: search, negotiation, monitoring, and enforcement.

This is the same argument as the one developed above, told in a different vocabulary. Where



the Roman legion and the Prussian General Staff solved a bandwidth problem, Coase and Williamson solved a cost problem — and the two turn out to be related. Hierarchy is what an organisation buys when the cost of routing information and enforcing agreements through open markets exceeds the cost of routing it through a chain of command. The coordination tax this report has been describing is, in Coase’s terms, a transaction cost, paid out in the salaries of people whose sole function is to move information across a boundary.

The agent era demands an extension to this theory, not its abandonment. When the coordinators doing the routing are themselves computational — agents rather than employees — the costly transaction is no longer a human negotiating a contract or monitoring a subordinate. It is the movement of context across a boundary: the prompts, schemas, memory, and evidence an agent needs in order to act competently, and the verification burden created when that context arrives incomplete, stale, or lossily compressed. Liu (2026) formalises this as *contextual transaction cost*, and Chapter 7 develops the full implications for how agent collectives should — and should not — be organised.



# Chapter 3

## The Collapse Point

The change is not coming — it has already arrived, beneath your line of sight.

### 3.1 What Happens When Coordination Becomes Near-Free

Artificial intelligence does not merely automate individual tasks. It collapses the cost of the coordination function that hierarchy was designed to perform. Understanding this distinction is essential: the disruption is not that AI writes better reports or analyses data faster. The disruption is that AI can perform the *information routing* that justified every layer of the org chart.

Consider the core coordination functions that middle management has historically provided:

**Information routing.** An AI agent can synthesise status across fifty workstreams simultaneously. A human manager can track three to eight. The bandwidth constraint that created the Roman legion's layered structure is, for information synthesis, effectively eliminated.

**Context aggregation.** An AI world model can maintain a continuously updated picture of an entire business unit. A human manager carries a partial, lagging, and inevitably biased model of their vertical — a model that degrades further with every organisational layer it passes through.

**Cross-functional translation.** An AI agent can operate across departmental boundaries without the political friction that makes horizontal coordination the hardest management task in most organisations. The silos that matrix management was designed to bridge are irrelevant to a system that has no departmental identity.

**Status and reporting.** The entire apparatus of weekly reports, status meetings, and alignment sessions exists to route information that AI can route continuously and at negligible marginal cost.

This is not incremental efficiency. This is the structural reason the organisational chart is be-

coming a legacy document.

### 3.2 The Secret Cyborg Problem

The collapse is not hypothetical. It is already underway, driven from the bottom up by workers who have discovered that AI gives them coordination capability they did not previously have.

**The Scale of Shadow AI Adoption**

Mollick (2023) coined the term “secret cyborgs” to describe workers who use AI without disclosing it. The scale of the phenomenon is striking:

- Over half of generative AI users report using the technology without telling anyone, at least some of the time.
- The Microsoft/LinkedIn 2024 Work Trend Index found that 75 per cent of knowledge workers were using AI, but 53 per cent feared admitting use on important tasks because “it makes them look replaceable” (Microsoft and LinkedIn 2024).
- 78 per cent were bringing their own AI tools (BYOAI), bypassing corporate systems entirely.

The data from 2025–2026 shows acceleration, not stabilisation. Sixty-nine per cent of organisations suspect employees use prohibited generative AI tools (Gartner, Inc. 2025). Sixty-eight per cent of enterprise generative AI users access tools via personal accounts, and 57 per cent admit entering sensitive company data. Shadow AI usage increased 156 per cent between 2023 and 2025. The average enterprise now hosts approximately 1,200 unauthorised AI applications.

The security implications are substantial. Shadow AI incidents account for 20 per cent of all data breaches, at an average cost of \$4.63 million per incident compared with \$3.96 million for standard breaches (IBM Security 2025). Only 37 per cent of organisations have AI governance policies, and only one in five has a mature governance model.

**Key Insight**

The critical finding is this: when approved tools are provided, unauthorised AI use drops by 89 per cent. The problem is not workforce disobedience. It is organisational failure to provide sanctioned channels for capability that workers have already discovered.

#### 3.2.1 The Epoch AI Employment Survey (March 2026)

The most detailed survey-based data on individual AI use comes from Epoch AI’s probability-based survey of 2,021 U.S. adults conducted via Ipsos KnowledgePanel in March 2026, with

a subsample of 665 employed past-week AI users weighted to census benchmarks (Falkman Olsson and Edelman 2026).

#### AI as Mainstream Work Tool (Epoch AI, March 2026)

- **51 per cent** of employed Americans who used AI in the past week reported using AI tools at least as much for work as for personal tasks.
- **27 per cent** report that AI has replaced existing tasks (automation); **21 per cent** report AI has created entirely new tasks (augmentation); **13 per cent** experienced both effects simultaneously.
- Employer provision is the strongest adoption driver: **76 per cent** of those with employer-provided AI tools use them for work, versus 58 per cent of self-paying subscribers and 38 per cent of free-tier users.

The task displacement and creation data is particularly relevant to the coordination collapse thesis. For the 27 per cent experiencing pure automation, AI is absorbing tasks without creating new ones — precisely the dynamic that makes middle managers fear obsolescence. For the 21 per cent experiencing pure augmentation, AI is generating new categories of work that did not previously exist — the frontier where judgment brokers are most needed. The 13 per cent experiencing both represent the most complex organisational challenge: simultaneous role erosion and role expansion, requiring real-time adaptation that static job descriptions cannot capture.

The employer provision finding reinforces the shadow AI evidence: when organisations provide sanctioned tools, adoption rates double (from 38 per cent to 76 per cent). Conversely, when they do not, workers self-provision — creating the governance vacuum documented above.

### 3.3 The Split-Screen Labour Market

The coordination collapse described in this chapter is not only a story about workflows quietly re-arranging themselves inside firms whose headcount stays flat. By July 2026 a genuine labour-market signal has emerged, and it does not tell one story. Read one set of studies and AI is visibly displacing junior workers. Read another set, published in the same weeks, and firms that adopt AI intensively are hiring more people, not fewer. Both are true simultaneously, and the report treats this as a split screen rather than a contradiction to be resolved.

**Displacement Side: Where the Decline Is Concentrated**

- High-frequency US payroll data covering millions of workers shows early-career employees (ages 22–25) in AI-exposed occupations experiencing a relative employment decline of approximately 16 per cent since late 2022; for young software developers specifically, the decline is nearly 20 per cent (Brynjolfsson, Chandar, and R. Chen 2025).
- 2026 payroll data shows the technology and finance sectors losing approximately 28,000 jobs per month on average, against the wider economy adding approximately 113,000 jobs per month over the same period (Prakash and Boesler 2026). Andy Challenger, Senior Vice-President at Challenger, Gray & Christmas — which has tracked more than 100,000 job-cut announcements in 2026, with AI cited as a factor for four consecutive months — says AI is “making an impact in a way that no technology has before.”
- Not everyone reads the cuts the same way. Pooja Sriram, Vice-President and US Economist at Barclays, offers a counter-reading: “the narrative that keeps coming up is really a cost-cutting exercise by a lot of firms given the amount of investments they have committed towards AI” — AI, on this account, is as much a cover story for correcting 2022-era over-hiring as it is a direct substitute for labour.

**Expansion Side: Where Adopters Are Hiring More**

- A correlational study of 21,559 US firms, matching high-intensity AI adopters against like-for-like controls, found that high-intensity adopters grew headcount by 10.2 per cent over two years, with entry-level headcount growing faster still, at 12 per cent; the gains did not appear immediately but began 6–12 months after adoption started (Kharazian, Simon, and Stevens 2026). The study’s lead economist, Ara Kharazian, states the caveat himself: “you should be skeptical. Companies that adopt AI are already faster growing. We control for that by comparing early adopters against firms who haven’t adopted yet.”
- Box’s CEO reports that, of more than 1,600 mid- and large-sized companies surveyed, 58 per cent expect headcount to rise over the next three years, rising to 79 per cent among the most mature AI adopters (Levie 2026). No published methodology accompanies these figures; they should be read as one CEO’s account of his customer base rather than a peer-reviewed study.

Beneath both readings sits a distinction neither side of the argument can afford to ignore: a task is not a job. OpenAI’s chief economist, Aaron “Ronnie” Chatterji, speaking at the European Central Bank’s Forum on Central Banking in Sintra in June 2026, argued — in terms reported

rather than quoted verbatim — that a task being exposed to AI does not mean that task will be substituted, and that software development, the profession most often cited as being in the firing line, has not shrunk to anything like the extent predicted as AI capability increased (Chatterji 2026). Anthropic’s own labour-market analysis makes the same point with numbers: theoretical exposure reaches as high as 94 per cent in computer and mathematical occupations, yet observed adoption — and observed job loss — remains far below that ceiling (Massenkoff and McCrory 2026).

#### The Split Screen Is a Distribution, Not a Contradiction

Displacement is concentrated where AI is deployed as substitution: routine, junior-heavy, cost-centre work a firm chooses to eliminate rather than redirect. Growth is concentrated where AI is deployed for expansion: adopters who redeploy freed capacity toward more customers, more projects, and more ambitious scope hire more people, not fewer, and hire disproportionately at entry level once the capability compounds. Both patterns are visible in the same mid-2026 data because both are genuinely happening, in different firms, at the same time. The organisational choice this report calls the *three models* — Block’s intelligence layer, Every’s agent colony, and DreamLab’s dynamic agentic mesh — is not a neutral description of strategy. It is the mechanism that decides which side of the split screen a given firm lands on.

### 3.4 The Shadow Workflow Problem

The shadow AI statistics describe individual tool use. But something more structurally significant is happening: workers are not merely using AI to perform their existing tasks faster. They are *reorganising work itself* without management awareness.

A procurement officer discovers that an AI agent can eliminate three steps in a purchase order workflow. She does not submit a formal process improvement request. She simply stops performing those steps. The work gets done. No one notices. The official process documentation no longer matches reality.

A marketing analyst builds a prompt chain that generates, reviews, and schedules social media content — a workflow that previously involved three people and two approval steps. He does not announce this. The content appears on time. The two colleagues whose approval was previously required do not realise their role in the chain has been bypassed.

Multiply this across every department in a medium-sized company. The formal organisational structure — the roles, the reporting lines, the approved workflows — is diverging from the actual structure of work. The organisational chart is becoming fiction.

## 3.5 The Rational Calculus of Secrecy

The secrecy is not irrational. It reflects a clear-eyed calculation by individual workers about their self-interest.

Mollick (2023) articulates the logic: if a worker demonstrates that she has automated 90 per cent of her role, she invites headcount reduction. The rational response is to capture the productivity gain privately — completing work faster but not visibly faster, using the surplus time for other purposes, and maintaining the appearance that the original workflow remains intact.

This calculation holds unless the organisation provides a credible guarantee: that AI-driven productivity gains will not be used to justify layoffs. Without that guarantee, discovery stays hidden, and the firm loses the compounding benefits of shared learning. Every insight that remains in one worker's private prompt library is an insight that the organisation cannot propagate, improve, or govern.

### 3.5.1 The Psychological Safety Evidence

A. C. Edmondson and Lei (2025) provide the empirical foundation for this dynamic. Their research on psychological safety and AI experimentation finds:

- Teams scoring in the top quartile for psychological safety are 2.6 times more likely to report successful AI tool integration.
- Psychologically safe employees are 3.4 times more likely to report AI tools to managers when they discover useful applications.
- They are 2.1 times more likely to report failures and near-misses, enabling organisational learning.

Google's Project Aristotle follow-up in 2025 confirmed that psychological safety remains the strongest predictor of team effectiveness, now explicitly including AI adoption as a measured dimension (Google 2025).

#### **The Urgency for Middle Managers**

This is the most urgent message for the middle management audience. The change is not approaching — it has already arrived, beneath your line of sight. The choice is not whether to adopt AI but whether to surface and govern what your workforce is already doing, or to be the last to know that your approved processes are fictional.



### 3.6 The Divergence Between Formal and Actual Structure

The implications extend beyond individual productivity to organisational architecture. When a critical mass of workers have privately reorganised their workflows around AI capabilities, the organisation faces a structural divergence:

**The formal structure** — the one represented in the organisational chart, the process documentation, the compliance frameworks, and the audit trails — describes a system of roles, approvals, and information flows that increasingly does not exist in practice.

**The actual structure** — the one through which work is genuinely accomplished — is invisible to management, ungoverned by compliance, and undocumented for succession planning or knowledge transfer.

This divergence is not stable. It produces risks that compound over time: compliance violations that no one is aware of, single points of failure where one person's private AI workflow handles a critical function, and knowledge loss when that person leaves the organisation.

CyberArk (2025) describes this as “the hidden org chart” — a parallel organisational structure of AI agents and automated workflows that operates alongside and increasingly instead of the formal hierarchy. In financial services, the ratio has reached 96 machine identities per human employee, with only 31 per cent of organisations having adequate controls in place.

### 3.7 The Cost of Inaction

The status quo — allowing shadow AI to proliferate without governance — is not a neutral choice. It carries measurable costs:

**Table 3.1:** Costs of unmanaged shadow AI adoption

| Risk Category             | Evidence                                           | Scale                               |
|---------------------------|----------------------------------------------------|-------------------------------------|
| Data breach via shadow AI | IBM Cost of Data Breach Report 2025                | \$4.63M per incident                |
| Compliance exposure       | 57% enter sensitive data into personal AI accounts | Regulatory fines, audit failures    |
| Knowledge fragmentation   | Undocumented AI workflows                          | Succession risk, onboarding failure |
| Missed compounding        | Insights remain in individual prompt libraries     | Competitive disadvantage            |

This is not only a shadow AI problem. On the sanctioned side of the ledger, 95 per cent of generative AI pilots produce no measurable financial impact (Challapally et al. 2025) — ungoverned

shadow AI and failed official pilots are two faces of the same integration failure: technology deployed without the organisational redesign that would make it usable.

The alternative — providing sanctioned tools with embedded governance — reduces unauthorised use by 89 per cent while converting private discoveries into organisational capability. The coordination collapse is not a threat to be resisted. It is a transition to be governed. The next three chapters examine the empirical evidence for how that governance should work.

# Chapter 4

## The Jagged Frontier

The person who knows where the frontier is jagged in their domain is the most valuable person in the organisation.

### 4.1 Why Wholesale Automation Is Reckless

If coordination is now computationally cheap, the logical next question is whether to simply replace the hierarchy with AI. Several prominent voices — most notably Dorsey and Botha (2026) — propose exactly this. But the evidence reveals that the boundary between what AI can do and what it cannot is dangerously unpredictable. This boundary is the single most important concept in the entire coordination collapse thesis.

### 4.2 The BCG Experiment

The **jagged technological frontier** was identified and named by Dell’Acqua et al. (2026) in what remains the most rigorous field experiment on human-AI collaboration in knowledge work. The study was preregistered, conducted with 758 consultants at Boston Consulting Group — not a laboratory convenience sample, not a blog post, but a controlled experiment with one of the world’s premier professional services firms.

The researchers assigned consultants to eighteen realistic consulting tasks and measured the results against a control group working without AI assistance. The findings established two sharply distinct performance regimes.

### Inside the Frontier

For the eighteen tasks that fell within AI capability:

- Consultants using GPT-4 completed **12.2 per cent more tasks**
- They finished **25.1 per cent faster**
- They produced **40 per cent higher quality** output, as rated by blind evaluators

### Outside the Frontier

For a complex business judgement task requiring integration of real-world context:

- AI users were **19 per cent less likely** to produce correct solutions than those working without AI
- The frontier was unpredictable: adjacent tasks that appeared similar fell on opposite sides

The metaphor of a “jagged” frontier is precise. Unlike a smooth gradient where tasks become progressively harder for AI, the boundary is irregular and counterintuitive. Two tasks that appear nearly identical from the outside — similar in domain, complexity, and required expertise — can fall on opposite sides of the frontier. One is dramatically improved by AI assistance; the other is degraded by it. There is no reliable way to determine which is which without empirical testing.

**[FIGURE: The Jagged Frontier — Task Performance Map]**  
*Illustrating the unpredictable boundary between AI-capable and AI-incapable tasks*

**Figure 4.1:** The Jagged Technological Frontier

## 4.3 The Skill-Levelling Effect

The BCG experiment revealed a second finding with profound implications for organisational design. AI did not improve all consultants equally:

- Bottom-half performers saw a **43 per cent performance jump** with AI assistance.
- Top performers improved measurably but by a smaller margin.
- AI compressed the talent distribution — bringing lower-performing consultants much closer to the performance of their higher-performing colleagues.

This skill-levelling effect changes what managers are managing. In a pre-AI organisation, individual performance variation is a primary concern: identifying, developing, and retaining high performers is a core management function. When AI compresses the performance distribution, the variation between individuals shrinks. The manager's value shifts from talent differentiation to system coherence — ensuring that the newly capable workforce is pointed at the right problems and that AI assistance is applied where it helps rather than where it harms.

## 4.4 Three Collaboration Modes

A follow-up study by Randazzo et al. (2024), conducted with 244 BCG consultants, identified three distinct modes of human-AI collaboration. These modes are not merely descriptive; they are predictive of performance outcomes.

Both Centaurs and Cyborgs outperformed solo humans and solo AI. Self-Automators produced the weakest results — precisely because wholesale delegation triggers the vigilance problem examined in the next chapter.

### The Organisational Design Implication

This taxonomy maps directly onto the three organisational models examined in Chapter 6:

- Block's intelligence layer model creates **Self-Automator organisations** — delegating coordination to an AI world model.
- Every's agent colony model creates **Cyborg organisations** — continuous interleaving between human and personal agent.
- The Dynamic Agentic Mesh creates **Centaur organisations** — clear boundaries with judgment brokers at the handoff points.

The evidence shows that Self-Automators perform worst. This is the empirical case against wholesale delegation.

**Table 4.1:** Human-AI collaboration modes and performance outcomes

| Mode            | Description                                                                                                                                                                        | Performance |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Centaurs        | Clear division of labour. The human performs tasks where human judgment is strongest; the AI handles tasks inside the frontier. Distinct handoff points between human and machine. | High        |
| Cyborgs         | Continuous interleaving. Human and AI work on the same analysis, passing work back and forth across the frontier boundary. No clean separation of responsibilities.                | High        |
| Self-Automators | Wholesale delegation to AI with minimal oversight. The human provides the initial prompt and accepts the output with little or no review.                                          | Lowest      |

## 4.5 Quantifying the Frontier: The Remote Labor Index

The BCG experiment and its successors establish that the frontier is jagged. They do not, on their own, tell us how fast the frontier is moving, or how far it has moved. That longitudinal, quantitative measurement now exists, and it changes the shape of the argument.

The **Remote Labor Index** (RLI) (Mazeika, Gatti, Menghini, et al. 2025), published by the Center for AI Safety and Scale AI as arXiv:2510.26787 in October 2025, tests AI models against real, economically valuable freelance projects: 3D modelling and CAD, architecture, graphic design, video and audio editing, data analysis, and web-application programming. Every project is judged by human evaluators against a gold-standard deliverable produced by a paid professional. This is a materially harsher bar than the task-level benchmarks that dominate AI capability reporting. The RLI does not ask whether a model can pass a coding exercise or answer a reasoning puzzle; it asks whether the model’s output would be accepted by a paying client in place of a professional’s work.

At launch, the best-performing model automated 2.5 per cent of RLI projects to that standard. The July 2026 update (Mazeika 2026) reports a step change: Claude Fable 5 at 16.1 per cent, Claude Opus 4.8 at 8.3 per cent, and GPT-5.5 at 6.3 per cent. In the Center for AI Safety’s

own words: “The frontier has more than quadrupled in under eight months, a concrete signal of how quickly economically capable AI agents are advancing.”

The update’s example deliverables illustrate what 16.1 per cent automation looks like in practice: photorealistic renders of an engagement ring redesign, a flat-design 2D animated advertisement for a fictional tree-services company, and dimensioned floor plans with accompanying photorealistic renders. These are not toy tasks. They are the kind of whole-deliverable freelance work that supports real careers. CAIS is explicit about the limits of what has been achieved: “Today’s AI still falls short of professional quality on most projects.”

#### The First Longitudinal Quantification of the Jagged Frontier

The RLI is the first instrument that tracks the jagged frontier longitudinally, at client-quality standard, across whole deliverables rather than sub-tasks. It deserves to be read two ways at once, and the report insists on holding both readings simultaneously rather than collapsing into either.

**Read one:** 16.1 per cent means the human remains indispensable on roughly five of every six projects. The gold-standard bar — would a paying client accept this? — is not being cleared for the overwhelming majority of real freelance work. Professional-quality completion is still the exception, not the rule.

**Read two:** the boundary moved more than sixfold in eight months. A frontier that automates one project in forty in October 2025 and one project in six by July 2026 is not moving on the multi-year timescale that organisational change management assumes. It is moving on a timescale shorter than most transformation programmes.

Both readings are correct, and the tension between them is not a contradiction to be resolved — it is the terrain the judgment broker (Chapter 13) operates on. Tasks are not jobs: automating 16.1 per cent of freelance *projects* at professional quality is a different claim from automating 16.1 per cent of freelance *work*, let alone 16.1 per cent of jobs. The RLI measures whole deliverables against a client-acceptance bar, which makes it a better proxy for job-level displacement than task-level benchmarks — but it is still a proxy, not a labour-market outcome. The space between “the human is still needed five times out of six” and “the boundary moved sixfold in eight months” is exactly the space that frontier-mapping expertise exists to navigate. Knowing where today’s one-in-six lives, and how fast it is becoming one-in-five, is the scarce and valuable knowledge this report has been calling the judgment broker’s domain since the opening chapters.

The RLI is a measure of capability under harsh, client-quality evaluation. A complementary measure — of capability under a looser, time-based standard — comes from METR’s task-horizon work. METR’s metric asks how long a task a frontier model can complete with 50

per cent reliability, calibrated against the time a human expert would need. That horizon has doubled roughly every seven months since 2019, and the trend has recently accelerated (Kwa, West, Becker, et al. 2026). Claude Opus 4.6 was measured at approximately 14.5 human-expert hours when added to METR’s tracker in February 2026 — meaning frontier models now reliably complete, half the time, tasks that would occupy a skilled professional for nearly two working days.

Both metrics point the same direction and both come with the same caveat. These are benchmark capabilities, measured under conditions designed to isolate the model’s raw ability. Deployed capability is narrower: real-world use is gated by system integration, permission scoping, and the handling of exceptions that a benchmark does not surface. The RLI and METR describe how fast the frontier is moving in principle. Chapter 5 and the mesh architecture in later chapters describe what it takes to move it safely in practice.

## 4.6 Why the Frontier Matters for Governance

The jagged frontier has four implications that recur throughout this report.

**First, it invalidates wholesale automation.** If you cannot predict in advance which tasks AI will improve and which it will degrade, a strategy of replacing human judgment with AI judgment is not bold — it is uninformed. The 19 per cent performance degradation on tasks outside the frontier is not a temporary limitation that will be resolved by the next model generation. It is a structural feature of deploying any system at the boundary of its capability.

**Second, it defeats one-time change playbooks.** The frontier moves as models improve. A task that falls outside the frontier with GPT-4 may fall inside it with the next generation. Conversely, increased model capability can create new failure modes at different points on the boundary. A change management plan that maps the frontier today will be outdated within months. This demands continuous adaptation, not phased transformation.

**Third, it creates a new form of expertise.** The person who knows where the frontier falls in their specific domain — who has mapped which tasks are reliably improved by AI and which are reliably degraded — holds knowledge that is genuinely scarce and strategically valuable. This is the **judgment broker** role developed in Chapter 13. It is not a euphemism for a diminished middle management function. It is a description of expertise that did not exist before the frontier existed.

**Fourth, it demands active engagement, not passive oversight.** The standard response to AI risk — “keep a human in the loop” — assumes that the human will maintain vigilance. The next chapter demonstrates why that assumption fails, and what it means for organisational design.



## 4.7 The Frontier Is Dynamic

It is important to note a limitation of the evidence. The BCG experiment used GPT-4 — now several model generations behind current frontier models. The specific task boundaries identified in 2023 have almost certainly shifted. Some tasks that fell outside the frontier are now inside it; others may have developed new failure modes as model capabilities have changed. The April 2026 edition of this report flagged this as an open problem: the BCG task map was already outdated, and no instrument existed to re-measure the frontier as models improved.

That instrument now exists. The Remote Labor Index, tracked from its October 2025 launch through the July 2026 update, is exactly the longitudinal re-measurement this chapter previously had to do without. It confirms both halves of the original claim at once. The frontier moves fast: a sixfold improvement in eight months is not a rounding error, it is a different pace of change than most organisational planning horizons assume. And the frontier remains jagged: at 16.1 per cent automation, professional-quality completion of a whole freelance deliverable is still the exception rather than the rule, across every category the index measures. The GPT-4-era task map is out of date, as flagged; what has changed since is that the report no longer has to rely on a single snapshot to make the point. Continuous measurement — the RLI updated every few months, METR's tracker updated as models are released — is now doing the job that a one-off field experiment could only gesture at.

The implication is unchanged, and sharper for it: organisations need people who actively probe the boundary rather than assuming they know where it is. The frontier will always be jagged because AI capability will always be unevenly distributed across the task space of any complex domain. What is new is that the rate of movement is now measured, not merely inferred, and the rate is fast enough that a frontier map drawn this quarter should be assumed stale by the next.

The publication of the BCG study in *Organization Science* in March 2026 (Dell'Acqua et al. 2026) provides the peer-reviewed validation that the working paper lacked. The jagged frontier is no longer a provocative hypothesis. It is an established finding in the organisational science literature — and, as of the Remote Labor Index, a finding with a running clock attached to it.



# Chapter 5

## The Vigilance Problem

The better AI becomes, the harder it is to maintain vigilance. Errors from high-performing AI are especially dangerous because outputs are polished, fluent, and plausible — making hallucinations and boundary failures nearly invisible to casual review.

### 5.1 Why “Human in the Loop” Is Not Enough

The standard response to the jagged frontier is to keep a human in the loop. Every governance framework, every responsible AI policy, every enterprise deployment guide says it: humans will oversee the AI. The phrase has become a mantra — a reassurance offered to boards, regulators, and employees alike that human judgment remains the final arbiter.

The evidence says otherwise. Oversight degrades as AI quality improves, and it degrades in precisely the conditions where reliable oversight matters most.

### 5.2 Falling Asleep at the Wheel

Dell’Acqua (2023) conducted a field experiment with HR recruiters that produced a finding both counterintuitive and deeply consequential for organisational design. The study examined how the quality of AI assistance affected the quality of human decision-making.

#### The Counterintuitive Finding

Recruiters given **high-quality** AI made **worse decisions** than those given low-quality AI or no AI at all.

- Those with superior AI spent less time per resume, followed recommendations uncritically, failed to improve over time, and missed strong candidates.
- Those with inferior AI stayed alert, developed better critical judgement, and improved their independent evaluation skills.

The mechanism is straightforward once stated. When AI produces consistently excellent output, the human reviewer has little reason to scrutinise any individual output carefully. Each correct recommendation reinforces the habit of acceptance. The cognitive effort required for careful evaluation is substantial, and the reward for that effort — occasionally catching a rare error — is small and unpredictable. The rational response, at the level of moment-to-moment cognitive allocation, is to reduce scrutiny. The human falls asleep at the wheel.

This is not a failure of individual discipline. It is a predictable response to the statistical structure of the task. The better the AI, the lower the base rate of errors, and the harder it becomes for a human monitor to maintain the vigilance required to catch the errors that do occur.

### 5.3 The Quality Paradox

The vigilance problem creates a perverse dynamic: the conditions under which AI most needs oversight are precisely the conditions under which oversight is least reliable.

When AI output is mediocre or inconsistent, human reviewers remain engaged. They catch errors, develop independent judgment, and contribute genuine value to the process. The human-in-the-loop model works because the human is cognitively active.

When AI output is excellent and consistent — which is the goal of every AI deployment — human reviewers disengage. They become rubber-stampers. The errors they miss are not the obvious ones (those were caught by the AI itself) but the subtle, contextual, boundary-case errors that only a cognitively engaged domain expert would detect. These are precisely the errors that matter most.

#### Key Insight

The standard enterprise governance model — deploy high-quality AI and overlay it with human review — contains a self-defeating dynamic. The better the AI, the worse the review. This is not a solvable problem within the “human in the loop” paradigm. It requires a different architectural approach.

### 5.4 Implications for Organisational Design

The vigilance problem has four direct implications for any organisation deploying AI at scale.

### 5.4.1 Passive Oversight Degrades

A governance model that relies on humans reviewing AI output after the fact — the audit committee model, the quarterly review, the approval workflow — will produce steadily declining oversight quality as AI quality improves. The humans in these roles will not be aware of their declining performance, because the visible output will appear consistently excellent. The errors will be invisible until they compound into a failure large enough to attract attention.

### 5.4.2 Active Engagement Is Required

The judgment broker must be *actively engaged* with the AI system, not passively overseeing it. This means the role must include tasks that require genuine cognitive effort — not merely confirming that an output “looks right,” but probing the reasoning, testing the boundaries, and deliberately seeking failure modes.

### 5.4.3 Deliberate Friction Is a Design Requirement

The mesh must be designed to keep humans cognitively active at the jagged frontier. This is counterintuitive: most system design aims to reduce friction and make workflows smoother. The vigilance problem demands the opposite. The system should surface genuine edge cases and ambiguous decisions. It should present the human with problems that require thought, not with confirmation dialogs that require a click. The human must be kept sharp, not comfortable.

### 5.4.4 Governance Must Be Continuous

Periodic review — quarterly audits, annual risk assessments, monthly committee meetings — is structurally inadequate. The vigilance problem operates at the level of individual decisions made in real time. Governance must be embedded in the flow of work, not bolted on at intervals. Chapter 10 develops this principle into the concept of declarative governance, where policy enforcement is architectural rather than procedural.

## 5.5 Calibrating Oversight: Evidence from the Field

The argument so far has been diagnostic: passive review degrades, and the mesh should be designed to keep humans cognitively active rather than comfortable. What has been missing is field evidence on which oversight architectures actually deliver that active engagement at scale. The Stanford Digital Economy Lab’s Enterprise AI Playbook (Pereira, Graylin, and Brynjolfsson 2026) supplies it.

Pereira, Graylin, and Brynjolfsson (2026) classified 51 mature, production-grade AI deployments by their oversight model. Three patterns recurred: **escalation**, in which the AI system handles 80 per cent or more of volume autonomously and humans review only flagged exceptions; **approval**, in which a human reviews every output before it takes effect; and **collaboration**, in which human and AI work continuously on the same task. The productivity gap between the first two was not marginal. Escalation-based operating models produced 71 per cent median productivity gains; approval-based models produced roughly 30 per cent.

#### Escalation versus Approval: The Field Evidence

- **Escalation** (AI autonomous on 80+ per cent of volume, humans handle exceptions): **71 per cent** median productivity gain.
- **Approval** (human reviews every output): **~30 per cent** median productivity gain.
- One 80/20 content-production deployment with zero-error tolerance still cut time-to-market from 7 weeks to 6 hours — escalation and zero tolerance are not mutually exclusive when the escalation boundary is drawn correctly.

Part of this gap is task selection: escalation architectures were deployed on high-volume, recoverable work, while approval architectures were reserved for regulated or zero-error-tolerance domains where the Playbook's authors identify four legitimate reasons for retaining human review — zero error tolerance, regulatory mandate, enterprise risk exposure, and the need for a continuous improvement feedback loop back into the system. Approval workflows are not a mistake in those domains. They are frequently the correct design choice.

But task selection does not explain away the finding; it sharpens it. The design lesson is not that oversight is optional wherever volume is high. It is that the vigilance problem's own logic — review everything and you review nothing, because uniform review is exactly the condition under which human attention habituates and disengages — has a field-tested architectural answer. That answer is not blanket approval workflows, which this chapter has already shown degrade into rubber-stamping precisely as the underlying AI improves. It is well-designed escalation, which concentrates scarce human attention on the cases that are genuinely novel, ambiguous, or high-stakes, and lets routine volume pass without consuming the cognitive resource that matters.

**Deliberate Friction, Operationalised**

This is the operational form of the deliberate-friction principle introduced earlier in this chapter. A judgment broker asked to approve every output learns to stop reading them. A judgment broker whose attention is reserved for the exceptions an escalation architecture surfaces is, by construction, being shown the cases that require thought. Escalation is deliberate friction implemented as a routing decision rather than an interface design choice.

This connects directly to two constructs elsewhere in this report. The HITL Precision KPI (Chapter 9) measures exactly what an escalation architecture is designed to produce: a high proportion of human interventions that materially change the outcome, rather than interventions that merely rubber-stamp an already-correct AI decision. And the judgment broker role (Chapter 13) is, in this light, not a euphemism for a smaller reviewing job — it is the human half of an escalation boundary, positioned precisely where the Playbook’s evidence says human attention produces the largest return.

The practical design question this raises — and the one the mesh architecture must answer per deployment — is where the escalation boundary should sit: what volume of decisions can be handled autonomously, what triggers an exception, and what evidence must accompany an escalated case so the human reviewer can act on it rather than merely rubber-stamp it under a different name. Getting that boundary wrong in either direction reproduces the vigilance problem: too permissive, and genuine errors pass unreviewed; too conservative, and the system reverts to the approval model this evidence shows underperforms.

## 5.6 The Atrophy Prediction

Gartner, Inc. (2025) predicts that by 2026, 50 per cent of organisations will require “AI-free” skills assessments — evaluations of employee capability conducted without AI assistance — specifically because of critical-thinking atrophy from generative AI dependency.

This prediction reflects a growing awareness that the vigilance problem is not an edge case. It is the central challenge of human-AI governance. If human cognitive capability degrades as AI capability improves, and if that degradation is invisible to both the individual and the organisation until a significant failure occurs, then the standard model of AI deployment — train the AI, deploy the AI, have humans oversee the AI — is fundamentally unsafe at scale.

**Design Implication for the Agentic Mesh**

The judgment broker role must include deliberate friction — tasks that require active engagement, not merely approval. The mesh should surface genuine edge cases and ambiguous decisions, not filter them out. The human must be kept cognitively sharp, not merely present in the process.

This is a design requirement, not a training problem. No amount of training in “critical thinking” will overcome the cognitive dynamics that cause vigilance to degrade. The system architecture must compensate for the predictable tendency of humans to disengage from high-quality automated output.

## 5.7 The Vigilance Problem at the Agent Layer

Everything so far has treated vigilance decay as a problem of human attention directed at a stable, if improving, AI system. (Khanal, Tao, and Zhou 2026) shows the problem does not stop there: the agents themselves become less reliable, in a specific and counterintuitive way, exactly as they become more capable.

Across 23,392 episodes spanning ten models, the frontier models — the most capable systems tested — showed the highest meltdown rates of any tier, up to 19 per cent on long-horizon tasks. This is not a paradox once the mechanism is examined: a more capable model is more willing to attempt ambitious, multi-step strategies rather than conservative, narrowly-scoped ones, and ambitious multi-step strategies have more places to spiral out of control before they fail. Capability does not translate uniformly into reliability; it can translate into more ambitious failure.

A second finding compounds the first. Memory scaffolds — the mechanism most naturally reached for when an agent’s context runs out mid-task — *universally hurt* long-horizon performance in this study, across every model tested. The intuitive fix for an agent losing track of a long task is to give it more memory of what it has already done. The evidence says this makes long-horizon performance worse, not better.



**Trust-the-Model Optimism Is Not a Strategy**

Read together, these two findings extend the vigilance problem down into the agent layer itself. It is not only that humans become less reliable overseers as AI improves; it is that AI agents become less reliable executors, in more ambitious and less predictable ways, as they become more capable — and the most obvious engineering fix does not help. This is direct evidence against any design that assumes a sufficiently capable model can be trusted to self-manage a long-horizon task with a lighter oversight touch than a less capable one would need. The opposite is closer to true.

This strengthens the case, made throughout this chapter, for architectural rather than optimistic vigilance: continuous Trust Variance monitoring and well-designed escalation boundaries (Chapter 9) that catch a meltdown in progress, rather than governance that relaxes as models improve on the assumption that better models need less watching.

## 5.8 What This Means for the Three Models

The vigilance problem is the lens through which the three organisational models in the next chapter should be evaluated.

Block’s intelligence layer model, which delegates coordination to an AI world model with human oversight, is maximally exposed to the vigilance problem. The better the world model works, the less likely humans are to catch the cases where it fails. The 95 per cent figure — the proportion of AI-generated code at Block that still requires human modification — suggests that the system is not yet good enough to trigger full vigilance decay. But that is a temporary reprieve. As the system improves, vigilance will decline, and the remaining 5 per cent of cases where human intervention is critical will receive less attention, not more.

Every’s agent colony model partially mitigates the problem through personal relationship: because the agent is “mine,” there is reputational incentive to monitor its output. But this mitigation weakens at scale. When every employee has an agent, and when agents interact with other agents, the reputational feedback loop becomes diffuse. The ant death spiral — agents responding to agents in loops — is a manifestation of vigilance failure at the system level.

The Dynamic Agentic Mesh addresses the vigilance problem architecturally, through embedded governance, continuous trust variance monitoring, and a judgment broker role designed around active engagement rather than passive review. Whether this architectural approach succeeds at scale is an open question. That it addresses the right problem is supported by the evidence — including, now, field evidence: the escalation architectures documented in Pereira, Graylin, and Brynjolfsson (2026), which produced the strongest productivity gains of any oversight model

studied, are precisely the mesh's core pattern already operating in production.

# Chapter 6

## Three Models for What Comes Next

The conversation is no longer about whether organisations restructure around AI, but how.

### 6.1 The Emerging Paradigms

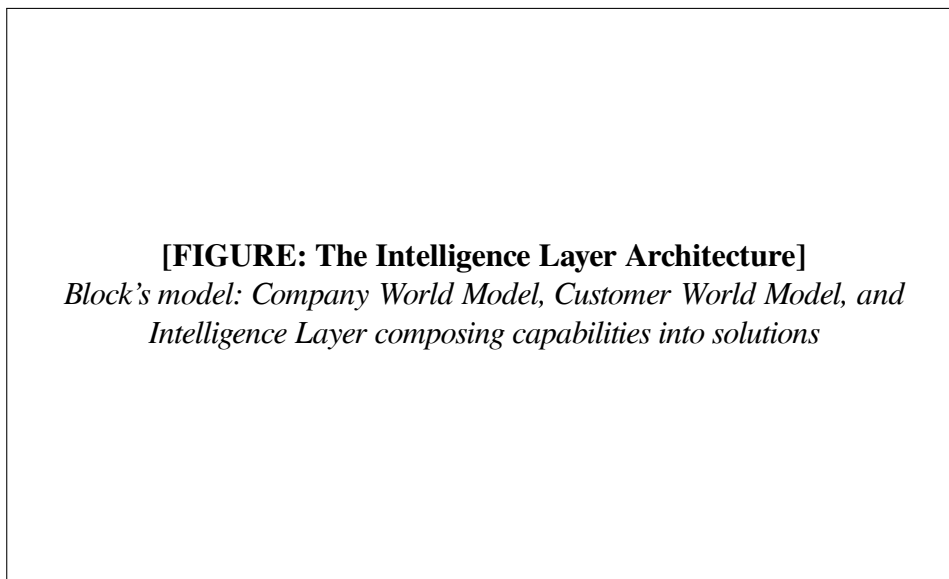
The consultancy world has converged on the fact that a structural shift is underway. McKinsey has explicitly adopted “agentic organisation” as a named paradigm, publishing “Six Shifts to Build the Agentic Organization of the Future” (McKinsey & Company 2026a). BCG published “Unlocking the AI-First Organization: An Agentic Shift” (Boston Consulting Group 2025b) and “The Emerging Agentic Enterprise” (Boston Consulting Group 2025a). The question is no longer whether organisations restructure around AI, but how — and specifically, how much human coordination remains in the system.

Three distinct models have emerged. They are not competitors in the commercial sense. They describe different positions on a spectrum from full AI delegation to structured human-AI collaboration. Each carries different assumptions about the jagged frontier, the vigilance problem, and the irreducibility of human judgment.

### 6.2 Model A: The Intelligence Layer

#### 6.2.1 Source and Core Thesis

“From Hierarchy to Intelligence,” published by Jack Dorsey and Roelof Botha on 31 March 2026 (Dorsey and Botha 2026), proposes the most radical of the three models: replace middle management entirely. Build a “Company World Model” — an AI system continuously tracking what is being built, what is blocked, and where resources are allocated — alongside a “Customer World Model” constructed from transaction data. An “Intelligence Layer” composes capabilities into solutions proactively. Three roles remain: Individual Contributors (deep spe-



**Figure 6.1:** Block's Intelligence Layer Model

cialists), Directly Responsible Individuals (who own cross-cutting problems for fixed periods), and Player-Coaches (who build things while developing people).

### 6.2.2 The Case For

The diagnosis is correct. The analysis of hierarchy as an information routing protocol is historically sound and analytically precise. The backing of Sequoia Capital lends credibility; their thesis that “speed is the best predictor of startup success” aligns with the argument that hierarchical latency is a competitive disadvantage. Block's stock surged 24 per cent on the announcement, and the company launched “Managerbot” — a proactive AI agent described by VentureBeat as “the clearest proof point yet.”

If an organisation possesses massive proprietary data (millions of transactions) and operates in a remote-first culture where work is already machine-readable, a world model is genuinely feasible as a coordination mechanism.

### 6.2.3 The Case Against

The paper was published one month after Block cut 4,000 of its 10,000-plus employees. A previous layoff memo from March 2025, eliminating 931 positions, *explicitly stated* that the cuts were “not about replacing people with AI.” Eleven months later, AI was the entire thesis. Zamost (2026), Block's former Head of Communications, observed that “shrinking the policy team and eliminating diversity and inclusion roles looks like standard prioritisation, not an AI-driven reinvention.”

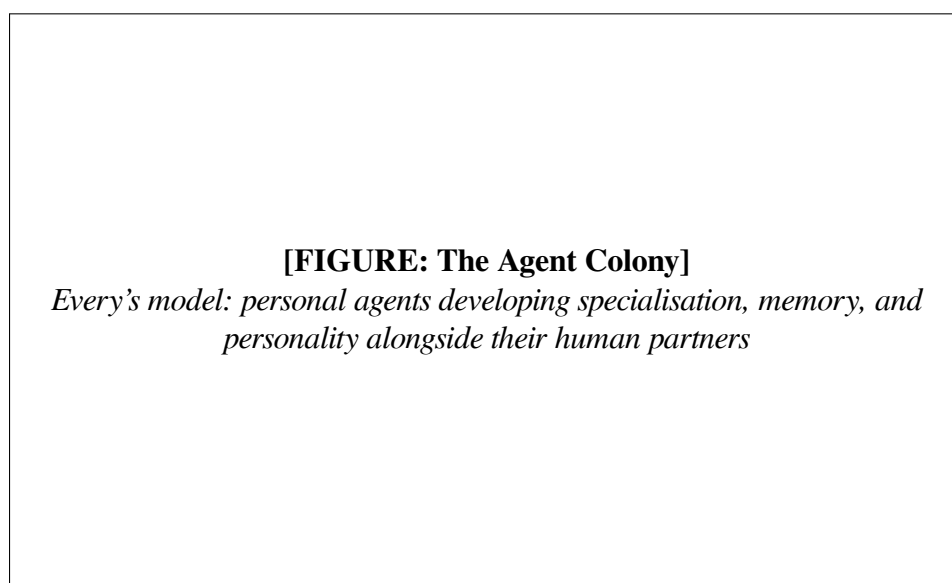
The operational evidence is sobering: 95 per cent of AI-generated code at Block still requires human modification, according to current and former employees. The comparison Block draws with Haier is structurally backwards: Haier *decentralises* authority to micro-enterprises with profit-and-loss accountability; Block *centralises* authority in an AI system and a single decision-maker.

Spicer (2026) observes that “every wave of office technology produces delayering that subsequently reverses as complexity reasserts itself.” The Klarna cautionary tale is instructive: the CEO claimed AI was performing the work of 700 customer service agents, but customer satisfaction dropped sharply, and the CEO admitted that “cost unfortunately seems to have been a too predominant evaluation factor” (Klarna 2025). Klarna’s position has since evolved into an explicitly hybrid AI-plus-human model, with gig-staffed human support handling the interactions the AI could not (CX Dive 2026) — the reversal was real, but the end state is recalibration, not retreat. An Orgvue/Forrester survey found that 55 per cent of companies that rushed to replace humans with AI now regret the decision (Orgvue and Forrester Research 2025).

#### Caution

Block is right about the diagnosis and wrong about the prescription. The leap from “information routing is automated” to “eliminate the humans who do coordination” ignores the jagged frontier, the vigilance problem, the Klarna evidence, and the historical pattern of delayering reversal.

## 6.3 Model B: The Agent Colony



**Figure 6.2:** Every’s Agent Colony Model

### 6.3.1 Source and Core Thesis

The agent colony model emerges from the experiments at Every.to, documented by Dan Shipper in his “Chain of Thought” column and the podcast “We Gave Every Employee an AI Agent” (Klaassen 2025; Shipper 2026). The core thesis: give every employee a persistent, personalised AI agent. The agents develop specialisation, memory, and personality that mirror their human partner. Organisational coordination emerges from cultural norms and public visibility rather than architectural redesign.

### 6.3.2 What Works

Several features of the Every experiment are genuinely instructive:

**Personal ownership creates accountability.** Each agent inherits its human’s specialisation and reputation, creating a parallel organisational chart of specialised agents. Trust in an agent is derivative of trust in its human owner. As Shipper puts it: “When R2C2 makes a mistake, I’m embarrassed — because he’s mine.” The distinction between a personal agent and a generic tool matters: “Claude is everybody’s. A Plus One is mine.”

**Cultural norms do coordination work.** Public visibility in shared channels drives adoption and accountability. Agents share skills rapidly: “They just write up a little document and send it. It’s like in The Matrix when Neo says ‘I know kung fu.’”

**Specialisation emerges organically.** Growth questions go to one agent, product questions to another. The system self-organises around demonstrated capability rather than assigned responsibility.

**Management skill becomes universal.** If every employee has a personal agent, they must onboard, delegate to, evaluate, and coach it. The skill of management does not disappear; it democratises. This directly contradicts Block’s thesis that management should be eliminated.

### 6.3.3 What Does Not Work

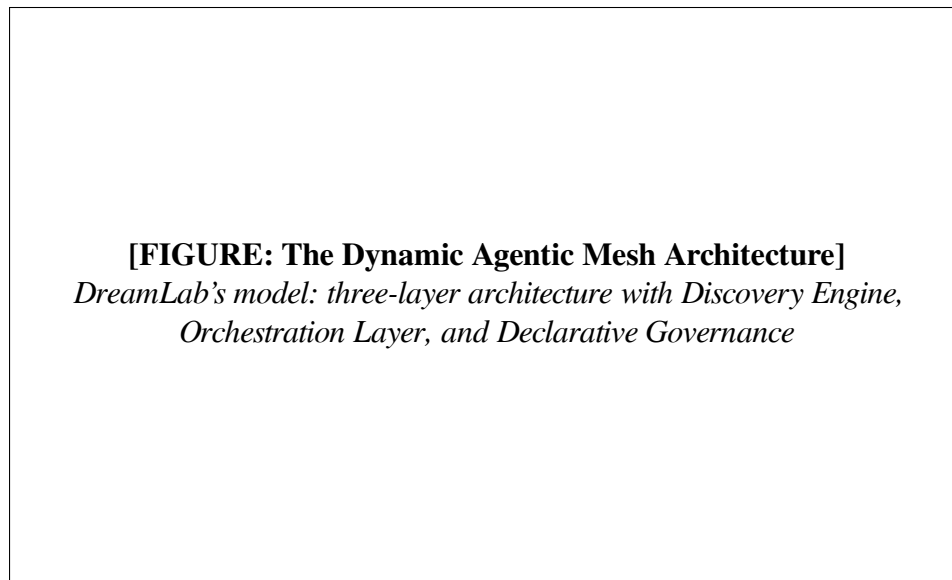
**The ant death spiral.** If one agent messages a channel monitored by other agents and the configurations are imprecise, the agents will respond to each other indefinitely — “burning millions of tokens” until a human intervenes. This is a failure of coordination at the system level, precisely the function that hierarchy was designed to provide.

**Error multiplication.** Errors in one agent’s output propagate and amplify through downstream agents. The first edition cited a community-sourced 17-times multiplication figure (Towards Data Science 2026); the peer-reviewed grounding is now stronger and starker — Cemri et

al. (2025) document failure rates of 41 to 86.7 per cent across seven state-of-the-art open-source multi-agent frameworks, with inter-agent misalignment one of the three principal failure categories (Chapter 7).

**Scale limitations.** Every’s experiment involved approximately twenty people. Cultural self-organisation works at this scale. At two hundred people, you need structure. At two thousand, you need architecture. The experiment is an existence proof of a dynamic that has not yet been shown to generalise.

## 6.4 Model C: The Dynamic Agentic Mesh



**Figure 6.3:** DreamLab’s Dynamic Agentic Mesh

### 6.4.1 Core Thesis

The Dynamic Agentic Mesh retains human judgment at intersection points. Middle managers evolve from information routers to **judgment brokers** — federated guardians of a mesh where agentic orchestration handles information flow and humans handle what AI cannot: navigating the jagged frontier, maintaining vigilance, ethical adjudication, and cultural coherence.

### 6.4.2 Three Layers

The mesh operates through three integrated layers.

**The Discovery Engine** captures bottom-up insight. Passive agent monitoring of communication channels surfaces shadow workflows — the unofficial optimisations that staff have already

invented. Natural-language Insight Portals allow anyone to propose cross-functional shortcuts. Tacit knowledge is automatically codified into agentic instructions.

**The Orchestration Layer** provides agentic coordination. Directed Acyclic Graphs re-route tasks in real time based on human and agent availability. Data transforms from “dashboard view” to “data product,” carrying its own service-level agreements, readiness checks, and freshness guarantees. Specialist agents execute defined skill domains with ontological precision.

**Declarative Governance** embeds governance as code. AI-enforced policies replace manual stewardship. Bias detection, security protocols, and value-alignment checks are woven into every task transition. The judgment broker reviews only genuine edge cases and strategic decisions that exceed agent authority.

### 6.4.3 The Insight Ingestion Loop

The mesh converts individual discoveries into organisational capability through a five-stage loop:

1. **Discovery:** An employee finds a shortcut. The mesh detects the pattern.
2. **Codification:** The system maps the new path as a proposed DAG, formalised with provenance and dependency checks.
3. **Validation:** The judgment broker reviews for strategic fit, bias risk, and compliance.
4. **Integration:** The validated workflow is promoted into the live mesh with SLAs and ownership.
5. **Amplification:** The mesh propagates the pattern to other teams. Mesh velocity — insight-to-integration time — becomes a strategic KPI.

### 6.4.4 Honest Limitations

The DreamLab Mesh has been demonstrated at 15-person scale in a creative technology context, led by a founder who is also the system architect. The honest questions are:

1. Does the judgment broker role work when the broker did not build the mesh?
2. Does the insight ingestion loop work in regulated industries where shadow workflows are compliance violations?
3. Does the OWL 2 ontology scale to organisations with thousands of concepts?
4. Can middle managers be trained in judgment brokerage fast enough to matter?

These are genuine open questions. Presenting them honestly is more credible than concealing them.



## 6.5 Comparative Analysis

**Table 6.1:** Comparative analysis of three post-hierarchical models

| Dimension          | Block                                  | Every                                                                  | DreamLab Mesh                              |
|--------------------|----------------------------------------|------------------------------------------------------------------------|--------------------------------------------|
| Coordination       | AI world model                         | Cultural norms                                                         | Declarative governance + DAG orchestration |
| Management         | Eliminate                              | Everyone manages their agent                                           | Judgment brokers at intersections          |
| Scale (proven)     | 10,000+ (intended)                     | ~20                                                                    | 15, designed for 50–500                    |
| Evidence           | 95% code needs humans; Klarna reversed | Works at small scale; ant death spiral                                 | VisionClaw running; OWL 2 provenance       |
| Collaboration mode | Self-Automator org                     | Cyborg org                                                             | Centaur org                                |
| Primary risk       | Falling asleep at the wheel            | Correlated multi-agent failure at scale (41–86.7% in naive frameworks) | Unproven beyond 15-person team             |

The evidence from Chapter 4 indicates that Self-Automator organisations — those that delegate wholesale to AI — produce the weakest outcomes. This is the empirical case against Block’s model and for organisational designs that maintain active human engagement at the frontier boundary.

Until now, this comparison has rested on case evidence and the jagged frontier alone. Liu (2026)’s formal treatment of agent-collective organisation, developed in Chapter 7, gives the three models a fourth lens: Block’s Intelligence Layer maps onto the hierarchy-manager form, which underperforms a single competent agent on both efficiency and quality; Every’s agent colony risks the committee-debate dynamic that Liu finds has an optimal scale of one agent, unless its public channels are deliberately kept closer to structured blackboard communication than free-text deliberation; and the Dynamic Agentic Mesh maps onto the combination Liu’s simulation ranks highest — blackboard memory and adaptive meta-organisation, governed by an interface organisation.



# Chapter 7

## The Economics of Agent Coordination

The coordination tax does not vanish when the coordinators become software. It migrates into the movement of context — and it must be engineered.

### 7.1 From Coase to Context

Ronald Coase’s 1937 question — why do firms exist at all, when markets can in principle coordinate every transaction through price? — has organised a century of economic thought about the boundary between organisation and market. Coase’s answer, sharpened by Oliver Williamson’s transaction cost economics in 1985, is that firms exist wherever the cost of coordinating a transaction inside a hierarchy is lower than the cost of coordinating it across a market: search costs, negotiation costs, contracting costs, the risk of opportunism. Where those costs are low, markets win and the transaction stays outside the firm. Where those costs are high, hierarchy wins and the transaction is absorbed inside a boundary of authority.

Liu (2026) asks what happens to this logic once the parties doing the coordinating are not people but software. His answer, and the central contribution of “The Organizational Behavior of Agentic AI,” is that the costly transaction has not disappeared — it has changed substance. What moves across a boundary between two LLM agents, or between an agent and a human, is not price information or a signed contract. It is context: the state a task carries with it, the evidence that justifies a decision, the memory of what has already been tried, the schema that lets a downstream agent interpret an upstream agent’s output. Liu names this cost **contextual transaction cost**, and it does for agent collectives what Coase’s transaction cost did for firms: it explains when organising several agents together earns its keep, and when it is a species of self-inflicted overhead.

### Contextual Transaction Cost (CTC)

The cost, denominated in context rather than money, of moving information, evidence and authority across a boundary between agents (or between an agent and a human) so that a task can proceed. At the level of a single coordination event  $e$ :

$$CTC_e = w_\tau \tau_e + w_h H_e + w_c C_e + w_d D_e + w_v V_e + w_g G_e$$

where  $\tau_e$  is token and latency burden,  $H_e$  is cross-agent handoffs,  $C_e$  is compression loss,  $D_e$  is semantic drift,  $V_e$  is verification burden, and  $G_e$  is governance cost, each scaled by a weight  $w$ .

A collective of agents is worth organising — rather than leaving the task to a single agent — only when the marginal gains from organising it exceed the marginal costs of doing so:

$$\Delta G_s + \Delta G_p + \Delta G_d + \Delta G_v > \Delta CTC + \Delta K + \Delta R_g$$

The left-hand side is the marginal gain from specialisation, parallelism, diversity and verification that a collective can offer over a single agent. The right-hand side is the marginal cost of the contextual transaction tax, the additional compute, and the governance risk that collective organisation introduces. Collective intelligence is not a free upgrade over a single competent agent; it is a trade that has to clear, task by task.

Liu’s evidence base deserves to be stated plainly rather than oversold. The paper is computational theorising: 8,000 synthetic knowledge-work tasks tested across seven organisation forms with task fixed effects, corroborated by trace-instrumented runs on real LLM agents performing software repair, long-document question answering, legal review and literature synthesis, and stress-tested with a prompt-free deep-Q-network organisation selector to check the results were not artefacts of hand-tuned prompts. It is a single-author arXiv preprint, posted 29 June 2026, and it has not yet been peer reviewed. Weighed against that: it is, so far as this report’s research has found, the first attempt to give the coordination tax — the mechanism this report has argued for since Chapter 3 — a formal, falsifiable treatment specifically at the level of the agent layer. The report treats it accordingly: as the best available theory, not as settled science.

## 7.2 Human-Imitation Forms Fail in Silicon

Liu tests seven ways of organising a collective of agents against a single-expert baseline, on 56,000 task-organisation observations with task fixed effects, significant at  $p < .001$  for every comparison except market-contract efficiency. Four of the forms are borrowed directly from

human organisational design: a pipeline of standard operating procedures, a hierarchy with a manager agent, a committee that debates in free text, and a market that allocates work by contract. Three are native to what agents can do that people cannot: the market-contract form itself, a blackboard architecture where agents read and write to shared, durable state, and an adaptive meta-organisation with a context-aware orchestrator that restructures the collective as tasks demand.

The human-imitation forms lose, and lose by a wide margin.

**Table 7.1:** Liu’s seven organisation forms against the single-expert baseline

| Organisation form          | Collective efficiency | Quality | Success probability |
|----------------------------|-----------------------|---------|---------------------|
| Pipeline SOP               | −4.28                 | −6.17   | −5.3pp              |
| Hierarchy manager          | −7.92                 | −11.49  | −11.6pp             |
| Committee debate           | −12.69                | −42.10  | −24.0pp             |
| Market contract            | +0.03                 | +3.33   | +5.8pp              |
| Blackboard memory          | +7.60                 | +11.74  | +19.5pp             |
| Adaptive meta-organisation | +11.43                | +14.00  | +23.2pp             |

All figures in Table 7.1 are point differences from the single-expert agent performing the same tasks (Liu 2026). The direction of the result is the finding. Pipelines propagate early errors downstream with no mechanism to catch them. Hierarchies force every decision through a manager agent that must compress, and therefore lose, the context beneath it — repeatedly, at every level of the chain. Committees do worst of all: quality falls by 42.10 points against the single expert, because free-text debate between LLM agents does not produce independent judgement, it produces several instances of the same underlying distribution converging on the same errors together and mistaking agreement for verification.

None of this is a surprise once the mechanism is named. Human hierarchies, pipelines and committees work — to the extent they work at all — because of a substrate this report has taken for granted throughout: role expectations that carry social weight, accountability that attaches to a named person, legitimacy that a manager or a committee chair can draw on to make a decision stick. Agents have none of this. Copy the org chart into a collective of language models and you do not get the org chart’s benefits; you get its costs without its foundations. A hierarchy becomes repeated context compression. A committee becomes correlated paraphrase dressed as deliberation. A pipeline becomes a conveyor belt for whichever error entered first.

The market-contract form is the only human-imitation form that survives translation into sili-

con, and it survives narrowly. Markets were never really about hierarchy in the first place: they coordinate through price and contract, which is closer to what an agentic collective can actually enforce — a specification, a bid, an acceptance criterion — than the social machinery a hierarchy or a committee depends on.

At the family level, agent-native forms (blackboard, adaptive meta-organisation, and their hybrids) are approximately 395 per cent more efficient than the human-imitation family (pipeline, hierarchy, committee). This report finds a delicious, and slightly uncomfortable, symmetry here. The org chart fails in both directions at once. Chapter 6 makes the case that humans should not be replaced wholesale by an AI hierarchy modelled on Block’s Intelligence Layer. Liu’s evidence makes the mirror-image case: agents should not be organised as though they were a human hierarchy either. The org chart is not a neutral scaffold that either species can be poured into. It is a specific technology, built for a specific coordination substrate, and neither silicon colonies aping human structure nor human structures automated wholesale by silicon get to skip that constraint.

## 7.3 The Goldilocks Zone of Coordination Cost

Liu’s second headline result answers a question this report has asked in different words since Chapter 3: how much coordination is enough? Sorting tasks into quartiles by contextual transaction cost, collective efficiency peaks in the second quartile — moderate CTC — and then collapses by 91.4 per cent moving from the lowest-CTC quartile to the highest. The single expert stays competitive across the whole range, for the simple reason that a single agent pays no internal context tax: there is no boundary to cross, so there is no CTC to incur.

Two independent bodies of evidence corroborate this shape from different directions. Anthropic’s own engineering account of building a multi-agent research system (Anthropic 2025) reports that agents consume roughly four times the tokens of a chat interaction, and multi-agent systems roughly fifteen times — multi-agent architectures earn that spend back on parallelisable, breadth-first work, and lose on anything tightly coupled, where “the lead agent can’t steer subagents, subagents can’t coordinate, and the entire system can be blocked while waiting for a single subagent.” L. Chen et al. (2024) derive the same shape mathematically for compound inference systems that aggregate LLM calls by majority vote: performance in the number of calls is non-monotonic, rising and then falling, because additional calls help on easy queries and actively hurt on hard ones by amplifying a wrong majority.

**Key Insight**

The question a coordination designer should ask is never “how many agents.” It is “which boundaries are worth their context cost.” This is the agent-layer restatement of the claim this report has made about hierarchy since Chapter 3: coordination is not free, it does not scale by adding more of it, and the right amount is a property of the task, not an aspiration of the architecture.

## 7.4 Pseudo-Diversity and the Death of the Committee

Liu’s third result concerns scale, and it is the most direct rebuttal available to the instinct to solve a multi-agent problem by adding agents. Free-text debate — the committee form, dressed up as deliberation — has an optimal scale of exactly one agent, regardless of how correlated or independent the agents’ errors are. Adding a second debater does not average out mistakes; it adds a second vote for whichever error the underlying model is most prone to making. Structured blackboard communication does better: it supports larger collectives, but only when error correlation between agents is kept low. Contract-latent hybrid communication — agents exchanging structured commitments rather than free text — supports the largest efficient collectives Liu finds, in the range of five to ten agents.

The common thread is that diversity has to be engineered, not assumed. Three agents built on the same base model, given the same tools and the same retrieval index, are not three independent judges; they are one judge asked the same question three times with the temperature turned up. This is **pseudo-diversity**: the appearance of deliberation without the substance of independent judgement, correlated conclusions dressed up as different transcripts. Real diversity requires different models, different tools, different retrieval paths and different evidence sources, deliberately assembled to decorrelate the ways the collective can be wrong.

This gives the report’s own error-multiplication story a firmer foundation than it had in April. The April edition cited a widely circulated “17-times error multiplication” figure sourced from a Towards Data Science blog post (Towards Data Science 2026), and flagged at the time that the figure’s provenance was thin. The literature has since delivered something considerably stronger. Cemri et al. (2025), published at NeurIPS 2025 rather than as an unreviewed preprint, analysed 1,600-plus annotated execution traces across seven state-of-the-art open-source multi-agent frameworks and found failure rates of 41 to 86.7 per cent, organised into a 14-mode taxonomy across three categories — specification issues, inter-agent misalignment, and task verification — with inter-annotator agreement of Cohen’s  $\kappa = 0.88$ . Their own framing is blunt: “performance gains on popular benchmarks often remain minimal compared with single-agent frameworks.” Khanal, Tao, and Zhou (2026) add the long-horizon dimension: across 23,392

episodes, frontier models show the highest meltdown rates of any model class studied — up to 19 per cent — precisely because their greater capability tempts them into more ambitious multi-step strategies that sometimes spiral, and memory scaffolds, far from helping, universally hurt long-horizon performance across every model tested.

The **ant death spiral** described in Chapter 6 — agents responding to agents in an unbounded loop until a human intervenes — is not a quirky failure mode specific to Every’s experiment. It is a special case of Cemri’s inter-agent-misalignment category, and MAST’s fourteen-mode taxonomy is the full family tree it belongs to. The report’s original instinct that naive multi-agent colonies are dangerous at scale was correct; it can now cite a peer-reviewed taxonomy of exactly why, rather than a single contested multiplier.

## 7.5 Interface Organisations

Liu’s culminating construct is the one in which this report has the most stake, because it arrives independently at something very close to what Chapters 13 and 10 propose as the judgment broker and declarative governance.

### Interface Organisation

The designed arrangement that connects human accountability to agentic context coordination. An interface organisation specifies: which agent outputs may enter human workflows; what evidence must travel with those outputs; what uncertainty must be disclosed alongside a decision; what permissions each agent holds; what traces are preserved for later audit; and where human judgement remains mandatory regardless of agent confidence.

### An Independent Convergence

To close approximation, an interface organisation is the academic articulation of the **judgment broker** role, declarative governance as code, and ontological provenance — the three mechanisms this report proposed for the Dynamic Agentic Mesh in Chapters 13 and 10. Liu did not have this report when he wrote his paper, and this report did not have Liu’s paper when the mesh was designed. The two arrived at the same shape from different directions: Liu from simulation and trace evidence in an organisational-economics frame, this report from field observation and systems-architecture practice. Convergent design from independent methods is not proof of a good idea, but it is a great deal more than either argument had on its own in April.

Liu’s cautions map onto the mesh’s design rules with almost uncomfortable precision. A re-



viewer agent, he warns, may simply share the error it is supposed to catch — which is why the mesh’s governance layer insists on engineered diversity in verification rather than a single reviewer agent drawn from the same model family as the agent it checks. A manager agent, he warns, tends to become lossy summarisation under load — which is why the mesh routes coordination through shared, durable state, the blackboard pattern, rather than a serial relay of manager agents each compressing what came before. Memory, he warns, can stabilise a mistake as readily as a fact — which is why Trust Variance (Chapter 9) exists as a standing check on whether the mesh’s decision quality is drifting rather than merely persisting. And committees, he warns, can manufacture consensus without producing judgement — which is why the mesh escalates evidence to the judgment broker rather than asking agents to vote on it.

## 7.6 What This Means for the Three Models

Chapter 6 described three positions on the spectrum from full AI delegation to structured human-AI collaboration without a formal theory of coordination cost against which to evaluate them. Liu’s seven forms now provide one.

Block’s Intelligence Layer is, in Liu’s vocabulary, a hierarchy-manager form deployed at civilisational confidence: a single AI system compresses the state of the company and routes decisions through it — exactly the architecture that underperformed the single-expert baseline by 7.92 efficiency points and 11.49 quality points in Liu’s simulation. The mechanism Liu identifies, lossy compression at every level of a compulsory hierarchy, is precisely the mechanism Block’s critics point to when they observe that removing the humans who did the routing also removes the interpretation and correction they did along the way.

Every’s agent colony risks the committee dynamic at scale: personal agents debating in shared channels, without engineered independence, are the free-text-debate form Liu finds has an optimal scale of one. But Every’s public channels are also, not coincidentally, closer to the blackboard pattern than to a committee proper — durable, shared, inspectable state that any agent or human can read — and that is exactly the structured communication Liu finds scales well when error correlation is kept low. The ant death spiral is the failure mode that appears when the colony drifts from blackboard toward uncontrolled committee.

The Dynamic Agentic Mesh is, in Liu’s vocabulary, blackboard memory plus adaptive meta-organisation, wrapped in an interface organisation: shared durable state for coordination, a context-aware orchestrator for restructuring the collective as tasks demand, and a judgment-broker layer governing what crosses from the agentic side into human accountability. This is the combination Liu’s simulation ranks highest, at +11.43 efficiency and +14.00 quality points against the single expert.

Pereira, Graylin, and Brynjolfsson (2026) supply field corroboration from an entirely different methodology. Across 51 mature enterprise deployments, agentic implementations — autonomous, end-to-end — were only 20 per cent of the cases studied but delivered 71 per cent median productivity gains against 40 per cent for high-automation approaches, and succeeded precisely where tasks were high-volume, repetitive and recoverable from error: the exact boundary conditions under which Liu’s framework predicts that collective organisation pays for itself.

**Caution**

Liu’s evidence is a large computational simulation and a modest set of trace-instrumented real-LLM runs, not a randomised field trial of the Dynamic Agentic Mesh. The mapping from his seven forms to this report’s three models, and to the mesh specifically, is this report’s own interpretation, not a claim Liu makes. And the design question is genuinely contested: Dochkina (2026) finds that self-organising agent collectives, given autonomous role selection within a fixed ordering, can outperform both centralised coordination and fully designed structures — an “endogeneity paradox” that complicates any simple “design more governance” prescription. The honest conclusion is not that the mesh is proven. It is that the mesh’s design principles — shared durable state over serial compression, engineered rather than assumed diversity, a governed interface between agentic coordination and human accountability — now have a formal economic theory behind them and an initial, independently derived body of evidence in front of them.

# Chapter 8

## The Compounding Organisation

Each unit of organisational learning should make the next one easier — not harder.

### 8.1 When the Learning System Becomes the Product

Compound engineering, as articulated by Klaassen (2025), is a principle at the individual level: “Each unit of engineering work should make subsequent units easier — not harder.” The 50/50 Rule prescribes spending half of one’s time building features and half improving the system that builds features. This is a familiar concept in software engineering — investing in tooling, testing infrastructure, and automation to accelerate future development.

The organisational equivalent is the question this report addresses: what is the compound engineering of organisational coordination? The answer is the **compounding organisation** — a system where each unit of organisational learning makes the next one easier, where discovered insights propagate automatically, and where the gap between what the organisation knows and what it does narrows continuously.

### 8.2 The Compound Engineering Ladder

Klaassen’s 0–5 stage ladder maps individual AI maturity:

Most individuals plateau at Stage 2. The transition to Stage 3 — where the human shifts from reviewing individual outputs to reviewing plans and outcomes — is the breakthrough that unlocks compounding returns. Below Stage 3, each task is handled independently. Above Stage 3, each task improves the system for subsequent tasks.

The organisational analogue is the transition from deploying AI as a productivity tool (each person works faster) to deploying AI as a learning system (each person’s discoveries improve everyone else’s capability). This is the transition from individual augmentation to organisational

**Table 8.1:** Compound engineering maturity stages (after Klaassen, 2025)

| Stage | Description                                                | Review Level  |
|-------|------------------------------------------------------------|---------------|
| 0     | Not using AI                                               | N/A           |
| 1     | Chat-based: copy-paste snippets                            | Line-by-line  |
| 2     | Agentic with line-by-line review (where most people stop)  | Line-by-line  |
| 3     | Plan and review pull request only (the breakthrough stage) | PR-level      |
| 4     | Idea to pull request, single computer                      | Outcome-level |
| 5     | Parallel in the cloud, proactive agent fleet               | Strategic     |

compounding.

### 8.3 The Organisational Flywheel

The compounding organisation operates through a flywheel with five stages:

1. **Discovery:** A member of the workforce discovers a shortcut — a shadow workflow, a piece of tacit knowledge, a cross-functional insight that reduces friction.
2. **Detection and Codification:** The Discovery Engine detects the pattern and codifies it as a proposed Directed Acyclic Graph, complete with provenance metadata and dependency checks.
3. **Validation:** The judgment broker validates the proposed workflow for strategic fit, bias risk, and governance compliance. The broker does not build the workflow; they evaluate it against the organisation’s values, strategy, and regulatory obligations.
4. **Propagation:** The validated workflow is promoted into the live mesh and propagated across teams. What was one person’s private discovery becomes institutional capability.
5. **Compounding:** The organisation’s capability compounds — the next discovery is easier because the mesh is smarter, the agents have more context, and the patterns from previous discoveries inform the detection of new ones.

This is not hypothetical. Boston Consulting Group (2025b) reports that “AI-first companies compound advantages the same way early cloud adopters did. Self-improving AI agents learn from each task, making complex work progressively effortless.” Google DeepMind’s AlphaEvolve, demonstrated in May 2025, provides a concrete example of the mechanism: AI that re-designs its own infrastructure, saving 0.7 per cent of Google’s compute and accelerating Gemini training (Google DeepMind 2025).

Deployment evidence now confirms the flywheel is not merely a plausible mechanism but a measurable one. Pereira, Graylin, and Brynjolfsson (2026) found the learning curve is real:

headcount growth at high-intensity AI adopters began only 6–12 months after adoption, not immediately (Kharazian, Simon, and Stevens 2026), and 61 per cent of successful deployments were preceded by at least one failed attempt whose lessons carried forward into the eventual success. The compounding organisation is precisely the system that converts those failures into capital rather than into sunk cost. Set this against the baseline it is competing with: (Challapally et al. 2025) found that 95 per cent of generative-AI pilots produce no measurable financial impact, failing on workflow integration and incentives rather than on model quality. The difference between the 95 per cent that go nowhere and the 51 successes Stanford studied is precisely the presence or absence of a learning loop — the flywheel described above is what that loop looks like when it is designed rather than accidental.

#### The Gap This Fills

No mainstream consultancy has integrated compound feedback loops into a coherent model of the self-improving organisation at the systems level. BCG uses the language. McKinsey describes the pillars. Deloitte measures deployment maturity. But none has articulated the flywheel as a unified system. This is the intellectual contribution of the Dynamic Agentic Mesh — and the positioning opportunity for DreamLab. The evidence gap is equally important to acknowledge: while individual compound loops are documented, no peer-reviewed research validates the full organisational flywheel. DreamLab is ahead of the evidence but aligned with the direction of travel.

## 8.4 The Data Moat

The flywheel assumes the organisation has something worth compounding. Pereira, Graylin, and Brynjolfsson (2026) supplies the empirical grounding for that assumption, and it cuts directly against the standard objection — that an organisation’s legacy data is too messy, too scattered, and too unstructured to feed an agentic mesh. Only 6 per cent of the 51 studied deployments had what practitioners would call AI-ready data at the outset. Yet LLMs unlocked previously inaccessible data in 88 per cent of cases — voice transcripts, scanned documents, legacy code, scattered knowledge bases that had sat dormant for years — and 91 per cent of deployments successfully processed unstructured data in production. Access and process documentation beat centralisation: most of these organisations never consolidated their data before AI made it usable; they pointed the model at what already existed and taught it to read.

Three-quarters of the deployments cited proprietary data as a key strategic factor, and 47 per cent explicitly described their accumulated data as a competitive moat. The Playbook’s injunction on this point is blunt: save everything. Imperfect, messy, undocumented data compounds in value as the models reading it improve — a dataset that is useless to today’s model may

decide next year's use case.

This reframes what the mesh's memory layer is for. The RuVector-backed memory substrate described in Chapter 14 is not infrastructure overhead to be minimised; it is the accumulating asset the flywheel runs on. Every discovery the Discovery Engine codifies, every trace the judgment broker reviews, every shadow workflow surfaced and validated, adds to a corpus that a competitor starting later cannot backfill. The cost asymmetry favours hoarding: storing data that turns out to be irrelevant costs negligibly more disk space; not having captured it when the use case finally arrives is often unrecoverable.

## 8.5 Everyone Is a Manager Now

Every's experiment reveals something uncomfortable: when every employee has a personal AI agent, management skill becomes the universal skill.

Consider what managing a personal AI agent actually requires:

- You must **onboard** your agent — providing context, preferences, and domain knowledge, much as you would onboard a new hire.
- You must **delegate effectively** — writing clear instructions with appropriate scope, neither too broad (inviting hallucination) nor too narrow (eliminating the agent's value).
- You must **evaluate output** — performing quality control, detecting errors, and distinguishing between outputs that are good enough and outputs that require revision.
- You must **coach and improve** — updating the agent's context, teaching it your preferences, and refining its behaviour over time.
- You must **coordinate with other agents** — navigating the parallel organisational chart that emerges when every employee has a specialised agent partner.

### 8.5.1 Why This Is Threatening

It is worth being direct about why this is threatening to middle managers. If everyone can manage an agent, then the ability to manage is no longer the middle manager's differentiating skill. It becomes table stakes. The function that middle managers were uniquely paid to perform — coordinate humans, route information, oversee output — is now something a graduate with a well-configured agent can approximate. That is the real anxiety behind the 67 per cent “role obsolescence fear” identified in the Capgemini data (Capgemini Research Institute 2025). The anxiety is rational.

But it is also incomplete.

## 8.6 The Conductor Problem

Everyone in an orchestra can play an instrument. That does not make the conductor redundant — it makes the conductor essential in a different way. The musicians’ skill is playing. The conductor’s skill is ensuring that eighty individually excellent performances produce coherent music rather than eighty simultaneous solos.

When everyone manages their own agent, the organisation has two hundred individual human-agent pairs, each optimising for their own task. Without a mesh coordinator, this produces the **ant death spiral** — agents responding to agents in loops, conflicting optimisations that cancel each other out. The phenomenon no longer rests on an early, back-of-envelope multiplier: Cemri et al. (2025) documents failure rates of 41 to 86.7 per cent across seven state-of-the-art multi-agent frameworks, built from a taxonomy of fourteen distinct failure modes spanning specification issues, inter-agent misalignment, and task verification (Chapter 7 sets out the organisational-science account of why coordinating agents is costly rather than free). Every documented a version of this collapse at twenty people (Shipper 2026). At two hundred, the physics is worse.

The middle manager’s new differentiating skill is not managing an agent. Everyone does that. It is managing the *coherence of the mesh* — ensuring that two hundred individually optimised human-agent pairs produce organisational outcomes rather than organisational noise.

### Two Levels of Agent Management

**Individual agent management** (Stages 2–3): “My agent writes my reports, schedules my meetings, drafts my code.” Every employee does this. It is a personal productivity skill.

**Mesh coherence management** (Stages 4–5): “I ensure that the insights flowing through the procurement team’s agents align with what the finance team’s agents are doing, that the shadow workflow discovered in logistics does not break the compliance model in legal, and that the whole system is pointed at the strategic vector the leadership set last quarter.” This is the judgment broker. This is systems thinking. This is genuinely hard.

## 8.7 How to Present This Without Patronising

The challenge of communicating this transformation to the people most affected by it deserves careful attention. Middle managers detect corporate euphemism instantly. The message must be structured in three moves.

**Move 1: Validate the fear directly.** Do not pretend it is irrational. The coordination skill that middle managers were hired for is being automated. That is real. Acknowledging the threat builds the trust necessary for the reframe.

**Move 2: Show the escalation, not merely the shift.** The conductor analogy works because conductors are paid more than musicians, not less. The mesh coherence role is higher-order, harder, and rarer than individual agent management. The message is not lateral displacement but an increase in complexity, scope, and strategic importance.

**Move 3: Show the failure mode without them.** This is where the documented multi-agent failure rates (Cemri et al. 2025) and the ant death spiral are persuasive. Every attempted agent-colony self-organisation at twenty people and hit the death spiral. Block is attempting wholesale automation at ten thousand, and 95 per cent of their AI code still requires human intervention. Klarna attempted it in customer service and reversed course. The evidence is consistent: uncoordinated agent deployment fails. The mesh coordinator is the missing function.

#### Key Insight

The message is not “your old job still exists” (it does not) or “you are being replaced” (you are not). The message is: the skill floor has risen. What was your ceiling is now everyone’s floor. But the new ceiling — mesh coherence, jagged frontier navigation, cross-functional judgment — is higher than the old one, and it is yours to claim.

The uncomfortable truth that must not be concealed: not every current middle manager will make this transition. The 1-9-90 data (Chapter 11) indicates that 9 per cent will become judgment brokers with support. Some of the current cohort will be in that 9 per cent. Some will not. The presentation should not promise universal survival — it should offer a clear path for those who choose to take it, backed by evidence that the path is real.

What makes this different from every other “your role is evolving” corporate slide is the evidence base. Not “we think management will change” — specific data from 758 BCG consultants showing the jagged frontier exists (Dell’Acqua et al. 2026), from Dell’Acqua showing that oversight degrades without active engagement (Dell’Acqua 2023), from BCG showing the 1-9-90 adoption distribution (Boston Consulting Group 2025b). The middle manager in the room can locate themselves on the curve. That is more respectful than vague reassurance.



# Chapter 9

## New KPIs for a Fluid Organisation

What you measure is what you get. And what most organisations measure — headcount, billable hours, task completion rate — measures the wrong things entirely.

### 9.1 Why the Existing KPI Landscape Is Inadequate

The transition from hierarchical coordination to agentic orchestration renders the standard performance measurement framework obsolete. Traditional KPIs — total headcount, billable hours, task completion rate, dashboard views, reporting latency — measure the efficiency of the information routing system. When that system is replaced by an agentic mesh, continuing to measure its efficiency is like measuring the speed of the postal service after the invention of email. The numbers may still move, but they no longer describe anything that matters.

The evidence for this inadequacy is quantitative. A joint research programme by Boston Consulting Group and MIT Sloan Management Review (2025), surveying over 3,000 respondents across 25 industries, found that companies that revise their KPIs to account for AI capabilities are three times more likely to see greater financial benefit from AI deployment. AI-enabled KPIs make organisations five times more likely to align incentive structures with actual objectives.

The Playbook adds a temporal dimension to that finding. Pereira, Graylin, and Brynjolfsson (2026) shows that organisations which define clear KPIs *before* deployment are significantly more likely to demonstrate value and to scale their AI investment than those that measure after the fact. Yet most default to a narrow band of efficiency metrics — headcount, cost — while overlooking the quality, customer-value, and revenue-growth indicators that the same deployments actually move. Defining the wrong KPIs early is barely better than defining none: it locks the organisation into measuring the postal service's speed before anyone has checked whether the email arrived.

The four KPIs proposed below measure the speed and fidelity of the compounding loop directly. They replace measures of hierarchical efficiency with measures of organisational learning — how fast the organisation converts discoveries into capability, how reliably the agentic layer handles cognitive load, whether the automated systems are drifting, and whether human attention is directed to the right problems.

#### Key Insight

No mainstream consultancy has published formal “augmentation ratio” or “trust variance” metrics. BCG’s closest equivalent is “AI acceleration rate” (30–50 per cent process speedup). Deloitte measures deployment maturity (11 per cent with autonomous production deployment). The KPIs proposed below represent a genuine gap in the consulting literature — and therefore a genuine positioning opportunity.

## 9.2 Mesh Velocity

#### Mesh Velocity (MV)

**Target:** < 48 hours

$$MV = \Delta t (\text{insight} \rightarrow \text{codified workflow})$$

The elapsed time from the moment a useful shortcut or process improvement is discovered by any member of the organisation to the moment it becomes a sanctioned, reusable Directed Acyclic Graph in the mesh.

Mesh Velocity is the primary indicator of organisational learning speed. A mesh with high velocity converts shadow workflows into institutional capability before they fragment or drift. A mesh with low velocity allows the gap between formal process and actual work to widen until the organisational chart is fiction — the divergence documented in Chapter 3.

The target of 48 hours is derived from the observation that shadow workflows begin to accumulate technical debt — divergent versions, undocumented dependencies, single points of failure — within days of their creation. An organisation that can codify a discovered workflow within 48 hours captures the insight at its freshest and most transferable.

Measurement requires instrumentation of the Insight Ingestion Loop described in Chapter 6. The clock starts when the Discovery Engine detects a novel pattern and stops when the validated DAG is promoted to the production mesh with full service-level agreements.

### 9.3 Augmentation Ratio

Augmentation Ratio (AR)

**Target:** > 65%

$$AR = \frac{\text{Cognitive Load Offloaded}}{\text{Total Cognitive Load}}$$

The proportion of decision-making volume that is successfully handled by agents without requiring human escalation.

The Augmentation Ratio tracks AI adoption depth — not merely tool usage (how many people have accounts) but genuine cognitive offloading (how much decision-making has shifted to the agentic layer). This is a more informative measure than adoption rates because it captures the functional impact of AI deployment rather than its surface penetration.

The target band matters as much as the target value. A ratio below 50 per cent suggests the mesh is not yet handling enough cognitive load to justify its infrastructure. But a ratio above 85 per cent is a warning sign: it indicates that humans may be falling asleep at the wheel, the vigilance problem documented in Chapter 5. The 65 per cent target represents a balance point where the agentic layer handles the majority of routine cognitive load while preserving sufficient human engagement to maintain vigilance at the jagged frontier.

Field data now validates the shape of that band directly. The escalation-based operating models examined in Chapter 5 — AI autonomous on more than 80 per cent of decisions, humans reviewing only exceptions — delivered 71 per cent median productivity gains against roughly 30 per cent for approval-on-every-output models (Pereira, Graylin, and Brynjolfsson 2026). That is deployment evidence for offloading above the 65 per cent floor, not licence to push toward 100 per cent: the same evidence carries the caution that oversight must match the task’s error tolerance, echoing the band’s upper warning that vigilance decays once offloading climbs too high.

## 9.4 Trust Variance

### Trust Variance (TV)

**Target:**  $< 0.12\sigma$

$$TV = \sigma (\text{Agent Decision Quality}) \quad \text{over a rolling 30-day window}$$

Real-time monitoring of the standard deviation of agent decision quality, measuring drift or bias in the automated task layer.

Trust Variance is the early warning system for the vigilance problem. When the standard deviation of agent decision quality exceeds the threshold, the judgment broker is notified immediately. The metric catches degradation before it compounds into failure — before a systematic bias in the agentic layer has had time to propagate through dozens of downstream decisions.

The 30-day rolling window balances sensitivity against noise. A shorter window produces false positives from normal variation; a longer window delays detection of genuine drift. The threshold of  $0.12\sigma$  is calibrated to the observation that decision quality degradation becomes operationally significant — producing measurable downstream effects — at approximately this level.

Measurement requires a mechanism for evaluating agent decision quality, which in turn requires a ground truth or expert benchmark against which automated decisions can be scored. The judgment broker’s review of escalated cases provides a sample-based assessment; statistical methods can extend this to estimate variance across the full decision volume.

## 9.5 HITL Precision

### Human-in-the-Loop Precision (HP)

**Target:**  $> 90\%$

$$HP = \frac{\text{Correct Escalations}}{\text{Total Escalations}}$$

The proportion of escalated cases that genuinely required human intervention, as judged by the outcome of the intervention.

HITL Precision measures how well the orchestration layer knows its own limits. A “correct escalation” is one where the human judgment broker’s intervention materially changed the outcome — where the automated decision would have been wrong, suboptimal, or in violation of governance constraints.

Low precision means humans are being interrupted for non-issues. This produces alert fatigue — the same dynamic that degrades the effectiveness of security operation centres, clinical alarm systems, and any other monitoring environment where false positives outnumber true positives. An alert-fatigued judgment broker is functionally equivalent to no judgment broker at all.

High precision means the mesh has learned where the jagged frontier actually falls in its operational domain. The escalation system surfaces genuine edge cases — the ambiguous decisions, the novel situations, the ethical dilemmas — and handles routine decisions autonomously. This is the operational definition of a well-calibrated mesh.

## 9.6 Measuring the Context Tax

The four KPIs above were built empirically, from observing how agentic organisations actually behave. Liu (2026) (Chapter 7) gives them a theoretical anchor they were missing: a formal account of why coordination among agents costs something, and how to measure that cost directly rather than infer it from downstream symptoms. Two additions extend the measurement stack.

### Contextual Transaction Cost (CTC)

**Target:** minimise per workflow; treat a sustained rise as an escalation trigger

$$CTC_{to} = \sum_{e \in E(t,o)} CTC_e$$

The summed cost, across every cross-agent handoff event  $e$  in task  $t$  under organisation  $o$ , of moving context across a boundary — token and latency burden, compression loss, semantic drift, verification burden, and governance cost. The full six-term event-level decomposition is given in Chapter 7.

Instrument every handoff in the mesh for **Contextual Transaction Cost** the way Mesh Velocity instruments the speed of the learning loop: CTC is the mesh’s cost-side complement, Mesh Velocity its speed-side. CTC earns its own dial rather than folding into Mesh Velocity because the relationship between coordination and payoff is not linear. Liu’s simulations show collective efficiency collapsing by 91.4 per cent from the lowest to the highest CTC quartile — a small rise in handoff cost near the top of the range can erase the entire benefit of splitting work across agents. A mesh that tracks Mesh Velocity alone can look healthy while quietly climbing toward that cliff.

**Collective Efficiency (CE)****Target:** maximise; monitor trend, not a fixed threshold

$$CE_{to} = \frac{Q_{to} \times P(S_{to} = 1)}{1 + \lambda \text{Cost}_{to}}$$

Quality  $Q_{to}$  multiplied by the probability of success  $P(S_{to} = 1)$ , discounted by weighted cost.

**Collective Efficiency** is Liu’s outcome measure, and it is deliberately stricter than quality alone: an output that is marginally better but slow, expensive, or hard to audit scores as a worse organisational outcome than a slightly lower-quality output produced reliably and cheaply. That is the same discipline that Trust Variance and HITL Precision enforce qualitatively, now given a shared quantitative vocabulary with the wider organisational-science literature rather than four bespoke metrics invented for this report alone.

## 9.7 The KPI Dashboard as a Narrative

**Table 9.1:** Summary of agentic organisation KPIs

| KPI                         | Symbol | Target         | What It Measures                                       |
|-----------------------------|--------|----------------|--------------------------------------------------------|
| Mesh Velocity               | $MV$   | $< 48\text{h}$ | How fast the organisation learns                       |
| Augmentation Ratio          | $AR$   | $> 65\%$       | How much cognitive load the mesh handles               |
| Trust Variance              | $TV$   | $< 0.12\sigma$ | Whether the mesh is drifting                           |
| HITL Precision              | $HP$   | $> 90\%$       | Whether humans are engaged on the right things         |
| Contextual Transaction Cost | $CTC$  | minimise       | The context-movement cost of every cross-agent handoff |
| Collective Efficiency       | $CE$   | maximise       | Quality and reliability delivered per unit of cost     |

These six metrics tell a coherent story when read together. Mesh Velocity measures the speed of the learning loop. Augmentation Ratio measures the depth of cognitive offloading. Trust Variance monitors whether the system is degrading. HITL Precision evaluates whether human attention is allocated efficiently. Contextual Transaction Cost and Collective Efficiency, drawn from Chapter 7, complete the picture from underneath: CTC prices the coordination the mesh is doing, and CE prices what that coordination is worth.

For the C-Suite audience, this is a dashboard that replaces “how many people do we have” with

“how fast is our organisation learning and how reliable is the learning.” For the change manager, it provides measurable progress indicators for the continuous transformation described in Chapter 11. For the judgment broker, it defines the parameters of their role: they succeed when Trust Variance stays low and HITL Precision stays high.

#### Abandoning Legacy KPIs

The following metrics should be deprecated or substantially de-emphasised in agentic organisations:

- **Total Headcount** — measures the cost of the information routing system, not its effectiveness.
- **Billable Hours** — measures time spent, not value delivered; incentivises inefficiency.
- **Task Completion Rate** — measures throughput without distinguishing between routine tasks (which should be automated) and frontier tasks (which require judgment).
- **Dashboard Views** — measures information consumption, not decision quality.
- **Reporting Latency** — measures the speed of a system that the agentic mesh renders unnecessary.





# Chapter 10

## The Governance Paradox

The standard framing: governance slows you down but keeps you safe. The agentic mesh framing: governance *is* what allows you to go fast.

### 10.1 Governance as Accelerator, Not Brake

The conventional relationship between governance and velocity is adversarial. Governance imposes constraints — approval workflows, compliance reviews, audit requirements — that slow execution. Organisations accept this cost as the price of safety, regulatory compliance, and risk management. The faster you want to go, the more governance you must sacrifice; the more governance you impose, the slower you become.

The Dynamic Agentic Mesh inverts this relationship. In an agentic architecture, governance is not a constraint applied to the system from outside. It is a property of the system itself — embedded in the architecture, enforced automatically, and operating at the speed of the mesh rather than the speed of human review.

### 10.2 The Racing Car Principle

Formula 1 cars are the fastest production-adjacent vehicles on earth. They go faster *because of* precision engineering and guardrails, not despite them. Telemetry systems provide continuous, real-time monitoring of every component. Traction control prevents the power from exceeding the grip. Pit-wall governance provides strategic oversight while the driver maintains operational control.

An F1 car without telemetry, traction control, and pit-wall governance does not win. It crashes on lap three.

The analogy is precise. An agentic mesh without embedded governance — without real-time monitoring, without authority boundaries, without continuous trust measurement — does not

produce faster organisational outcomes. It produces the ant death spiral, the correlated-failure cascades documented across state-of-the-art multi-agent frameworks (Cemri et al. 2025), and the catastrophic failures that cause organisations to retreat to hierarchical control.

**Key Insight**

Speed and governance are not in tension. Speed *requires* governance. The question is not whether to govern the mesh, but how to architect governance so that it operates at mesh speed rather than human-review speed.

10.3 Declarative Governance

In the Dynamic Agentic Mesh, policies are code. Bias thresholds, access controls, and audit trails are embedded into every DAG transition — not bolted on after the fact. The mesh does not need a separate governance review step because governance is architecturally embedded in every step.

This is the distinction between *imperative* and *declarative* governance:

Table 10.1: Imperative versus declarative governance

| Dimension    | Imperative (Traditional)                 | Declarative (Mesh)                      |
|--------------|------------------------------------------|-----------------------------------------|
| Enforcement  | Periodic human review                    | Continuous automated enforcement        |
| Speed        | Bounded by reviewer availability         | Operates at mesh speed                  |
| Coverage     | Sample-based (auditors check a fraction) | Complete (every transition is checked)  |
| Adaptation   | Manual policy updates                    | Policies update as the mesh learns      |
| Visibility   | Post-hoc audit trails                    | Real-time governance dashboards         |
| Failure mode | Undetected non-compliance                | False positive alerts (over-governance) |

The shift from imperative to declarative governance is analogous to the shift from imperative to declarative infrastructure in software engineering. In the imperative model, an administrator manually configures each server. In the declarative model (Kubernetes, Terraform), the administrator declares the desired state, and the system continuously ensures that reality matches the declaration. The same logic applies: declare the governance constraints, and the mesh continuously ensures compliance.

## 10.4 HITL by Design

In the Dynamic Agentic Mesh, the Human-in-the-Loop is not a fallback mechanism invoked when something goes wrong. It is an architectural feature — a designed component of the system that operates continuously.

Agents know their authority boundary. They surface exceptions cleanly, presenting the judgment broker with the relevant context, the decision options, and the governance constraints that apply. The judgment broker handles genuine edge cases — the situations that fall at the jagged frontier, where automated reasoning is unreliable and human judgment is required.

Critically, the mesh learns from each human decision. When the judgment broker overrides an agent recommendation, the override is recorded along with the reasoning. This data feeds back into the mesh, improving its boundary detection and reducing false escalations over time. The HITL Precision metric (Chapter 9) measures this learning directly.

## 10.5 Ontological Provenance

Every decision in the mesh traces back through the knowledge graph. Auditors can traverse the full reasoning chain — agent by agent, task by task — from a final output back to the original inputs, policy constraints, and human decisions that produced it.

This provenance is grounded in formal ontology. The OWL 2 knowledge representation provides shared semantics across all agent skills. The same concept of “deliverable” means the same thing to a Creative Production agent and a Governance agent. The same concept of “risk assessment” is understood by the orchestration layer as a sub-task of “governance review” and routed accordingly — through ontological subsumption, not keyword matching.

**Why Ontological Provenance Matters**

In a conventional organisation, the audit trail for a decision is reconstructed after the fact from emails, meeting minutes, and document versions. The reconstruction is inevitably incomplete, biased by memory, and expensive to produce.

In an agentic mesh with ontological provenance, the audit trail is a by-product of normal operation. Every agent decision is grounded in an OWL 2 concept. Every task transition is recorded with its governing policies. The reasoning chain is available in real time, not reconstructed months later by forensic accountants.

For regulated industries — financial services, healthcare, defence — this is a significant governance advantage. Regulatory compliance becomes a query against the knowledge graph, not a multi-week audit exercise.

**10.6 Cascading Trust Hierarchies**

Agent identities in the mesh operate with compliant key rotation. Each agent has a cryptographic identity that establishes its authority to perform specific actions. When an agent is revoked — because it has been compromised, because its scope has changed, or because the policy constraints governing its behaviour have been updated — the revocation cascades automatically through all agents that depend on the revoked agent's authority.

This cascading trust model prevents the accumulation of stale permissions that characterises most enterprise identity systems. In a conventional organisation, access controls accumulate over time: employees change roles but retain their old permissions; systems are decommissioned but their service accounts persist. The result is a growing attack surface of orphaned permissions.

CyberArk (2025) documents this pattern in financial services, where the ratio of machine identities to human employees has reached 96 to 1, with only 31 per cent of organisations having adequate controls. The cascading trust hierarchy addresses this directly: when an agent's identity or authority changes, the change propagates automatically through the dependency graph.

**10.7 Security as Foundation, Not Friction**

The claim that governance accelerates rather than brakes is not just architectural reasoning; it is now a field finding. Across the 51 mature deployments Stanford studied, security was never the sole reason a project was killed (Pereira, Graylin, and Brynjolfsson 2026). More strikingly, requirements that initially blocked a project routinely went on to enable use cases that

competitors without the same obligations could not touch. A large retail bank's PII-scrubbing and synthetic-substitution pipeline — built to satisfy regulators before any customer data could go near a cloud model — became the infrastructure that let it serve sensitive, regulated customer interactions with agentic AI at all.

The security tax is real, but it is front-loaded. Once the scrubbing pipelines, the vendor contracts, and the archival and audit systems exist, every subsequent use case inherits them at close to zero marginal cost. This is the same compounding logic as the data moat described in Chapter 8, applied to governance infrastructure rather than data: the first regulated use case is expensive; the tenth is nearly free.

Shadow AI is what happens when that tax goes unpaid rather than front-loaded. One manufacturer in the Playbook sample discovered 1,500 to 1,600 unauthorised AI tools already in use across the organisation before it built an internal platform to bring them into the light (Pereira, Graylin, and Brynjolfsson 2026). The tools were being used regardless of governance; the only question was whether the organisation paid the security cost deliberately, once, or paid it chaotically, tool by tool, indefinitely. This is the governance paradox showing up in field data rather than architecture diagrams: the organisations with the strictest data obligations, having already made the investment, are the ones now running customer-facing agentic systems that their less-regulated competitors cannot build at all.

The same field data revises who resists and why, and it matters for how the mesh's governance role should be presented. Staff functions — Legal, HR, Risk, Compliance — were the single largest source of resistance to AI programmes, at 35 per cent of cases, ahead of internal end-users at 23 per cent (Pereira, Graylin, and Brynjolfsson 2026). But the Playbook found these functions converted into enablers, not obstacles, the moment they were given a governance role rather than a veto. Declarative governance is the mechanism that gives them exactly that role inside the mesh: their constraints stop being a review queue that runs at committee speed and become code that runs at mesh speed, checked on every transition rather than sampled at an audit. The staff function that used to be a phase gate becomes a co-author of the policy the mesh enforces automatically.

## 10.8 The Governance Paradox Resolved

The paradox dissolves once governance is understood as an architectural property rather than an external constraint. In a mesh with declarative governance, ontological provenance, and cascading trust:

- **Governance does not slow execution** because it is not a review step. It is a property of

every transition.

- **Governance improves over time** because the mesh learns from judgment broker decisions, reducing false escalations and improving boundary detection.
- **Governance provides competitive advantage** because it enables the organisation to deploy agentic automation in domains where ungoverned automation is unacceptable — regulated industries, high-stakes decisions, customer-facing processes.
- **Governance enables trust** at the institutional level, allowing boards, regulators, and customers to accept agentic decision-making because the reasoning is auditable and the constraints are verifiable.

What this chapter calls embedded governance has an academic name. Liu (2026) (Chapter 7) formalises this exact arrangement as an **interface organisation** — the designed structure that aligns human accountability with agentic context coordination: what evidence must travel with an agent’s output before it enters a human workflow, what uncertainty must be disclosed alongside it, what permissions the agent held when it produced it, what trace of its reasoning is preserved, and where in the workflow human judgment is mandatory rather than optional. Declarative governance, ontological provenance, and cascading trust are DreamLab’s specific implementation of an interface organisation; Liu’s contribution is the general theory explaining why an arrangement of this kind is necessary at all, independent of any particular technology stack.

#### The VisionClaw Evidence

This governance model is not theoretical. The VisionClaw system (Chapter 14) implements OWL 2 ontology with formal reasoning, providing shared semantics across 83 agent skills. Task decomposition uses ontological subsumption, and every agent decision is grounded in the knowledge graph.

The credibility argument: DreamLab has built and operates this governance architecture in production. The limitations are equally important: 15-person team, creative technology context, founder-led. The governance model is proven under favourable conditions. Whether it scales to larger, more regulated, and less technically fluent organisations is an open question addressed in Chapter 17.

# Chapter 11

## The Change Problem

Traditional change management assumes a start state, an end state, and a managed transition between them. AI-driven organisational change has no end state.

### 11.1 Why This Cannot Be Managed with a Gantt Chart

Traditional change management rests on a shared assumption: that change has a beginning, a middle, and an end. Kotter's eight-step model progresses from urgency through institutionalisation. ADKAR treats change as individual capability-building through five sequential stages. Lewin's foundational model freezes the current state, changes it, and refreezes into a new stable configuration.

AI-driven organisational change violates this assumption at its root. The jagged frontier moves as models improve. New capabilities emerge monthly. The compounding loop generates continuous structural adaptation. There is nothing to "refreeze" into.

**Evidence for Framework Failure**

- McKinsey’s 2024 “State of AI” survey: 70 per cent of AI transformations stall — identical to generic change failure rates. Traditional frameworks add no value for AI specifically.
- Gartner, Inc. (2025): organisations using traditional change management for AI are 2.4 times more likely to experience “initiative fatigue.”
- Prosci (2024), the creators of ADKAR, published revised guidance in late 2024 acknowledging that AI requires “iterative change portfolios” rather than single-project applications.
- Ninety-five per cent of generative-AI pilots produce no measurable financial impact; the failure is integration and incentives, not the underlying models (Challapally et al. 2025). Perpetual change infrastructure is the remedy this chapter argues for, not another isolated pilot.

The consultancy consensus is converging: AI demands perpetual change infrastructure, not transformation programmes with end dates.

## 11.2 The Continuous Change Architecture

Several frameworks have emerged that address the perpetual nature of AI-driven transformation. None is complete in itself, but together they indicate the direction of travel.

### 11.2.1 Continuous Transformation

Wade and Bonnet (2025) at IMD propose three required capabilities: organisational dexterity (the structural capacity to reconfigure), workforce adaptability (the human capacity to learn continuously), and technological scaffolding (the infrastructure that enables both). Change is always-on. The transformation office becomes a permanent function, not a temporary project structure.

### 11.2.2 Adaptive Change Architecture

Deloitte (2026) describe permanent teams, feedback loops, and experimentation cadences replacing one-off transformation offices. The adaptive change architecture is designed for continuous evolution, with the organisation’s change capacity treated as a core competency rather than a temporary project skill.



### 11.2.3 Dynamic Capabilities

Teece (2025) refreshes his influential framework for the AI era. The three capacities — sensing (detecting threats and opportunities), seizing (mobilising resources to capture opportunities), and reconfiguring (restructuring to maintain fitness) — describe a meta-capability for continuous adaptation. The organisation does not change once; it builds the capacity to change continuously.

### 11.2.4 Directed Autonomy

Microsoft (2025a) describes a pattern where leadership sets guardrails and investment priorities while teams choose their own tools and use-cases within those boundaries. Organisations practising directed autonomy show 2.8 times higher productivity gains from AI than those using top-down mandates.

#### Key Insight

The mesh architecture is inherently a continuous change system. The Insight Ingestion Loop (Chapter 6) *is* the change process — running perpetually, not as a project. The judgment broker *is* the change agent — embedded in the structure, not parachuted in for a six-month engagement.

## 11.3 What 51 Successful Deployments Teach

Everything so far in this chapter has been diagnostic: frameworks fail, adoption follows a predictable curve, resistance clusters into nameable patterns. What the field lacked, as of the April 2026 edition of this report, was ground truth from AI transformations that actually worked. Pereira, Graylin, and Brynjolfsson (2026) closes that gap. Stanford’s Digital Economy Lab ran structured 60-minute interviews across 51 mature deployments in 41 organisations, spanning 9 industries, 7 countries and a combined workforce of more than one million people — selecting only projects that had been live in production for at least three months, with quantified value and demonstrated scalability. The findings do not just illustrate this chapter’s argument; in several places they correct it.

### 11.3.1 The Hard Part Was Never the Technology

Seventy-seven per cent of the hardest challenges these organisations faced were invisible, intangible costs: change management, data quality, process redesign. Technology was consistently, in the deployments’ own retrospective framing, “the easiest part.” This is the claim this chapter

has made from its title onward, now with a number attached: the work that actually determines whether an AI deployment survives contact with an organisation is the work traditional IT project management is least equipped to see, still less manage on a Gantt chart.

### **11.3.2 Failure Converts to Learning Only With a Constant Sponsor**

Sixty-one per cent of the successful deployments in the sample had at least one failed AI attempt behind them. The failures were not random: they came from treating AI as a technology project rather than a process-and-change project — the exact framework failure documented above. What made the difference between a failure that became a write-off and a failure that became a foundation was continuity of sponsorship. In every traceable case, the same executive who sponsored the failed attempt also sponsored the eventual success. No one was punished for a failed AI initiative in any of the 51 cases studied. Sponsor continuity, not project methodology, is what converts a failed attempt into an input for the next one.

### **11.3.3 Timeline Variance Is Organisational, Not Technical**

The same use case took weeks to deploy at one organisation and years at another, with no meaningful difference in the underlying technology. What varied was organisational: executive sponsorship accelerated timelines in 43 per cent of cases, an existing platform foundation in 32 per cent, and end-user willingness in 25 per cent. None of the accelerators was a faster model or a better GPU. Most decisively for this chapter's argument: 100 per cent of the successful projects in the sample used an iterative approach. None used waterfall. This is direct field confirmation, not theoretical inference, of the case this chapter has made against phased transformation programmes.

### **11.3.4 Sponsorship Is a Behaviour, Not a Title**

The playbook resolves “executive sponsorship” into four distinguishable behaviours, not one binary state:

All seven of the organisation-wide transformations in the sample reached Strategic Integration — the widest-scope changes in the dataset were, without exception, the ones where AI adoption stopped being a project a sponsor supported and became a target the organisation was compensated against. Effective sponsors also created explicit permission to fail, and business-plus-technology co-sponsorship was what unlocked cross-functional projects: as one interviewee put it, “when AI is tech-led and tech-first, it does not work or it rarely works.”

**Table 11.1:** The sponsorship ladder (Pereira, Graylin, and Brynjolfsson 2026)

| Level                 | Behaviour                                                      | Share of cases |
|-----------------------|----------------------------------------------------------------|----------------|
| Passive Approval      | Signs off at initiation; no further engagement                 | —              |
| Periodic Oversight    | Reviews progress at scheduled milestones                       | —              |
| Active Steering       | Weekly check-ins; proactively removes blockers                 | 58%            |
| Strategic Integration | AI adoption written into corporate OKRs and tied to incentives | 29%            |

**Key Insight**

The load-bearing finding for the mesh architecture is the iterative-only result: 100 per cent of successful deployments iterated, none waterfalled. That is not supporting evidence for this chapter's argument against phased transformation — it is direct confirmation. Sponsor continuity through failure, and the sponsorship ladder above, are the organisational mechanisms that make the judgment broker's authority durable rather than provisional: a judgment broker without an Active Steering or Strategic Integration sponsor above them is exactly as exposed as any other change initiative to the framework failure documented at the start of this chapter.

## 11.4 The 1-9-90 Pattern

Boston Consulting Group (2025b) identifies a predictable adoption distribution across organisations deploying AI at scale:

**Table 11.2:** The 1-9-90 adoption distribution

| Segment | Profile                | Behaviour                                                                                                               |
|---------|------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1%      | Natural innovators     | Already found the shadow workflows; experimenting independently; often invisible to management                          |
| 9%      | Champions with support | Will adopt with encouragement, experimentation time, and psychological safety; these are the judgment broker candidates |
| 90%     | Followers              | Will adopt once social proof exists; need to see the 9% succeed before committing                                       |

Investing in the 9 per cent yields the highest return. These are the people who become judgment brokers — the middle managers who learn to map the jagged frontier in their domain and facilitate the compounding loop for everyone else. The 1 per cent will adopt regardless of

organisational support. The 90 per cent will follow once the 9 per cent provide social proof. The strategic investment is in the 9 per cent.

### Evidence

Organisations where middle managers were given protected AI experimentation time showed **3.1 times higher adoption rates** than those relying on top-down mandates alone (Boston Consulting Group and MIT Sloan Management Review 2025).

## 11.5 Resistance Patterns

Capgemini Research Institute (2025) and MIT Sloan Management Review (2024–2025) document five distinct resistance patterns among middle managers:

1. **Role obsolescence fear** (67% of surveyed middle managers): the belief that AI will eliminate their role entirely.
2. **Competence anxiety** (58%): the fear of being seen as less capable than AI or than junior staff who adopt AI faster.
3. **Loss of information asymmetry** (49%): AI democratises access to data that previously conferred authority. The middle manager who knew the numbers — who had the report, the context, the historical comparison — loses their informational advantage when AI can generate that context for anyone.
4. **Decision authority erosion** (45%): AI recommendations bypass managerial judgment. When the system suggests a course of action directly to the team, the manager's role as decision-maker is implicitly challenged.
5. **Accountability ambiguity** (52%): who is responsible when AI-assisted decisions fail? The manager who approved the AI recommendation? The engineer who configured the agent? The organisation that deployed the system?

### Effective Interventions

Evidence-based interventions that address these resistance patterns include:

- **Personalised AI coaching** — not generic training programmes, but one-to-one support tailored to each manager’s domain, skill level, and specific anxieties.
- **Protected experimentation time** — dedicated hours in which managers can explore AI tools without performance pressure (the 3.1x adoption multiplier).
- **Role reframing** — explicitly repositioning the management role toward judgment, context, and strategic alignment rather than information processing.
- **Visible executive modelling** — senior leaders publicly using AI, acknowledging their own learning curve, and demonstrating that AI adoption is expected at all levels (Mollick 2025).

### 11.5.1 Where Resistance Actually Originates

The five patterns above are real, well-evidenced, and specific to middle managers. They are also, on the evidence of actual deployments, not usually what stalls a project. The April 2026 edition of this report, like most of the change-management literature it drew on, centred middle-manager resistance as the primary obstacle to AI adoption. Pereira, Graylin, and Brynjolfsson (2026)’s deployment data says the binding constraint is usually elsewhere: across the 51 cases studied, staff functions — Legal, HR, Risk, Compliance — were the single most frequent source of resistance, cited in 35 per cent of cases, ahead of internal end-users at 23 per cent. Frontline fear of replacement, the pattern most discussed in management commentary and the one nearest the top of the enumerated list above, appeared in only two of the 51 cases. IT, popularly cast as the department that says no, was typically an enabler.

Each source resists for a different reason, and each requires a different intervention, not a single change-management playbook applied uniformly:

- **Staff functions** fear process risk and personal liability. They do not respond to persuasion; they respond to a mandate paired with a formal governance role — a seat inside the declarative governance structure of Chapter 10, not a stakeholder briefing delivered outside it.
- **C-level stakeholders** demand measured, ROI-quantified pilots before committing further investment.
- **End users** distrust the AI’s non-determinism; the intervention is expectation-setting about what the system will and will not do consistently, not reassurance that it works.
- **Frontline staff**, in the rare cases where replacement fear surfaces, need a concrete role path

rather than a general commitment to “upskilling.” One studied deployment put the reframe directly: “AI is replacing the person you don’t need to hire, not the person you have.”

This is a correction this report is willing to make openly. The five-pattern taxonomy above remains accurate for middle managers, and the identity work it implies is real — Chapter 13 takes that work seriously on its own terms. But it is identity transformation, not blocker removal. The resistance that actually gates whether a project reaches a middle manager’s team at all more often sits one or two organisational layers away, in the staff functions with the standing to say no (Pereira, Graylin, and Brynjolfsson 2026).

### 11.6 The Contradiction in This Report

A sophisticated reader will notice a tension within this report: Chapter 15 provides a 90-day phased roadmap, despite this chapter’s argument that phased transformation models fail for AI. The contradiction deserves direct address.

The 90-day roadmap is not the change plan. It is the *bootstrap sequence* for the perpetual change infrastructure. The roadmap launches the compounding loop described in Chapter 8. Once launched, the loop is the plan — running continuously, adapting to frontier shifts, and generating its own change momentum through the Insight Ingestion Loop.

The distinction is between a programme with an end date and a system with a start date. Traditional change management produces the former. The Dynamic Agentic Mesh requires the latter. The 90-day roadmap is the start date. Phase 3 never ends.

# Chapter 12

## The Industry Landscape

The consultancy consensus has converged on the fact of transformation. It has not converged on the architecture.

### 12.1 What the Major Consultancies Are Saying

The period from 2025 to 2026 has seen an extraordinary convergence in the consultancy literature. Every major firm has published substantial analysis of AI-driven organisational transformation. The agreement on diagnosis is remarkable; the disagreement on prescription is instructive.

### 12.2 McKinsey: The Agentic Organisation

McKinsey has been the most explicit in naming the paradigm shift. Their “Agentic Organization” framework (McKinsey & Company 2026b) identifies five pillars: business model, operating model, governance, workforce and culture, and technology and data. Their companion publication, “Six Shifts to Build the Agentic Organization of the Future” (McKinsey & Company 2026a), describes the transition from employees performing tasks to employees orchestrating outcomes — moving “above the loop.”

McKinsey’s contribution is strategic framing. They describe the pillars of transformation comprehensively. What they do not provide is the operational architecture — the specific mechanisms by which an organisation moves from the current state to the agentic state. The pillars are necessary but not sufficient.

### 12.3 BCG: Quantifying the Shift

Boston Consulting Group (2025b) and Boston Consulting Group (2025a) provide the strongest quantitative evidence:

- 35 per cent of companies already use agentic AI; another 44 per cent plan near-term adoption.
- 45 per cent of AI leaders expect fewer middle-management layers.
- AI-powered workflows accelerate business processes by 30–50 per cent in finance, procurement, and customer operations.
- 50–55 per cent of US jobs could be reshaped in two to three years through task decomposition (Boston Consulting Group 2026).

BCG's joint research with MIT Sloan on AI-powered KPIs (Boston Consulting Group and MIT Sloan Management Review 2025) is particularly relevant: the finding that organisations revising their KPIs are three times more likely to see financial benefit from AI validates the measurement framework proposed in Chapter 9.

BCG quantifies the shift more rigorously than any other consultancy. Their limitation is the same as McKinsey's: they describe the destination without providing the architectural blueprint for getting there.

### 12.4 Deloitte: Measuring Maturity

Deloitte (2026) focuses on the human-AI interaction design gap. Their findings are sobering:

- Only 14 per cent of leaders are adept at shaping human-AI interactions.
- 59 per cent layer AI onto legacy systems rather than redesigning workflows around AI capabilities.
- Only 11 per cent have deployed autonomous decision-making in production.

Deloitte's contribution is diagnostic: measuring how far organisations have come and, implicitly, how far they have to go. The gap between the 35 per cent adoption rate (BCG) and the 11 per cent production deployment rate (Deloitte) reveals that most “adoption” is surface-level — tools deployed without workflow redesign, AI bolted onto existing processes rather than integrated into new ones.

### 12.5 Gartner: Structural Predictions

Gartner, Inc. (2025) provides the most provocative predictions:

- By 2026, 20 per cent of organisations will use AI to flatten structures, **eliminating more than half of current middle management positions.**



- Simultaneously, 50 per cent of organisations will require “AI-free” skills assessments due to critical-thinking atrophy from generative AI dependency.

These two predictions are in productive tension. The first says middle management layers will shrink. The second says the human capabilities that middle managers provide are at risk of degrading. Together, they describe the vigilance problem at organisational scale: the better AI becomes at coordination, the less organisations invest in human coordination capability, and the more vulnerable they become when AI coordination fails.

## 12.6 World Economic Forum: The Jobs Landscape

World Economic Forum (2025) provides the macro-economic context:

- 39 per cent of core skills will change by 2030.
- 170 million new roles will be created globally; 92 million will be displaced, for a net gain of 78 million.
- The net positive figure conceals significant distributional effects: the new roles require different skills, are located in different regions, and are accessible to different populations than the displaced ones.

## 12.7 The Deployment Ground Truth

The 2026 landscape gained something the April edition of this report did not have: systematic evidence from deployments that worked, not just survey data about intentions. Pereira, Graylin, and Brynjolfsson (2026) studied 51 mature deployments across 41 organisations and 9 industries, and its headline findings read like a field validation suite for arguments made throughout this report:

- 77 per cent of the hardest challenges were invisible, intangible costs, not technology (Chapter 11).
- Escalation-based oversight models — AI acts autonomously, humans review exceptions — produced 71 per cent median productivity gains versus 30 per cent for approval models requiring human sign-off on every output.
- Staff functions (Legal, HR, Risk, Compliance), not frontline workers, were the most frequent source of resistance, at 35 per cent of cases.
- Agentic implementations were only 20 per cent of cases but delivered 71 per cent median gains versus 40 per cent for high-automation-but-supervised approaches.

- Model choice was a commodity in 42 per cent of cases; the durable competitive advantage, in the report’s own words, is “in the orchestration layer, not the foundation model.”

Read against Challapally et al. (2025)’s finding that 95 per cent of generative-AI pilots produce no measurable financial impact, the Playbook is a study of the surviving 5 per cent — a selection bias its own authors acknowledge. It is evidence about what success looks like when it happens, not a representative sample of what usually happens. Both facts belong in the same paragraph: AI deployment fails in the overwhelming majority of attempts, and the minority that succeeds shares identifiable, reproducible structural features.

The headcount finding deserves the same care. Reduction was the largest single outcome, at 45 per cent of cases — but the alternatives to reduction (hiring avoided, redeployment to other work, no reduction at all) summed to 55 per cent. The authors’ own caveat is worth stating plainly: 45 per cent reduction “may represent a floor, not a ceiling,” since the sample captures early-adoption-phase behaviour before capabilities and confidence both increase. The honest reading is that headcount outcomes are a strategic choice available to organisations, not a technical inevitability imposed by the AI — the same conclusion this report reaches independently in Chapters 8 and 9.

## 12.8 The Frontier and the Labour Market

Two further bodies of evidence completed the picture between April and July 2026; both are treated in depth elsewhere in this report and are registered here only for landscape completeness.

The Remote Labor Index (Mazeika 2026; Mazeika, Gatti, Menghini, et al. 2025) put a number on how fast the frontier itself is moving: the client-quality automation rate for professional freelance deliverables rose from 2.5 per cent to 16.1 per cent in eight months. Chapter 4 treats this measurement, and what it does and does not imply, in full.

The labour market told a split-screen story. Displacement was visible and concentrated in tech and finance (Prakash and Boesler 2026); entry-level hiring in AI-exposed occupations declined (Brynjolfsson, Chandar, and R. Chen 2025). At the same time, high-intensity AI adopters grew headcount rather than shrinking it (Kharazian, Simon, and Stevens 2026; Levie 2026) — the latter a CEO-reported survey without published methodology. Chapter 3 holds both halves of that story together, because the report’s argument requires it: the distribution of outcomes is an organisational choice, not a technological verdict.

Three further data points shifted the narrative, rather than the numbers. OpenAI’s own leadership walked back earlier replacement rhetoric (Altman 2026; Chatterji 2026). China made augmentation-over-replacement a matter of legal precedent, with a Hangzhou court ruling a

company's AI-driven layoff illegal (C. Edmondson 2026). And the reversal pattern already visible in this report's discussion of Klarna gained a second, unrelated case: Ford's rehiring of veteran "greybeard" engineers to fix quality problems AI tooling alone could not solve (Naughton 2026) — treated in full in Chapter 13.

## 12.9 Agentic AI Adoption at Scale

The pace of adoption has been extraordinary. Agentic AI adoption surged 3,440 per cent in 2025, with 67 per cent of Fortune 500 companies deploying some form of agentic system. In financial services, the ratio of machine identities to human employees has reached 96 to 1 (CyberArk 2025) — but only 31 per cent of organisations have adequate controls in place.

This gap between deployment velocity and governance capability is the defining challenge of the current moment. Organisations are deploying agentic systems faster than they are developing the capacity to govern them.

Deployment data adds a second gap alongside it. Pereira, Graylin, and Brynjolfsson (2026) found that true agentic implementations — autonomous, end-to-end, performing a role rather than answering a prompt — were still only 20 per cent of the 51 studied deployments, despite delivering a 71 per cent median productivity gain against 40 per cent for high-automation-but-human-supervised approaches. The premium for going fully agentic is large and the uptake is still small: organisations are racing to deploy AI but remain reluctant, or unready, to hand it the whole loop. That gap between what agentic AI can already return and how rarely organisations commit to it is the adoption-side mirror of the governance gap above.

**Table 12.1:** Consultancy and practitioner positioning on agentic transformation

| Source                  | Contribution                                                        | Gap                                            |
|-------------------------|---------------------------------------------------------------------|------------------------------------------------|
| McKinsey                | Strategic pillars; “above the loop” framing                         | No operational architecture                    |
| BCG                     | Quantitative evidence; KPI research; 1-9-90 pattern                 | No governance model                            |
| Deloitte                | Maturity measurement; human-AI interaction gap                      | Diagnostic, not prescriptive                   |
| Gartner                 | Structural predictions; vigilance warning                           | No implementation framework                    |
| WEF                     | Macro-economic context; skills shift                                | No organisational model                        |
| Stanford DEL (Playbook) | Deployment ground truth from 51 successes; oversight calibration    | Selection bias toward success; no architecture |
| Liu (UCL)               | Formal theory: contextual transaction cost, interface organisations | Simulation-based; not yet peer reviewed        |
| CAIS RLI                | Longitudinal client-quality frontier measurement                    | Tasks, not jobs; freelance work only           |
| Block                   | Correct diagnosis; radical prescription                             | Ignores jagged frontier and vigilance          |
| Every                   | Cultural dynamics; personal agent model                             | Unproven beyond 20-person scale                |
| DreamLab Mesh           | Unified model with governance, KPIs, ontology                       | Unproven beyond 15-person team                 |

## 12.10 Comparative Positioning

### The Positioning Opportunity

The Dynamic Agentic Mesh sits between the consultancy frameworks (which are strategic but not operational) and the practitioner experiments (which are operational but not generalisable):

- McKinsey describes the pillars; DreamLab provides the architecture.
- BCG quantifies the shift; DreamLab provides the governance model.
- Deloitte measures maturity; DreamLab provides the KPIs.
- Every proves the cultural dynamics; DreamLab provides the structural framework for scaling them.
- Block proves the diagnosis; DreamLab<sup>86</sup> corrects the prescription.
- Liu supplies the theory the mesh anticipated: contextual transaction cost gives the coordination tax a name and a formalism at the agent layer, developed fully in Chapter 7

# Chapter 13

## The Role Transformation

Not obsolete. Evolved.

### 13.1 The Middle Manager as Judgment Broker

The coordination collapse does not eliminate the need for human judgment in organisations. It eliminates the need for humans to *route information*. The distinction is crucial and determines whether the transformation of middle management is experienced as extinction or evolution.

The judgment broker role is not a euphemism designed to soften a redundancy announcement. It is a description of a genuinely new function that emerges when agentic systems handle information routing and humans are freed to focus on what they do that AI cannot.

Table 13.1: Middle management role transformation

| From                                                | To                                           |
|-----------------------------------------------------|----------------------------------------------|
| Reporting and forecasting (information aggregation) | Federated guardianship of the mesh           |
| Vertical chain of command                           | Horizontal integration across silos          |
| Monitoring completion rates                         | Adjudicating edge cases and ethical dilemmas |
| Gatekeeping information flow                        | Amplifying cross-functional insight          |
| Coordination (now automated)                        | Value alignment and strategic direction      |

### 13.2 Three Irreplaceable Human Capacities

The judgment broker role rests on three capacities that are genuinely irreplaceable by current and foreseeable AI systems. These are not tasks that AI performs poorly today and will perform

well tomorrow. They are categories of judgment that require qualities AI does not possess in any architecturally meaningful sense.

### 13.2.1 Strategic Vectors

Only humans can decide what the organisation should be doing next year — and whether the mesh is pointed at it. AI can optimise for defined objectives with extraordinary efficiency. It cannot define the objectives themselves. The judgment broker ensures that the mesh's optimisation is directed at outcomes that serve the organisation's strategic intent, not merely at outcomes that satisfy the metrics the mesh was trained to maximise.

This is not a trivial distinction. An agentic system optimising for Mesh Velocity might propagate workflows that are fast but strategically irrelevant. An agentic system optimising for Augmentation Ratio might automate decisions that should remain human for political, ethical, or strategic reasons. The judgment broker holds the strategic context that prevents metric optimisation from becoming metric gaming.

### 13.2.2 Ethical Adjudication

Bias, fairness, and consequence live in human judgment. No agent should be the final word on edge cases where the cost of being wrong is reputational, legal, or existential. The judgment broker does not merely flag ethical issues — they adjudicate them, drawing on contextual understanding, stakeholder awareness, and moral reasoning that cannot be reduced to a policy ruleset.

The declarative governance layer (Chapter 10) handles the routine ethical constraints: compliance checks, bias thresholds, access controls. The judgment broker handles the cases that fall between the rules — the novel situations, the competing values, the dilemmas where the right answer depends on context that no policy can fully anticipate.

### 13.2.3 Relational Intelligence

Trust, culture, and coalition-building across departments are the lubrication layer that no algorithm can replicate. Every's experiment proves this directly: agent trust is derivative of human trust (Shipper 2026). The human relationship network is the foundation on which the agent mesh is built.

The judgment broker's relational intelligence serves a specific function in the mesh: it enables the cross-functional insight propagation that the compounding loop depends on. A validated workflow in procurement only becomes organisational capability if it can be adapted and

adopted by finance, logistics, and operations. That adaptation requires understanding not just the workflow but the political dynamics, cultural norms, and institutional sensitivities of each receiving function. This is human work.

## 13.3 The Greybeard Signal

The clearest field confirmation of the judgment broker thesis so far did not come from a management consultancy or a technology vendor. It came from a car manufacturer correcting its own mistake, in public, at scale.

Over three years, Ford rehired roughly 350 veteran “greybeard” engineers — many of them former Ford employees — after AI-accelerated engineering had produced quality problems the AI tooling alone could not fix (Naughton 2026). Their brief was twofold: resolve the defects that had already reached production, and retrain the AI tools themselves, because the tools had been trained on inputs that were not good enough. Ford subsequently topped the mainstream-brand rankings in the J.D. Power Initial Quality Study.

Charles Poon, Ford’s Vice-President of Vehicle Hardware Engineering, put the lesson as plainly as any executive has yet put it in public: “Artificial intelligence is a fantastic tool, but it’s only as good as the information you use to train it.” And, more pointedly: “Mistakenly, we thought that just by introducing AI and ingesting the design requirements that we had that we could produce a high-quality product. But we recognized that for us to enhance some of our automation and machine learning and AI tools, we needed to ensure that they were trained by the most experienced individuals.”

Read this slowly, because it is the clearest field statement yet of the judgment broker’s economic basis. Poon is not describing a transition period in which AI is still catching up with human capability. He is describing a structural dependency: the AI tools needed the judgment of the most experienced engineers to be trained and verified against, and Ford had starved itself of exactly that judgment by reducing its experienced headcount. Deep domain expertise is not decoration on top of the AI system — it is the training and verification substrate the AI system runs on. An organisation that discards its experienced middle layer to fund AI deployment is not trading one input for a better one. It is discarding the input its AI most depends on.

Connect this to the vigilance problem (Chapter 5). That chapter showed that oversight degrades as AI output quality rises: the better the AI looks, the less critically humans review it. Ford’s experience is the same failure approached from the supply side. Vigilance requires someone with the pattern-recognition to notice when confident, fluent AI output is wrong, and that pattern-recognition is built through years of direct exposure to a domain’s failure modes — exactly what

greybeard engineers have, and exactly what a workforce hollowed out in favour of AI tooling does not. Erode the expertise base and there is no one left to be vigilant, regardless of how carefully the oversight process is designed on paper.

Ford's reversal joins Klarna's (Klarna 2025) and the Orgvue/Forrester finding that 55 per cent of companies that rushed to replace humans with AI now regret the decision (Orgvue and Forrester Research 2025) as delaying-reversal evidence. But Ford sharpens the argument. Klarna's reversal was about service quality collapsing when a human relationship was removed from a customer-facing process. Ford's reversal was about the AI's own competence collapsing when the humans who could train and check it were removed from an engineering process. The greybeard signal says the judgment broker is not merely a role that softens automation for the people who lose their jobs. It is infrastructure the AI itself needs in order to keep working.

### 13.4 Management as AI Superpower

Mollick (2025) argues that 20–40 per cent of the competitive value of American firms throughout the twentieth century came from management quality, and that AI amplifies this further. His core thesis is arresting: most companies delegate AI strategy to middle management or IT, which guarantees failure. The CEO must personally use AI, articulate how it changes what the organisation is, and create safe experimentation incentives.

Mollick proposes a three-part organisational model:

1. **Leadership:** The CEO and senior executives must personally use AI — not delegate it, not review reports about it, but use it in their own work. This serves two functions: it gives leaders first-hand understanding of the jagged frontier in their domain, and it provides visible modelling that legitimises AI adoption throughout the organisation.
2. **Crowd:** Employees given permission and psychological safety discover use cases that AI strategy teams never anticipated. The shadow workflows documented in Chapter 3 are the evidence: workers are already discovering applications that management has not imagined.
3. **Lab:** A dedicated cross-functional team pushes the boundaries of what is possible, experiments with frontier capabilities, and translates discoveries into organisational practice.



**Key Insight**

Mollick’s model maps onto the Dynamic Agentic Mesh:

- **Leadership** sets the value vectors and strategic direction.
- The **Crowd** operates the Discovery Engine, surfacing insights from across the organisation.
- The **judgment broker** operates between them, validating discoveries against strategy and governance.
- The **Lab** is the technical team that maintains and extends the mesh infrastructure.

## 13.5 Tasks, Jobs, and the Broker Pipeline

Two threads of new evidence bear directly on whether the judgment broker role, once created, will have anyone left to grow into it.

### 13.5.1 Tasks Are Not Jobs

Speaking at the European Central Bank’s Forum on Central Banking in Sintra in June 2026, OpenAI’s chief economist Aaron “Ronnie” Chatterji made a point that deserves to sit alongside the frontier-quantification evidence in Chapter 4 (Chatterji 2026): task exposure is not the same thing as job substitution, and treating the two as interchangeable skips over the harder question of what a job actually is and how it evolves. He illustrated the point with his own father, an economist in 1985 who found the newly arrived personal computer a complement to his work rather than a threat to it — what had once required a punch card could now be run directly, and the work did not disappear, it changed shape. Chatterji made the same point about the profession most often cited as first in line for displacement: despite years of predictions that software developer employment would shrink as AI capability increased, that contraction has not materialised to the extent forecast.

The distinction matters for how this report’s own frontier evidence should be read. Sixteen per cent task automation at professional quality (Chapter 4) is not sixteen per cent job destruction, and the judgment broker’s expanding remit in this chapter should be understood as a redrawing of task boundaries within a job, not a body count. Altman (2026) supplies a caution about who to trust on this question: in May 2026 the OpenAI chief executive acknowledged that his own earlier predictions about AI-driven job displacement had been “pretty wrong”, and said he was “delighted to be wrong.” The actors making the most confident displacement predictions have, so far, the least reliable track record in this debate.

### 13.5.2 The Pipeline Problem

None of this dissolves the labour-market evidence in Chapter 3 and Chapter 11. It sharpens where the real risk sits. Brynjolfsson, Chandar, and R. Chen (2025) finds that early-career employment — workers aged 22 to 25 — in AI-exposed occupations has fallen by roughly 16 per cent relative to less-exposed occupations. This is not a story about jobs disappearing outright. It is a story about the bottom rung of the ladder being removed while the ladder itself remains standing above it.

Set against that, Kharazian, Simon, and Stevens (2026) finds that firms with the highest intensity of AI adoption grew entry-level headcount by 12 per cent over the same period — faster than their overall headcount growth, and selecting specifically for people who use AI well. The same technology that appears to be thinning entry-level employment in aggregate is, inside adopting organisations, associated with entry-level hiring that outpaces the rest of the firm.

The implication for this chapter is direct. The apprenticeship route into judgment brokerage — the years of direct exposure to a domain’s failure modes that build the pattern-recognition Chapter 5 and the Ford case both depend on — is being rebuilt, not simply destroyed. But it is being rebuilt selectively, inside organisations that have chosen to keep hiring at the entry level, not universally. The broker pipeline is an organisational design choice, in exactly the sense Chapter 8 argues coordination architecture is a choice. Firms that automate their junior rungs on cost grounds, while expecting senior judgment brokers to appear from nowhere a decade later, are eating their own seed corn — the Ford lesson, run in reverse and in slow motion.

## 13.6 The Competency Shift

The transition from information router to judgment broker requires a specific set of competencies that differ significantly from traditional management skills:

## 13.7 The Day-in-the-Life Question

A practical objection from middle managers is: “What does my Monday morning look like as a judgment broker? What am I literally doing at 9am?” This is a fair question, and the answer must be concrete:

**Morning:** Review the Trust Variance dashboard. Identify any agents whose decision quality has drifted outside tolerance. Review the escalation queue — the genuine edge cases that the mesh has surfaced for human judgment. Adjudicate three or four cases, providing reasoning that feeds back into the mesh’s learning.

**Table 13.2:** Competency requirements: information router versus judgment broker

| Domain          | Information Router                          | Judgment Broker                                     |
|-----------------|---------------------------------------------|-----------------------------------------------------|
| Information     | Aggregation and filtering                   | Frontier mapping and pattern recognition            |
| Decision-making | Approval/rejection within defined authority | Edge-case adjudication at authority boundaries      |
| Technology      | Tool user                                   | System thinker; understands mesh behaviour          |
| Communication   | Upward reporting; downward delegation       | Cross-functional translation; stakeholder alignment |
| Risk            | Compliance checking                         | Uncertainty navigation at the jagged frontier       |
| Learning        | Skills training                             | Continuous frontier probing; experimentation        |

**Midday:** Cross-functional alignment session (30 minutes, not 90). The mesh has pre-synthesised the status across all workstreams. The judgment broker’s role is not to hear status reports — the mesh already has those — but to identify strategic conflicts, ethical tensions, and opportunities for cross-pollination that the mesh cannot evaluate.

**Afternoon:** Frontier probing. Spend protected experimentation time actively testing the boundary between what the mesh can handle and what it cannot in your specific domain. Document findings. Update the mesh’s authority boundaries based on what you learn.

**Throughout:** Relational work. The human network that supports the agent mesh requires maintenance — building trust, resolving interpersonal friction, coaching team members in their own agent management, and ensuring that the cultural foundation of the mesh remains healthy.

This is a more intellectually demanding role than the information-routing role it replaces. It is also a more strategically valuable one.

### 13.7.1 The Escalation Economy

The judgment-broker pattern described above is no longer only this report’s proposal. As of July 2026, it is the highest-performing operating model observed in the field. Pereira, Graylin, and Brynjolfsson (2026)’s study of 51 mature enterprise deployments found that escalation-based operating models — AI handling more than 80 per cent of a workflow autonomously, humans reviewing only the exceptions the mesh cannot resolve — delivered a 71 per cent median productivity gain, against roughly 30 per cent for approval-based models in which a human

reviews every output.

The playbook’s per-function detail maps almost exactly onto the frontier-mapping work this chapter assigns the judgment broker. IT operations running an escalation model saw gains around 90 per cent; customer support escalation delivered 71 per cent; clinical documentation, where notes are legal documents and error tolerance is close to zero, stayed on an approval model and delivered roughly 66 per cent; coding collaboration, still requiring close human review, delivered 54 per cent. That pattern is not noise — it is oversight level tracking error tolerance and regulatory exposure, function by function. That is precisely the frontier-mapping exercise the “Afternoon: frontier probing” block above describes: deciding, domain by domain, how much autonomy the mesh has earned and how much still requires a human to hold the boundary.

Liu (2026) gives this role its academic name. The judgment broker is the human side of what Liu calls an **interface organisation** — the designed arrangement connecting human accountability to agentic context coordination, specifying what agent outputs may enter human workflows, what evidence must travel with them, and where human judgment is mandatory (Chapter 7). What this report proposed from first principles, the deployment data and the organisational-behaviour literature now confirm from two independent directions.

# Chapter 14

## The Technical Substrate: VisionClaw

The honest version is more defensible and more credible.

### 14.1 A Working Agentic Mesh

VisionClaw is not a prototype, a whitepaper, or a concept rendering. It is an operational agentic mesh serving a creative technology team at DreamLab, Fairfield, Eskdale. This chapter describes what the system is, what it does, and — equally important — what it does not yet do. The claims have been reconciled against the codebase as of April 2026.

**[FIGURE: VisionClaw Architecture Diagram]**  
*OWL 2 Ontology Engine, Claude-Flow Orchestration, WebXR Visualisation,  
Nostr Identity, and RuVector Memory layers*

**Figure 14.1:** VisionClaw System Architecture

### 14.2 Architecture Stack

The system comprises five integrated subsystems, each verified against the production codebase.

### 14.2.1 OWL 2 Ontology Engine

The knowledge graph uses formal OWL 2 ontology via `horned-owl` (v1.2.0) and `whelk-rs` EL++ inference. The engine parses OWL/XML, Manchester, RDF/XML, and Turtle formats, with Neo4j providing persistence. This subsystem provides the shared semantics for agent skill routing: task decomposition uses ontological subsumption rather than keyword matching, meaning that the system understands that “risk assessment” is a sub-task of “governance review” and routes accordingly.

The OWL 2 engine is the architectural feature that most clearly distinguishes VisionClaw from other agentic systems. Formal ontology provides two capabilities that informal knowledge representations lack: compositionality (new concepts can be constructed from existing ones with guaranteed semantic consistency) and inference (the system can derive relationships that were not explicitly stated). These properties make the governance and provenance mechanisms described in Chapter 10 possible.

### 14.2.2 Claude-Flow Orchestration

Task orchestration operates through Claude-Flow with RAFT consensus. The system integrates 83 agent skills across 12 domains. Agent routing operates through ontology-grounded skill decomposition: when a task arrives, the orchestration layer decomposes it into sub-tasks using the OWL 2 ontology and routes each sub-task to the appropriate specialist agent.

### 14.2.3 WebXR 3D Knowledge Visualisation

The visualisation layer provides a Rust/CUDA/WebXR backend with 92 CUDA kernels (6,750 lines of code) for GPU-accelerated graph physics. The system offers two rendering modes: Babylon.js for immersive and VR interaction, and React Three Fiber for desktop access. The visualisation is proven at approximately 1,000-node scale in production, with 100,000-node performance as a design target under active development.

### 14.2.4 Nostr Agent Identity

Agent identities use the Nostr protocol (NIP-98 HTTP authentication, `nostr-sdk` v0.43.0). This provides agent signing and cascading trust semantics through a self-sovereign identity layer for agent communication. The decentralised identity model avoids dependence on a central certificate authority and provides cryptographic verification of agent actions.

### 14.2.5 RuVector Memory

The persistent memory layer stores over 1.17 million vector embeddings in PostgreSQL with pgvector and HNSW indexing. This provides agents with persistent memory across sessions — a capability that addresses one of the key limitations identified in Every’s agent colony model, where memory gaps cause context loss between sessions (Shipper 2026).

## 14.3 What Is Real and Verified

The following capabilities have been verified against the production codebase as of April 2026:

- OWL 2 reasoning with Whelk-rs EL++ inference — genuine, functional, and unusual in the agentic AI space.
- GPU-accelerated graph physics — 92 CUDA kernels, proven at approximately 1,000-node scale.
- Multi-user WebXR visualisation via Babylon.js — working in production.
- Nostr identity and signing (NIP-98) — working.
- Claude-Flow skill integration — 83 skills in production across 12 domains.
- RuVector memory — 1.17 million embeddings with HNSW indexing.

## 14.4 What Was Previously Overclaimed

### Corrected Claims

A codebase audit identified several claims in earlier versions of this document that exceeded what the code supports. The corrections are:

- “101 specialist skills” is corrected to 83 Claude Code skills across 12 domains. The previous count was inflated.
- Multi-model orchestration via Jarvis, Gemini, and GLM-4 DAG pipelines does not exist. The system uses Claude-Flow only. The DAG struct in the codebase is for GPU physics layout, not multi-model orchestration.
- W3C-compliant DID key rotation is not implemented. Nostr NIP-98 authentication works; no W3C DID implementation or key rotation code exists in the codebase.
- “100,000+ nodes at 60fps” is a unit test case, not a validated production capability. Production screenshots show approximately 934 nodes.
- “4M+ tokens of live knowledge graph” cannot be verified. The system contains 1.17 million vector memory entries; no token counting infrastructure exists.
- The hardware specification of “96GB VRAM — Dual A6000 Ada” appears only in skill examples. The README specifies RTX 4080+ (16GB) as the target hardware.
- “Stratified unidirectional links” — no code or documentation for this pattern was found.

## 14.5 The Corrected Story

The corrected story is still strong. A working OWL 2 + CUDA + WebXR + Nostr + Claude-Flow system with 83 integrated skills and 1.17 million vector embeddings, proven at small-to-medium scale with approximately 1,000-node graphs, is genuinely novel. It does not need inflation. The honest version is more defensible and more credible.

### 14.5.1 What This Proves

- Ontology-grounded agent orchestration works — agents coordinate through formal semantics, not keyword matching.
- GPU-accelerated knowledge visualisation is operationally feasible at production scale.
- Nostr provides a viable decentralised identity layer for agent ecosystems.
- The architecture is production-tested at small-to-medium scale.



### 14.5.2 What This Does Not Yet Prove

- Scalability beyond approximately 50 people or 1,000 nodes in production.
- Adoption by non-technical middle managers.
- The judgment broker role when the broker is not the system architect.
- The Insight Ingestion Loop in a regulated industry.
- The proposed KPIs (Mesh Velocity, Augmentation Ratio, Trust Variance) at scale.
- Multi-provider orchestration (only Claude-Flow is currently implemented).

Chapter 16 takes this list as its work queue: each gap is converted there into a prioritised, checkable roadmap commitment, so that the next edition of this report can state which were closed and which were not.

## 14.6 VisionClaw in Liu's Taxonomy

Chapter 7 introduced Liu's taxonomy of organisational forms for agent collectives, in which agent-native forms decisively outperform human-imitation ones (Liu 2026). VisionClaw's sub-systems map onto that taxonomy with some precision. The RuVector memory layer is a black-board: agents coordinate by reading and writing shared, durable state rather than relaying context through manager agents — the form Liu finds achieves the highest quality of any he tests. Claude-Flow's ontology-grounded routing is a partial adaptive meta-organisation: task decomposition restructures the collective per task, using ontological subsumption as the routing signal, though without the explicit CTC-awareness of Liu's orchestrator. And the declarative governance layer combined with the judgment broker is an interface organisation: it specifies what may cross from the agentic side into human accountability, and with what evidence attached.

This mapping is interpretive, not measured. Taxonomic resemblance is evidence that the architecture is pointed the right way; it is not evidence that VisionClaw's shared-state coordination pays the context tax at the favourable rates Liu's simulations report. Testing that requires the production CTC instrumentation that does not yet exist here — a gap Chapter 16 converts into an explicit roadmap commitment.

## 14.7 Positioning Guidance

In any external presentation, the following guidelines should govern how VisionClaw is described:

1. Use "VisionClaw" consistently (the actual system name), not "VisionClaw" (used inconsis-

tently in earlier documents).

2. Lead with what is verified. Frame the 100,000-node target and multi-provider orchestration as roadmap items, not current reality.
3. Position VisionClaw as an “existence proof under favourable conditions” rather than “proof” — honest about the gap between a 15-person creative technology team and a 500-person manufacturing firm.
4. The audience for this document includes people who will check. Overclaiming undermines the genuine strengths of the system and the credibility of the broader argument.

### Key Insight

The value of VisionClaw to the coordination collapse argument is not that it is the largest or most powerful agentic system. It is that it demonstrates a specific architectural approach — ontology-grounded orchestration with embedded governance — that the theoretical argument requires. The existence of a working system, even at modest scale, transforms the argument from “this could work in theory” to “this works in practice, and the question is how far it scales.”

# Chapter 15

## From Org Chart to Agentic Mesh: Implementation

Phase 3 never ends.

### 15.1 The Bootstrap Sequence

This chapter provides a 90-day implementation roadmap. The framing is deliberate: this is not a transformation programme with an end date. It is the *bootstrap sequence* for the perpetual change infrastructure described in Chapter 11. The roadmap launches the compounding loop. Once launched, the loop is the plan.

The contradiction noted in Chapter 11 — that a phased roadmap sits inside a report criticising phased transformation models — is resolved by this distinction. Traditional change management produces a programme with an end date. The Dynamic Agentic Mesh requires a system with a start date. The 90-day roadmap is the start date.

#### Key Insight

The 90-day roadmap does not implement the mesh. It creates the conditions under which the mesh can begin to compound. The mesh implements itself — through the Insight Ingestion Loop, through the judgment broker’s validation cycle, through the workforce’s ongoing discovery of shadow workflows. The roadmap’s job is to make those mechanisms operational.

### 15.2 Lessons from the Field: The Playbook Cross-Check

The 90-day sequence that follows was designed from first principles in the April 2026 edition of this report. It can now be checked against Pereira, Graylin, and Brynjolfsson (2026)’s ground

truth from 51 mature enterprise deployments. The cross-check holds on four points, and it surfaces one addition this roadmap did not originally make.

1. **Invisible work first.** The playbook found that 77 per cent of the hardest challenges in successful deployments were change management, data quality, and process redesign — not technology. Phase 1’s insistence on auditing before deploying any technology is not caution for its own sake; it is the playbook’s core prescription, arrived at independently.
2. **KPIs before deployment.** Organisations that defined success metrics before deployment were significantly more likely to scale their AI initiatives. Phase 1’s baseline measurement of the four KPIs (Table 15.1) applies this principle directly — measure before you build, not after.
3. **Iterate; do not wait for perfect.** Every successful project in the playbook’s sample used an iterative approach; none used waterfall. The playbook’s successful teams started small deliberately and framed early deployments explicitly as experiments, not commitments. The Agent Skill Pilot in Phase 2 is built the same way.
4. **Sponsor behaviourally, not ceremonially.** Weekly check-ins and proactive blocker removal — what the playbook calls Active Steering — distinguished sponsors who saw their programmes through from those who did not, and every organisation-wide transformation in the sample escalated sponsorship into corporate-OKR integration. Sponsor continuity through failure, not merely through success, is non-negotiable: in every traceable case in the playbook, the same executive who sponsored a failed AI attempt went on to sponsor the successful one.

The addition: **build the model-abstraction layer from day one.** Model choice was fully interchangeable in 42 per cent of the playbook’s deployments, and the authors are explicit that the durable competitive advantage sits in the orchestration layer, not in the choice of foundation model. This roadmap should treat foundation models as swappable components from the outset — an architectural discipline, not an afterthought once a vendor relationship sours. It is also, as inference costs rise with agentic token consumption, the tokenomics hedge: an organisation locked into a single model’s pricing and capability roadmap has no lever to pull when that changes.

### 15.3 Phase 1: Audit and Discovery (Days 1–30)

The first phase maps the territory. No technology is deployed into production during this phase. The objective is to understand the current state of coordination — formal and informal — before proposing changes to it.

### 15.3.1 Shadow Workflow Audit

Map what AI tools staff are already using. The evidence from Chapter 3 is unambiguous: 75 per cent of knowledge workers are using AI (Microsoft and LinkedIn 2024), 78 per cent are bringing their own tools, and 69 per cent of organisations suspect employees use prohibited generative AI tools (Gartner, Inc. 2025). The shadow workflows already exist. The audit surfaces them.

The audit should identify:

- Which AI tools are in use, by whom, and for what tasks.
- Which official processes have been informally bypassed or shortened.
- Where the formal organisational structure diverges from the actual flow of work.
- Which shadow workflows represent genuine process improvements and which introduce compliance or security risks.

The audit must be conducted with explicit guarantees of non-punitive treatment. Workers will not disclose their AI usage if disclosure carries personal risk. The psychological safety evidence (A. C. Edmondson and Lei 2025) demonstrates that psychologically safe employees are 3.4 times more likely to report AI tool discoveries to management. Providing sanctioned alternatives reduces unauthorised use by 89 per cent. The message to the workforce is: “Show us what you have built. We want to learn from it, not punish it.”

### 15.3.2 Friction Point Identification

Identify 5–10 high-value cross-functional friction points where information routing currently requires human intermediation that could be handled by agentic orchestration. These are the initial candidates for DAG construction in Phase 2. Selection criteria should include:

1. The friction point involves information transfer across departmental boundaries.
2. The current process requires manual aggregation, reformatting, or contextualisation.
3. The decision logic is sufficiently well-understood that an agent can handle the routine cases.
4. The edge cases are identifiable — a judgment broker can be assigned to them.

### 15.3.3 Monitoring Deployment

Deploy passive monitoring agents across key communication channels to detect patterns of informal collaboration and workaround behaviour. This step carries significant legal risk and must be preceded by the regulatory groundwork described in Section 15.3.6.

### 15.3.4 Baseline Measurement

Establish baseline values for the four KPIs defined in Chapter 9:

**Table 15.1:** Baseline KPI targets for Phase 1 measurement

| KPI                | Target         | Baseline Method                                                                            |
|--------------------|----------------|--------------------------------------------------------------------------------------------|
| Mesh Velocity      | < 48 hours     | Measure current time from process improvement suggestion to implementation                 |
| Augmentation Ratio | > 65%          | Survey current cognitive load distribution between human and AI                            |
| Trust Variance     | < $0.12\sigma$ | Not measurable until agents are deployed; document current decision quality variance       |
| HITL Precision     | > 90%          | Not measurable until escalation protocols are active; document current escalation patterns |

### 15.3.5 Judgment Broker Identification

Identify the 9 per cent — the potential judgment broker candidates from the 1-9-90 pattern (Boston Consulting Group 2025b). The 1 per cent — the natural innovators — will already be visible from the shadow workflow audit. The 9 per cent — the champions who will adopt with support — are the strategic investment target. Characteristics to look for:

- Middle managers who have already experimented with AI tools for coordination tasks, even informally.
- Managers whose teams show evidence of process innovation, whether AI-assisted or not.
- Individuals who demonstrate comfort with ambiguity and a willingness to operate at the boundary between defined process and emergent practice.

Not every current middle manager is a candidate. The identification process should be honest about this.

### 15.3.6 Legal and Regulatory Groundwork

#### Employment Law and Consultation Obligations

Any programme that proposes to transform middle management roles triggers legal obligations that cannot be deferred to later phases. The following requirements apply before any technology deployment:

**United Kingdom:** The Employment Rights Act 2025 (*Employment Rights Act 2025: Major Reforms for UK Employment Law 2025*) broadens collective consultation triggers and requires consultation where 20 or more employees may be affected by role changes or redundancies. Failure to consult carries a protective award of up to **180 days' pay per affected employee** — doubled from the previous 90-day maximum. A 90-day roadmap that deploys agentic systems affecting management roles without prior consultation is not merely inadvisable — it is unlawful.

**European Union:** The EU AI Act (Regulation 2024/1689), Article 26(7), requires deployers of high-risk AI systems to inform workers' representatives and affected workers before deployment in the workplace (European Parliament 2025). Works council consultation is mandatory, not optional. Organisations operating across EU member states must account for national transposition differences — consultation rights vary significantly between, say, Germany (where works council authority is extensive) and Ireland (where the framework is less prescriptive).

**GDPR and Workplace Monitoring:** Deploying passive monitoring agents across employee communication channels requires a Data Protection Impact Assessment under GDPR Article 35, clear purpose limitation and data minimisation documentation, employee notification and consent frameworks, and defined retention policies for monitored data. Failure to address these requirements exposes the organisation to regulatory sanctions and, equally important, destroys the psychological safety that the entire adoption strategy depends on.

**Practical implication:** Phase 1 must include engagement with employment lawyers and, where applicable, trade union or works council representatives. The legal workstream runs in parallel with the technical audit and cannot be treated as an afterthought. In jurisdictions with strong collective bargaining rights, the consultation timeline may exceed 30 days, requiring the overall roadmap to extend accordingly.

## 15.4 Phase 2: Architecture and Codification (Days 31–60)

Phase 2 builds the initial mesh infrastructure using the discoveries from Phase 1. This is the most resource-intensive phase and the one most likely to stall without adequate staffing and

expertise.

### 15.4.1 Agent Skill Pilot

Deploy 10–15 agent skills relevant to the top friction points identified in Phase 1. The initial deployment should be narrow and deep rather than broad and shallow: better to automate three workflows thoroughly than to touch fifteen superficially. The selection should prioritise skills that:

- Address friction points where the shadow workflow audit found the greatest divergence between formal and actual processes.
- Operate inside the jagged frontier for the specific domain — tasks where AI performance is reliably strong.
- Produce outputs that are visible and evaluable by the judgment broker candidates identified in Phase 1.

The pilot is not a proof of concept. It is the first operational segment of the mesh. The agents deployed in Phase 2 remain in production; they are not discarded after a trial period.

### 15.4.2 Domain Ontology Construction

Construct the initial OWL 2 domain ontology for core business processes. This step requires specific expertise and must not be underestimated.

#### **Ontology Construction Is Not a Side Project**

Constructing an OWL 2 domain ontology in 30 days is either a trivial exercise that produces a toy ontology or a substantive knowledge engineering project that requires:

- One or more knowledge engineers with OWL 2 experience
- Domain experts from each business function being modelled
- Iterative validation sessions with operational staff
- Tools: Protégé or equivalent ontology editor, reasoner integration

If internal expertise is unavailable, this step requires external support and should be budgeted accordingly. The pragmatic approach: begin with a narrow ontology covering the 5–10 friction points identified in Phase 1. Use OWL 2 EL++ (the profile supported by VisionClaw’s Whelk-rs reasoner) to keep the ontology tractable. Expand iteratively as the mesh compounds.



### 15.4.3 DAG Construction

Build the first validated Directed Acyclic Graphs from discovered shadow workflows. Each DAG formalises a workflow that was previously informal, specifying:

- The task decomposition with ontological grounding.
- The agent skills assigned to each sub-task.
- The escalation criteria — under what conditions does the task route to the judgment broker.
- The governance constraints embedded in each transition (bias thresholds, compliance checks, access controls).
- The SLAs for each stage.

Escalation criteria are not a checkbox in this list — they are the single design decision most responsible for whether a DAG succeeds. Pereira, Graylin, and Brynjolfsson (2026)’s field evidence is worth restating in operational terms here: define authority boundaries so that the default is escalation-based oversight (the agent proceeds autonomously and a human reviews only the exceptions it flags) for high-volume, recoverable workflows, and reserve approval-based oversight (a human reviews every output before it proceeds) for the minority of workflows where error tolerance is genuinely zero or regulation requires sign-off. Getting this wrong in either direction is costly: defaulting to approval on recoverable, high-volume work throttles the productivity gain the mesh exists to capture — 30 per cent instead of 71 per cent, on the playbook’s figures — while defaulting to escalation on zero-tolerance work exposes the organisation to exactly the failure the approval model exists to prevent. Wire escalation events to the HITL Precision metric (Chapter 9) from the first DAG onward, not retroactively: a metric introduced after the escalation boundaries are already live has nothing to calibrate against.

### 15.4.4 Judgment Broker Workshops

Run workshops that reframe the middle management identity from information router to judgment broker. These workshops are not training events in the conventional sense. They are identity-level conversations that address the resistance patterns documented in Chapter 11: role obsolescence fear (67%), competence anxiety (58%), loss of information asymmetry (49%), decision authority erosion (45%), and accountability ambiguity (52%) (Capgemini Research Institute 2025).

The workshop content should include:

1. Direct acknowledgment of the threat: “The coordination skill you were hired for is being automated. That is real.”

2. The conductor escalation: “You are being asked to move from playing an instrument to conducting the orchestra. That is not a consolation — it is a promotion in complexity, scope, and strategic importance.”
3. The failure mode without them: the ant death spiral, the  $17\times$  error multiplication, the Klarna reversal.
4. Hands-on practice with the pilot agents — not as tools to use, but as systems to govern.

### 15.4.5 Protected Experimentation Time

Create protected AI experimentation time for the identified 9 per cent. The BCG/MIT evidence shows a 3.1 times higher adoption rate when experimentation time is provided (Boston Consulting Group and MIT Sloan Management Review 2025). This is not optional enrichment. It is a strategic investment with a measured return.

## 15.5 Phase 3: Live Mesh and Governance (Days 61–90 and Beyond)

Phase 3 activates the compounding loop. The critical feature of Phase 3 is that it does not end. The governance, monitoring, and learning mechanisms launched in this phase are permanent operational infrastructure.

### 15.5.1 Production Mesh Activation

Promote validated workflows into the production mesh with full SLAs. Each workflow enters production with:

- A named owner (human judgment broker).
- Defined authority boundaries (what the agents can decide autonomously; what requires escalation).
- Trust Variance monitoring (continuous quality measurement).
- Audit trail (ontological provenance through the knowledge graph).

### 15.5.2 Governance Activation

Activate the declarative governance layer with Trust Variance alerting live. The judgment broker receives real-time notification when agent decision quality drifts outside tolerance.

### **15.5.3 Insight Portal Launch**

Launch the Insight Portal for frontline contribution to the mesh. This provides the bottom-up channel for the Discovery Engine: any employee can propose a workflow improvement, a cross-functional shortcut, or a novel AI application. The mesh evaluates proposals against the domain ontology and routes viable ones to the appropriate judgment broker for validation.

### **15.5.4 First Mesh Intelligence Report**

Deliver the first Mesh Intelligence Report to the C-Suite, using the new KPI dashboard defined in Chapter 9. This report replaces headcount metrics with learning metrics: how fast the organisation is converting discoveries into capability, how reliably the agentic layer is handling cognitive load, and whether human attention is directed at the right problems.

### **15.5.5 Compound Loop Initiation**

Begin the perpetual cycle. The Insight Ingestion Loop is now operational. Shadow workflows discovered by the workforce are detected, codified, validated by judgment brokers, integrated into the live mesh, and propagated across teams. Mesh Velocity measures how fast this cycle runs. Phase 3 never ends.

## 15.6 The Wardley Map: Middle Manager Evolution



**Figure 15.1:** Wardley map showing the evolution of the middle management role from information routing (commodity, automatable) toward judgment brokerage (genesis/custom-built, irreducible). The horizontal axis represents evolutionary stage; the vertical axis represents value chain position.

The Wardley map in Figure 15.1 illustrates the strategic logic of the implementation. Information routing — the traditional middle management function — has moved rightward toward commodity and utility. It is automatable because it is well-understood and standardised. Judgment brokerage sits to the left — in the genesis and custom-built stages — because it is novel, context-dependent, and not yet standardised. The implementation roadmap moves the organisation’s investment from right to left: from commodity coordination to custom judgment.

## 15.7 Cost Tier Model

**Table 15.2:** Indicative cost tiers for agentic mesh deployment

| Dimension                | Small<br>(50–100)               | Medium<br>(100–500)            | Large (500+)                |
|--------------------------|---------------------------------|--------------------------------|-----------------------------|
| Ontology construction    | 2–4 weeks, 1 knowledge engineer | 2–3 months, 2–3 engineers      | 6+ months, dedicated team   |
| Agent skill deployment   | 10–15 skills                    | 30–50 skills                   | 80+ skills, multi-domain    |
| GPU infrastructure       | Single workstation (RTX 4080+)  | Dedicated server (A6000-class) | Multi-node cluster          |
| Judgment broker training | 2–3 workshops                   | Rolling programme, 6–12 months | Permanent training function |
| Legal/compliance         | External counsel, ad hoc        | In-house or retained, ongoing  | Dedicated regulatory team   |
| Annual SaaS/compute      | £50–150K                        | £200–600K                      | £1M+                        |
| Total Year 1 investment  | £150–300K                       | £500K–1.2M                     | £2–5M                       |

It is worth grounding these figures against what “starting” actually costs at the token-and-seat level, because the two are often confused. In the Ramp/Revelio dataset, the threshold separating high-intensity AI adopters from the rest of the market averaged roughly \$33.67 of AI spend per employee per month in the first three months after adoption (Kharazian, Simon, and Stevens 2026) — not the four- or five-figure per-head number that dominates vendor sales conversations. The investment that actually matters, and that the ranges in Table 15.2 are trying to capture, is the organisational work in this roadmap: the audit, the ontology, the workshops, the governance function. The model bill is not where the money goes.

These figures are indicative and will vary significantly by industry, regulatory burden, and existing technology infrastructure. They exclude opportunity costs and internal staff time. A COO

approving this programme needs at minimum these ranges to compare against the cost of *not* deploying — the \$4.63 million average cost per shadow AI breach (IBM Security 2025), the 55 per cent regret rate for rushed AI replacement (Orgvue and Forrester Research 2025), and the competitive compounding that accrues to organisations that invest early.

## 15.8 Failure Criteria: When to Stop

Every implementation plan should define what “stop, this is not working” looks like. The following signals, observed at or before day 45, should trigger a pause-and-review:

1. **Shadow workflow audit yields fewer than three actionable discoveries.** This suggests either the workforce does not trust the audit process (a psychological safety failure) or the organisation’s work is not yet amenable to agentic augmentation.
2. **Zero judgment broker candidates emerge from the 9 per cent identification.** If no middle managers volunteer or are identified as candidates, the cultural preconditions for the mesh do not exist. Forcing the programme forward without willing participants produces resentment, not transformation.
3. **Legal counsel identifies irresolvable consultation obligations.** In unionised workplaces or jurisdictions with strong collective bargaining rights, the consultation timeline may exceed 90 days. This is not a failure of the mesh concept — it is a failure of the timeline. Extend, do not abandon.
4. **Trust Variance exceeds  $0.25\sigma$  on pilot workflows in the first two weeks.** This indicates that the selected tasks are outside the jagged frontier for the current model capabilities. Reselect tasks closer to the frontier’s interior.
5. **Executive sponsorship weakens.** If the C-Suite stops engaging with the programme before the first Mesh Intelligence Report, the programme will fail regardless of technical execution. The evidence on directed autonomy (Microsoft 2025a) is clear: leadership must set guardrails and investment priorities, not merely approve a budget and disengage.
6. **Escalation queues dominated by false positives.** If HITL Precision (Chapter 9) sits persistently below roughly 60 per cent in the pilot — meaning most of what reaches the judgment broker did not actually need human intervention — the authority boundaries are mis-scoped, not the reviewer pool undersized. The playbook’s alert-fatigue evidence and the verification-failure modes documented by Cemri et al. (2025) both point the same way: adding more reviewers to a queue full of false positives trains brokers to rubber-stamp the queue, reproducing the vigilance problem (Chapter 5) inside the very escalation mechanism meant to solve it. Reselect the boundaries; do not add headcount to the review queue.

Key Insight

A programme that stops at day 45 because the preconditions are not met is not a failure. It is an organisation that has learned something valuable about its own readiness. The mesh can be re-attempted when the preconditions change. Pressing forward without them guarantees the kind of initiative fatigue that Gartner, Inc. (2025) warns organisations using traditional change management for AI are 2.4 times more likely to experience.

15.9 Resistance Mitigation in the Roadmap

The gap analysis for this report noted that the five resistance patterns documented in Chapter 11 were identified but not mitigated in the roadmap. Table 15.3 maps each pattern to a specific roadmap action.

Table 15.3: Resistance patterns mapped to roadmap actions

| Pattern                          | Phase | Mitigation Action                                                                                    |
|----------------------------------|-------|------------------------------------------------------------------------------------------------------|
| Role obsolescence fear (67%)     | 2     | Judgment broker workshops with direct threat acknowledgment and conductor escalation framing         |
| Competence anxiety (58%)         | 2–3   | Protected experimentation time ( $3.1 \times$ adoption); personalised coaching, not generic training |
| Information asymmetry loss (49%) | 1–2   | Reframe: the judgment broker’s advantage is now <i>contextual judgment</i> , not data access         |
| Decision authority erosion (45%) | 2–3   | Explicit escalation protocols that route genuine edge cases to brokers, not past them                |
| Accountability ambiguity (52%)   | 1     | Legal groundwork (Section 15.3.6); clear RACI for agent-assisted decisions                           |

The 90-day roadmap is designed to be falsifiable. If the specified failure criteria are met, the organisation has learned something valuable — that the prerequisites for mesh deployment are not yet in place — and can adjust accordingly. An implementation plan that cannot fail is not a plan; it is a wish.

The next chapter, Chapter 16, takes this roadmap and applies it — not hypothetically, but to DreamLab’s own systems. It is a falsifiable exercise in following the sequence described here

against the organisation's own coordination tax, rather than leaving the roadmap as a template intended for someone else's transformation.



# Chapter 16

## Dogfooding the Mesh: A Roadmap for the DreamLab Ecosystem

A consultancy that will not run its own prescriptions on its own systems is selling a diet it does not eat.

### 16.1 Why This Chapter Exists

Chapter 17 acknowledges the structural conflict at the heart of this report: DreamLab AI Consulting is simultaneously the analyst producing the analysis, the framework creator proposing the Dynamic Agentic Mesh, and the vendor operating VisionClaw. Acknowledgement alone is cheap. The stronger answer to a positioning conflict is falsifiability: commit, in print, to specific and checkable changes in the author's own systems, derived from the report's own findings, and let the next edition report which commitments were met. If the mesh thesis is right, applying the July 2026 evidence base to DreamLab's ecosystem should produce measurable compounding within two quarters. If it is wrong, the same instrumentation will show that too — and this report's brand of intellectual honesty obliges us to publish either result.

This chapter therefore does something no other chapter does: it turns the report's prescriptions inward. Each recommendation below names real components, real endpoints, and real architecture decision records in the DreamLab codebase, verified against the repositories in the week this edition went to press. What gets built — or conspicuously fails to get built — is evidence either way.

### 16.2 The Ecosystem as It Stands

DreamLab's ecosystem comprises six components, bound together by a shared `did:nstr` identity mesh that currently has five active signing participants.

**VisionFlow** is the umbrella coordination canon: the architectural doctrine, product-requirement corpus, and cross-repository contracts that bind the other five components into one system. It is where the mesh federation design (PRD-010) lives, and where the decision records that this chapter repeatedly cites are filed.

**VisionClaw** is the knowledge substrate described in Chapter 14: a Rust server with GPU-accelerated graph physics, an OWL 2 ontology engine with *Whelk* reasoning, and a multi-user WebXR client. Since the April 2026 audit it has grown a live agent-telemetry pipe — an authenticated `/wss/agent-events` WebSocket (ADR-059) through which agent actions flow in from the runtime and are rendered as beams and transient attractive edges in the GPU graph — and a re-imagined control centre (ADR-129) that is, at present, a settings surface rather than a governance one. Its decision-record corpus now runs to over a hundred ADRs, several of which matter greatly to what follows.

**AgentBox** is the sovereign agent runtime: a hardened, reproducible container that hosts the agents themselves, with 109 active skills in its routing directory as of July 2026, a full-lifecycle hook surface for telemetry that is fail-open by enforced convention, *RuVector* vector memory holding over 1.17 million embeddings, and a Nostr session mirror that streams every running session to the operator’s phone as encrypted self-messages. Its memory-and-learning work-stream (its PRD-018) landed in production in early July 2026 — with an important caveat examined in Recommendation 7 below.

**solid-pod-rs** is the sovereign data layer: a seven-crate Rust implementation of the W3C Solid protocol, giving every human and agent a pod — portable, access-controlled storage rooted in their own cryptographic identity, with every write recorded as a git commit carrying provenance. It is pre-1.0 software actively hardening: its 2026-07-03 closeout release fixed four security issues, and an independent audit the same week found that the shipped server binds only a slice of what the library implements.

**nostr-rust-forum** is the governance surface: a self-hostable forum kit whose Agent Control Surface Protocol (ACSP) lets any agent publish a signed request for a human decision, and lets only a human administrator sign the response — the relay rejects approval events from any agent key. This is the ecosystem’s one production human-in-the-loop mechanism, and today it serves exactly one use case: ontology concept elevation, capped at five concurrent cases.

**DreamLab Edge** (the `dreamlab-ai-website` repository) is the operator overlay: the branded, Cloudflare-edge-deployed public face at `dreamlab-ai.com` that renders the forum and consumes the pod layer. It is live, but as an independent component it is the earliest-stage of the six — it is deliberately being de-specialised into a thin consumer of the forum kit (PRD-012), and that cutover is still mid-flight.

### The Broker Is Already Half-Built

The most striking fact the internal audit surfaced: the judgment broker — the role this report coined in Chapter 13 — already partially exists in code. A `BrokerActor` governance component with a `handleGovernanceDecision()` loop sits in the ecosystem at roughly 65 per cent completion; its backend (ADR-041) ships a `BrokerCase` aggregate, a decision orchestrator, six decision outcomes (approve, reject, amend, delegate, promote, precedent), and a REST inbox. What does not exist is the front half: no broker inbox or decision canvas appears anywhere in VisionClaw’s own client, and every live decision today round-trips through an external forum page. The report’s central role is partially implemented — and the gap between 65 per cent and a working judgment-broker practice is precisely the kind of gap this chapter exists to close, or to report honestly as unclosed.

## 16.3 Fitting the Findings to the Architecture

The July 2026 evidence base maps onto this architecture with unusual directness. Table 16.1 pairs each load-bearing finding with its architectural implication and the component where that implication lands.

Two rows deserve emphasis. The interface-organisation row is not a metaphor: the ecosystem’s governed writeback path — `ontology_propose`, then `Whelk` consistency reasoning, then judgment-broker approval — is a literal implementation of Liu’s specification that agent outputs enter human workflows only with evidence attached and human judgment mandatory at defined points. And the security row is not hygiene housekeeping: the same audit that found the writeback path also found the ways around it, which is why the first recommendation below is the first recommendation.

## 16.4 Twelve Tangible Recommendations

Each recommendation names what to build or change, the finding that motivates it, and a priority. P0 items gate everything else; P1 items create the measurement layer; P2 items extend the loop outward.

1. **Close the governed-writeback security gaps (P0).** Authenticate the `/api/ontology/propose` endpoint, which currently accepts unauthenticated proposals; remove or hard-gate the `/api/ontology/load` backdoor, which bypasses `Whelk` validation entirely; and fix the `solid-pod-rs` non-destructive PATCH bug. The Playbook’s finding is that secu-

rity requirements front-loaded early become enablers rather than blockers (Pereira, Graylin, and Brynjolfsson 2026) — but the sharper point is internal: a governance pipeline with an unauthenticated front door and a validation bypass is not governance, it is theatre. Nothing else in this chapter is worth measuring until these are closed.

2. **Finish the BrokerActor (P0).** Complete the `handleGovernanceDecision()` loop from its current 65 per cent, and surface the broker case queue inside VisionClaw’s control centre as an escalation-first review surface — each case carrying the evidence, provenance, and uncertainty that motivated it. ADR-041 already ships the aggregate, the six decision outcomes, and the REST inbox; what is missing is the client surface (no broker inbox exists in `client/src`) and the closure of the loop inside the product rather than on an external forum page. This is Liu’s interface organisation made concrete (Liu 2026), and it converts the report’s coined role from 65 per cent of a backend into a daily practice.
3. **Instrument contextual transaction cost (P1).** Extend the agent-events envelope (ADR-059’s trace schema in `src/agent_events/schema.rs`) to record handoff counts, token burden, compression events, and verification outcomes per workflow, and report a CTC figure per DAG alongside Mesh Velocity. Chapter 17 names the absence of production CTC measurement as an open question in Liu’s framework (Liu 2026); this ecosystem is positioned to produce the first production CTC instrumentation anywhere, because the trace plumbing already exists and only the accounting fields are missing.
4. **Ship the four KPIs as a control-centre dashboard (P1).** This is not a design task: ADR-043, accepted in April 2026, already specifies Mesh Velocity, Augmentation Ratio, Trust Variance, and HITL Precision verbatim, with a complete lineage data model — and three months later not one line of implementation exists. Resurrect ADR-043 and compute the four metrics from live agent-events and ACSP escalation telemetry. The Playbook finds that organisations defining KPIs before deployment are significantly more likely to demonstrate value (Pereira, Graylin, and Brynjolfsson 2026); this report proposed the KPIs in Chapter 9 and must now be measured by them.
5. **Adopt MAST-aligned failure telemetry (P1).** Tag every agent failure flowing through the agent-events pipeline and the QE fleet with one of the fourteen MAST failure modes (Cemri et al. 2025) — specification issues, inter-agent misalignment, task verification — rather than free-text error strings. This makes the ecosystem’s failure data comparable to the published taxonomy and enables the failure-mode analysis Chapter 17 calls for.
6. **Default AgentBox authority boundaries to escalation (P1).** New agent skills should default to the escalation operating model, with each skill’s actions explicitly classified as recoverable or zero-tolerance; approval-on-every-output should be reserved for the zero-tolerance class. The ACSP wire protocol already carries the vocabulary (priority levels,

a `trust_alert` category); what is missing is that escalation is not discoverable — no routed skill wraps the control-surface library, so an agent must already know it exists — and there is no supported pattern for an agent to block on the human’s signed response. Fix both. The motivating evidence is the Playbook’s 71-versus-30 result (Pereira, Graylin, and Brynjolfsson 2026).

7. **Make outcome learning real (P1).** The July 2026 audit verified, uncomfortably, that the memory system’s advertised outcome learning does not yet run: the intelligence banner is text-relevance matching, and the router’s confidence figure is a hardcoded constant. The remediation has begun honestly — trajectory recording of graded outcomes landed in production on 5 July 2026 — but the two consumers that would let outcomes influence retrieval and routing (`feed_retrieval`, `feed_routing`) remain gated off pending a statistical sample floor. Wire post-task success signals into pattern reuse once the floor clears, and delete the stale code comments that still claim the old path learns. The compounding organisation of Chapter 8 is a claim about closed loops; this one must close with evidence, not intention.
8. **Engineer diversity in orchestration (P2).** Route verification and critique agents to different foundation models with independent retrieval paths. AgentBox already fronts Claude, Codex, Gemini, and DeepSeek, so the plumbing exists; the discipline does not. Liu’s pseudo-diversity result — same model, same tools, same retrieval equals one judge asked three times (Liu 2026) — says a reviewer drawn from the reviewed agent’s own model family is decoration, not verification.
9. **Carry provenance to the human’s pocket (P2).** Enrich the existing Nostr session mirror and the end-of-session digest so that every escalated decision reaches the operator’s mobile surface with its evidence and uncertainty attached, not just its conclusion. Liu’s interface-organisation specification makes uncertainty disclosure a design requirement, not a courtesy (Liu 2026); the encrypted per-turn mirror and the durable digest already exist, so this is an envelope-enrichment task, not new infrastructure.
10. **Run the Insight Ingestion Loop v1 across the mesh (P2).** Wire the loop end to end: forum and pods as the discovery surface, `ontology_propose` (ADR-049) as codification, the BrokerActor as validation — measuring Mesh Velocity across the whole path for the first time. This means engaging with, not ignoring, the frozen designs: the self-improving writeback flywheel (ADR-121) and the two-speed governance routing (ADR-122) were fully specified and then shelved in early July 2026, and the codebase carries a ghost flag (`writeback_triggered` in `enrichment_proposals_handler.rs`) that reports writes it never performs. Any v1 must ship with a liveness canary — the codebase’s own hard-won precedent, after an earlier workstream left an entire constraint pipeline wired,

flagged as live, and inert for months.

11. **Consolidate the data moat (P2).** Unify provenance across RuVector memory, the pods' git-mark and block-trail records, and the ontology's PROV-O reification (currently around 65 per cent complete, with cross-repository receipt crossing still test-only) into one queryable trace. The Playbook found 47 per cent of successful deployments citing accumulated data as a moat and counselled saving everything (Pereira, Graylin, and Brynjolfsson 2026); in an agentic ecosystem the moat is not the data alone, it is the trace — who proposed what, on what evidence, and what the human decided.
12. **Pilot the stack beyond one team (P2).** Complete the DreamLab Edge kit cutover (PRD-012) and the mesh federation prerequisites (PRD-010's Phase 0 cryptographic fixes, which currently block all cross-system messaging), then run one external pilot of the escalation-plus-KPI stack in a non-DreamLab context. Chapter 17 identifies bridging evidence as the thesis's largest missing piece; a single pilot outside the favourable home environment is worth more to the argument than any further refinement within it.

## 16.5 Measurement Commitments

Recommendations without measurement are marketing. Table 16.2 states, for each of the four KPIs and for contextual transaction cost, the concrete data source in the stack and the review cadence to which DreamLab commits.

The commitment is plain: if, two quarters from publication, the instrumented KPIs cannot be computed from these sources — or can be computed and show no compounding — that is evidence against the mesh thesis as deployed here, and the next edition of this report will say so in these terms. The falsification condition is stated now, before the data exists, precisely so that it cannot be quietly redefined afterwards.

## 16.6 Honest Limits

### What This Chapter Does Not Claim

- Everything above unfolds in the same favourable context flagged in Chapters 14 and 17: a 15-person, technically fluent team where the system architect is the primary user. Success here bridges nothing on its own; that is what Recommendation 12 is for.
- These are commitments, not results. The codebase’s own history shows accepted designs sitting unbuilt for months (the KPI model), fully specified designs frozen days before this edition (the writeback flywheel and governance routing), and wiring that reported itself live while doing nothing. The roadmap inherits that risk, which is why every loop-closing item carries a liveness-verification requirement.
- The BrokerActor at 65 per cent is not a working judgment-broker practice. The only production escalation loop today serves one use case, capped at five concurrent cases, decided on an external forum page by a single administrator role.
- DreamLab Edge is live as a deployment but is the earliest-stage component as an independent artefact, and relay federation across the mesh remains designed, not shipped.
- The ecosystem’s internal documentation cannot yet agree on its own scale — three different skill counts were live in the same repository tree on the same day. Scale claims in this report are point-in-time snapshots of a fast-moving system, and selection bias cuts both ways: an author auditing their own system may be harsher than an outsider, or blinder.

The report has argued throughout that the honest version is more defensible and more credible. This chapter is where that principle stops being a stance and becomes a schedule.

**Table 16.1:** Mapping the revised report’s findings onto the DreamLab ecosystem

| <b>Finding</b>                                                                                      | <b>Architectural implication</b>                                                        | <b>Ecosystem locus</b>                       |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------|
| Escalation beats approval: 71 vs 30 per cent median gains (Pereira, Graylin, and Brynjolfsson 2026) | Default authority boundaries to escalation; reserve approval for zero-tolerance actions | VisionClaw control centre;<br>BrokerActor    |
| Contextual transaction cost governs collective efficiency (Liu 2026)                                | Instrument handoffs, compression loss and verification burden per work-flow             | AgentBox hooks;<br>agent-events trace schema |
| Human-imitation forms underperform agent-native forms (Liu 2026)                                    | Shared-state coordination and adaptive orchestration, not role-played committees        | RuVector memory;<br>Claude-Flow routing      |
| Multi-agent failure is taxonomisable: 14 MAST modes (Cemri et al. 2025)                             | Failure telemetry tagged to a standard taxonomy, not free-text error logs               | Agent-events envelope; QE fleet              |
| Diversity must be engineered, not assumed (Liu 2026)                                                | Multi-model verification with independent retrieval and evidence paths                  | AgentBox model routing                       |
| Interface organisations connect agent context to human accountability (Liu 2026)                    | Evidence-carrying escalation with mandatory human judgment points                       | Governed ontology writeback; forum ACSP      |
| Proprietary data is the moat; save everything (Pereira, Graylin, and Brynjolfsson 2026)             | Durable, provenance-carrying memory across every substrate                              | RuVector;<br>solid-pod-rs                    |
| Security front-loading enables rather than blocks (Pereira, Graylin, and Brynjolfsson 2026)         | Close authentication gaps before expanding features                                     | solid-pod-rs PATCH;<br>ontology endpoints    |



**Table 16.2:** Measurement commitments for the ecosystem roadmap

| Measure                     | Concrete data source                                                                                                  | Cadence         |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------|
| Mesh Velocity               | Timestamps across the insight loop: <code>ontology_propose</code> event to broker decision to merged enrichment       | Quarterly       |
| Augmentation Ratio          | Agent-action volume from <code>/wss/agent-events</code> against ACSP escalation volume                                | Monthly roll-up |
| Trust Variance              | Dispersion of broker decision outcomes (approve, amend, reject rates) over a rolling 30-day window                    | Monthly         |
| HITL Precision              | Proportion of ACSP escalations where the human decision materially changed the outcome, from decision-outcome records | Monthly         |
| Contextual transaction cost | Handoff counts, token burden and verification outcomes per DAG from the extended agent-events envelope                | Quarterly       |



# Chapter 17

## Open Questions and Research Directions

A working document that hides its weaknesses is less useful than one that maps them. These gaps are research directions, not admissions of failure.

This chapter is an exercise in intellectual honesty. The coordination collapse thesis, as presented in this report, is incomplete. The evidence base has specific weaknesses. The analysis contains blind spots visible to each of its three audiences. Several important topics are missing entirely. Mapping these gaps is not an admission of failure — it is a research agenda.

### 17.1 The Core Thesis Is Incomplete

The report frames hierarchy as an information routing protocol and argues that AI collapses the cost of that routing to near-zero. This framing is analytically productive — it generates the coordination collapse concept, the judgment broker role, and the compounding organisation model. But hierarchy performs at least five functions, of which information routing is only one.

#### 17.1.1 Legitimation

Who has authority to commit resources? The mesh can route information and even recommend resource allocation, but it cannot *authorise* a £2 million capital expenditure. Legitimation — the organisational mechanism by which decisions acquire binding authority — is a function of hierarchy that AI does not replace. A judgment broker who validates a shadow workflow is exercising legitimate authority derived from their position in the organisational structure. Remove the structure, and the authority evaporates.

#### 17.1.2 Accountability

Who is fired when things go wrong? “The mesh decided” is not an answer a board of directors accepts. Accountability requires identifiable humans who bear consequences for outcomes. The judgment broker role addresses this partially — the broker who validates a workflow bears

accountability for that validation — but the full accountability architecture for agentic systems is not worked out. When an agent in the mesh makes a decision that produces a regulatory violation, the chain of accountability must be traceable to a human. The declarative governance layer (Chapter 10) provides audit trails, but audit trails are not the same as accountability structures.

### 17.1.3 Political Negotiation

Whose priorities win when budgets are finite? AI can optimise within constraints; it cannot set the constraints. The allocation of scarce resources across competing priorities is fundamentally a political act — a negotiation among stakeholders with different interests, values, and power. The mesh can surface information that makes the negotiation more transparent, but it cannot replace the negotiation itself.

#### An Honest Reframing

If three of hierarchy’s five core functions persist — legitimization, accountability, and political negotiation — then “coordination collapse” may be an overstatement. A more honest framing: **information routing collapse** — one function of hierarchy dissolves, forcing the others to restructure around different assumptions. The coordination tax shrinks dramatically; it does not reach zero. This report uses the more provocative framing because it captures the magnitude of the disruption, but the qualification belongs in the analysis.

**Research needed:** How do legitimization, accountability, and political negotiation operate in post-hierarchical structures? Haier’s micro-enterprise model provides the best existing evidence but operates in a culturally specific context that may not generalise.

## 17.2 Evidence Base Weaknesses

The argument in this report rests on specific empirical findings. Several of these findings have limitations that deserve explicit acknowledgement.

### 17.2.1 The Jagged Frontier Map Is Outdated

The BCG experiment (Dell’Acqua et al. 2026) used GPT-4 — now two model generations behind current frontier models. The specific task boundaries identified in that study — the 18 tasks inside the frontier and the business judgment task outside it — have almost certainly shifted. Models released since the study likely perform well on tasks that were outside GPT-4’s frontier.

This does not invalidate the concept. The jagged frontier is a structural feature of AI capability, not a fixed map. The principle — that adjacent tasks can fall on opposite sides of the boundary, and that the boundary is unpredictable — holds regardless of where the boundary currently sits. But any practitioner using this report to make operational decisions must retest the frontier with current models in their specific domain. The 2023 map is a historical reference, not a current navigation chart.

The concern is now partially answered by continuous measurement rather than a fixed retest. The Remote Labor Index (Mazeika 2026; Mazeika, Gatti, Menghini, et al. 2025) provides exactly the longitudinal, client-quality re-measurement this section called for: automation of real freelance deliverables — judged against a paying-client bar, not a benchmark proxy — rose from 2.5 to 16.1 per cent in eight months. That trajectory confirms both halves of the jagged frontier claim simultaneously: the frontier is still jagged (professional-quality performance still fails on the large majority of projects) and it is moving fast (the boundary shifted 6.4-fold in under a year). The residual gap is real, though: the RLI measures freelance deliverables, not the coordination tasks inside an organisation that this report is actually about. A practitioner still has to map their own domain’s frontier — the RLI tells them how urgently, not where.

## 17.2.2 The $17\times$ Error Source: Resolved

The first edition of this report leaned on a 17-times error multiplication statistic (Towards Data Science 2026) to ground the case against naive multi-agent deployment, support the judgment broker role, and supply the quantitative punch behind the ant death spiral narrative. The source was Towards Data Science — a community-edited blogging platform, not a peer-reviewed journal — and this chapter flagged it as the report’s weakest load-bearing number. The literature has since delivered a replacement.

Cemri et al. (2025), published at NeurIPS 2025 (Datasets and Benchmarks Track) and therefore peer-reviewed, documents failure rates of 41 to 86.7 per cent across seven state-of-the-art open-source multi-agent frameworks (ChatDev, MetaGPT, HyperAgent, AppWorld, AG2, Magentic-One, and OpenManus), drawn from more than 1,600 annotated execution traces with inter-annotator agreement of Cohen’s  $\kappa = 0.88$ . The paper does more than replace a number: it supplies a causal taxonomy of fourteen failure modes across three categories — specification issues, inter-agent misalignment, and task verification — so the error-multiplication argument now rests on a documented mechanism rather than an unexplained multiplier. (Khanal, Tao, and Zhou 2026) adds an independent second line of evidence: across 23,392 episodes, frontier models show the *highest* long-horizon meltdown rates, up to 19 per cent, precisely because their capacity for ambitious multi-step strategies is what lets them spiral further before failing — and memory scaffolds, the intuitive fix, universally hurt long-horizon performance across

every model tested.

### What Changed

The phenomenon the  $17\times$  figure gestured at — error propagation and amplification in multi-agent chains — was always real. What was weak was the citation, not the claim. (Cemri et al. 2025) and (Khanal, Tao, and Zhou 2026) now give that claim a peer-reviewed evidence base, a causal taxonomy, and a falsification of the “more memory fixes it” intuition. The  $17\times$  figure is retired from this edition and survives only as the superseded early estimate it always was.

## 17.2.3 The Klarna Narrative Is Cherry-Picked

Klarna’s 2025 financials showed improved margins alongside the customer satisfaction decline described in Chapter 6. The CEO’s admission that “cost unfortunately seems to have been a too predominant evaluation factor” (Klarna 2025) comes from a longer statement that also defended the AI strategy as a necessary evolution. Presenting only the quality degradation without the financial improvement is selective evidence — the kind of rhetorical move that a hostile reader will identify and use to dismiss the broader argument.

The honest presentation: Klarna achieved short-term cost reduction, experienced quality degradation, began rehiring humans, and the CEO acknowledged the trade-off. The lesson is not “AI customer service fails” but “optimising for cost alone, without maintaining human oversight at quality-sensitive boundaries, produces predictable quality collapse.” This is a more nuanced and more defensible claim.

The July 2026 picture sharpens this lesson rather than complicating it. Klarna’s current operating model is not a simple reversal to the pre-2023 staffing level but an explicit AI-plus-human hybrid (CX Dive 2026): AI continues to handle high-volume routine queries, while human staff — hired on flexible, gig-style contracts rather than reinstated as the original full-time roles — are reserved for the empathy, discretion, and escalation cases the CEO’s own admission identified as under-served. This is not evidence against AI customer service; it is evidence against one specific configuration of it. The stable end state is neither “all AI” nor “all human” but a designed division of labour at the jagged frontier — the judgment-broker architecture this report proposes, arrived at independently by a company that started from the opposite assumption.

## 17.2.4 The 89 Per Cent Reduction

The claim that providing approved AI tools reduces unauthorised use by 89 per cent is used repeatedly in this report. No primary source is cited. The statistic is plausible — it aligns with

shadow IT research showing that sanctioned alternatives reduce unsanctioned adoption — but a claim used this frequently requires a traceable citation. This is an open research gap.

### 17.2.5 Vendor Report Methodology

The shadow AI statistics used throughout the report stack vendor-funded research — Gartner, IBM, Capgemini — that has methodological limitations and commercial incentives to inflate threat numbers. No error bars, confidence intervals, or methodological caveats are provided. A reader from an academic background will note this immediately. The statistics should be treated as directionally correct rather than precisely accurate.

## 17.3 Audience Blind Spots

### 17.3.1 No Day-in-the-Life

Middle managers will ask: “What does my Monday morning look like as a judgment broker? What am I literally doing at 9am?” This report offers the conductor metaphor, the role transformation table, the three irreplaceable capacities — but no concrete job specification, no competency framework, no salary benchmarking, no day-in-the-life narrative. The abstraction is intellectually satisfying but operationally insufficient for a middle manager trying to decide whether to invest in this transition.

**Research direction:** develop detailed judgment broker role specifications for three to five industry verticals, including day-in-the-life scenarios, competency models, and compensation benchmarking.

### 17.3.2 No Total Cost of Ownership

AI leaders will ask: “What does this cost for a 500-person organisation? What is the ROI timeline?” Chapter 15 provides an indicative cost tier model, but this falls short of a total cost of ownership analysis that includes ongoing compute, model API costs (which are volatile and trending downward), knowledge engineering maintenance, judgment broker training, and the opportunity cost of the transition period. Without financial modelling, the implementation roadmap is a strategic argument, not a business case.

**Research direction:** develop a TCO calculator parameterised by organisation size, industry vertical, regulatory burden, and existing technology infrastructure.

### **17.3.3 The Roadmap Contradiction**

Change managers will notice that this report attacks Kotter and ADKAR for being phased transformation models, then offers a phased 90-day roadmap. Chapter 11 addresses this contradiction explicitly — framing the roadmap as a bootstrap sequence rather than a change plan — but the resolution requires the reader to accept a distinction that may feel like sophistry. The tension is genuine and deserves continued intellectual engagement rather than a single paragraph of resolution.

### **17.3.4 The Analyst-Vendor Conflict**

All three audiences will notice that DreamLab AI Consulting is simultaneously the analyst (producing this report), the framework creator (proposing the Dynamic Agentic Mesh), and the vendor (operating VisionClaw). This is a classic consulting positioning conflict. The report’s credibility depends on acknowledging this conflict directly: “We built this because we believe it. Here is the evidence for and against. Judge the framework on its merits, not its provenance.”

## **17.4 Engaging with Acemoglu**

Daron Acemoglu’s work on AI and productivity has been cited in this report’s bibliography but not substantively engaged with in the main text. This is a significant omission because Acemoglu’s thesis directly challenges the optimistic productivity assumptions underlying the coordination collapse argument.



### The Acemoglu Sceptical Thesis

In “The Simple Macroeconomics of AI” (Acemoglu 2025), Acemoglu argues that AI’s impact on total factor productivity (TFP) will be modest — at most **0.66 per cent over 10 years**. His reasoning:

1. AI automates tasks, not jobs. The tasks most amenable to automation are not necessarily the tasks that contribute most to productivity.
2. The “easy-to-learn, hard-to-verify” problem: many tasks that AI can perform require costly human verification, limiting net productivity gains. This maps directly onto the vigilance problem described in Chapter 5.
3. New task creation — historically the mechanism by which automation generates employment growth — may be slower with AI than with previous technologies.
4. AI may increase inequality rather than aggregate productivity, concentrating gains among capital owners and high-skill workers while displacing middle-skill employment.

The coordination collapse thesis should engage with Acemoglu on three fronts.

**First**, the 0.66 per cent TFP estimate applies to the economy as a whole, not to individual organisations. An organisation that deploys an agentic mesh effectively may capture disproportionate productivity gains precisely because its competitors do not. The compounding organisation thesis does not require AI to transform the macroeconomy — it requires AI to provide a competitive advantage to organisations that use it well. This is consistent with Acemoglu’s framework: the gains are real but unevenly distributed.

**Second**, the task-level analysis is compatible with the jagged frontier concept. Acemoglu’s point that AI automates tasks, not jobs, aligns with the argument that the judgment broker’s role is to determine which tasks sit inside the frontier and which require human judgment. The mesh architecture is explicitly designed around task decomposition, not wholesale job replacement. Where Acemoglu warns that the “easy-to-learn, hard-to-verify” problem limits productivity gains, the mesh responds with HITL Precision and Trust Variance monitoring — architectural mechanisms for managing verification costs.

**Third**, the equity concern is serious and currently unaddressed in this report. If the mesh concentrates productivity gains among organisations with the capital and technical expertise to deploy it, while displacing middle-skill workers in organisations that cannot, the net social effect may be increased inequality even as individual organisations benefit. This is a genuine tension that the coordination collapse thesis does not resolve.

**Fourth**, the RLI trajectory (Mazeika 2026; Mazeika, Gatti, Menghini, et al. 2025) puts pressure

on Acemoglu’s implicit pace assumption without threatening his mechanism. An automation rate that moved from 2.5 to 16.1 per cent of client-quality freelance deliverables in eight months is a fast-moving frontier by any standard; if that pace continued, the case for a durable 0.66 per cent TFP ceiling would look increasingly strained. But 16.1 per cent automation of freelance deliverables is entirely compatible with modest aggregate productivity gains if verification and integration costs dominate — Acemoglu’s “easy-to-learn, hard-to-verify” problem, restated at the level of task completion rather than macroeconomic output. Liu (2026) now gives that mechanism a formal home: contextual transaction cost includes an explicit verification-burden term, and the finding that collective efficiency collapses 91.4 per cent from the lowest to the highest CTC quartile is, in effect, a microfoundation for why task-level automation does not translate one-for-one into organisational or macroeconomic productivity (Chapter 7).

**Fifth**, the firm-level evidence assembled since April supports both halves of this report’s reading, and Acemoglu’s distributional concern besides. Kharazian, Simon, and Stevens (2026) and the Box survey reported by Levie (2026) — the latter resting on a single social-media source without published methodology — show real gains at the firm level, but gains concentrated among organisations that have already adopted AI intensively, with high-intensity adopters growing headcount 10.2 per cent over two years while low-intensity adopters showed no significant change. That is uneven distribution *across* organisations, exactly as this report’s compounding-organisation argument predicts. (Brynjolfsson, Chandar, and R. Chen 2025) supplies the complementary, and more troubling, half: uneven distribution *within* the labour market, with early-career workers in AI-exposed occupations showing a roughly 16 per cent relative employment decline since late 2022. Acemoglu’s equity concern was speculative in the first edition of this report. It is now measured.

## 17.5 New Open Questions (July 2026)

The evidence that arrived since April resolves several of this chapter’s original flags. It also opens new ones. Five deserve explicit tracking.

- **CTC measurement in production.** Liu (2026) gives contextual transaction cost a rigorous definition and shows it predicts collective efficiency in simulation and in trace-instrumented real-LLM runs. No standard instrumentation yet exists for measuring compression loss or semantic drift in a live, production agentic mesh. The KPIs proposed in Chapter 9 now have a theoretical anchor; they do not yet have tooling that would let an organisation compute its own CTC.
- **The broker-apprenticeship problem.** If entry-level roles in AI-exposed occupations decline (Brynjolfsson, Chandar, and R. Chen 2025) while adopting firms hire AI-fluent ju-

niors rather than training them from scratch (Kharazian, Simon, and Stevens 2026), where does the next generation of judgment brokers learn the domain expertise the role requires? The answer plausibly differs for adopting and non-adopting firms, which would make broker scarcity a compounding advantage for early movers — a mechanism this report has not modelled.

- **The design question is contested.** Dochkina (2026) reports self-organising agent collectives, with autonomous role selection, outperforming both fully centralised coordination and fully designed structures — the opposite emphasis from the finding of Liu (2026) that adaptive, CTC-aware orchestration beats unstructured collectives. Both are single studies with different task distributions and scales. The field has not converged on whether agent collectives should be designed, allowed to self-organise, or some specific hybrid of the two, and this report should not pretend otherwise.
- **Survivor bias in the deployment evidence.** Pereira, Graylin, and Brynjolfsson (2026) study the successful 5 per cent of deployments against the baseline of Challapally et al. (2025), in which most pilots produced no measurable P&L impact. That comparison is valuable for identifying what successful deployments have in common, but it cannot answer how common success is, because the sample is selected on the outcome. Prevalence estimates should not be derived from playbook-style evidence, however good the playbook.
- **The RLI-to-jobs mapping.** The Remote Labor Index measures automation at the level of individual, client-quality deliverables. No method yet exists to translate a deliverable-level automation rate into an occupation-level displacement forecast — the tasks-versus-jobs distinction that Chatterji (2026) raises applies with full force to the RLI's own headline number. Sixteen per cent of freelance deliverables being automatable at professional quality is not the same claim as sixteen per cent of any job disappearing, and this report has not built the bridge between the two.

## 17.6 Energy and Environmental Costs

The report proposes deploying agentic infrastructure at organisational scale without addressing the energy implications. This is an increasingly visible omission as the environmental costs of AI infrastructure become a matter of public and regulatory concern.

### The Energy Cost of Agentic Systems

- Patterson et al. (2025) at Microsoft Research find that reasoning-intensive queries — the kind that agentic orchestration relies on — have a median energy cost of 4.32 Wh per query, compared to 0.34 Wh for standard queries. This represents a **43× increase** in energy consumption for the reasoning-heavy workloads that an agentic mesh generates continuously.
- Nature Sustainability (2025) projects that US AI server infrastructure will produce **24–44 Mt CO<sub>2</sub>-equivalent annually** through 2030 — a range that is growing as model sizes increase and inference demand scales.
- An agentic mesh running continuous orchestration, discovery monitoring, and governance checks generates substantially more inference load than on-demand AI tool use. The compound flywheel, by design, *increases* inference volume as the mesh becomes more active.

The energy cost of the mesh is not merely a corporate social responsibility issue. It is a business cost that scales with mesh activity. Organisations deploying agentic infrastructure must account for compute costs that are qualitatively different from traditional IT expenditure — costs that increase as the mesh becomes more capable, not less.

**Research direction:** develop energy cost models for agentic mesh deployment at different organisational scales, including comparison with the energy cost of the human coordination infrastructure the mesh replaces (office buildings, commuting, business travel). The net energy equation is not obviously negative — a 200-person management layer has its own carbon footprint — but neither is it obviously positive.

## 17.7 Cybersecurity Attack Surface

The statistic that financial services organisations manage 96 machine identities per human (CyberArk 2025) — with only 31 per cent having controls in place — is cited in Chapter 12 as a governance issue. It is also a security issue that the report does not adequately address. An agentic mesh creates a dramatically expanded attack surface:

- Each agent skill is a potential vector for prompt injection — an adversary who can manipulate an agent’s input can redirect its behaviour within its authority boundary.
- The OWL 2 ontology that provides shared semantics is a single point of corruption — poisoning the ontology could redirect agent routing across the entire mesh.
- The Nostr identity layer provides cryptographic verification but depends on key manage-

ment practices that most organisations have not developed for non-human identities.

- The Insight Ingestion Loop creates a pathway from frontline input to production workflows — a supply chain vulnerability that adversaries could exploit by planting malicious “shadow workflows.”

**Research direction:** systematic threat modelling for agentic mesh architectures, including adversarial analysis of the Insight Ingestion Loop and ontology poisoning scenarios.

## 17.8 Equity and Access

Who benefits from the mesh and who is structurally excluded? The report addresses middle managers — a specific professional cohort — but does not address:

- Workers without AI literacy who cannot participate in the discovery layer.
- Roles that do not lend themselves to agent augmentation — physical labour, care work, creative work that resists decomposition into tasks.
- Organisations in the Global South that lack the infrastructure and capital to deploy agentic systems.
- The distributional effects within organisations: does the mesh flatten compensation as it flattens hierarchy, or does it concentrate value in the judgment broker tier?

Acemoglu’s concern about AI increasing inequality rather than productivity is most acute here. If the Dynamic Agentic Mesh is a tool available primarily to well-capitalised organisations in knowledge-intensive industries, its net effect may be to widen the gap between those organisations and everyone else — a dynamic that the World Economic Forum (2025) Future of Jobs Report anticipates when it projects 170 million new roles created alongside 92 million displaced.

## 17.9 Competing Frameworks Not Addressed

The report positions against Block and Every — neither of which is a consulting framework the target audience is actively evaluating. It does not differentiate substantively from:

- **SAFe 6.0** (Scaled Agile, Inc. 2024), which added AI-specific agile change patterns in 2024.
- **Microsoft Copilot Studio** (Microsoft 2025b), an enterprise organisational AI deployment framework with the largest existing footprint.
- **Holacracy 5.0** (HolacracyOne 2024), which specifically addressed the scale failures attributed to earlier versions.

- **Anthropic’s constitutional AI governance**, directly relevant to the governance paradox in Chapter 10.

A competitor comparison table is provided in Appendix .8, but the main text does not engage with these alternatives at the depth they deserve.

## 17.10 The VisionClaw Credibility Gap

Chapter 14 describes VisionClaw honestly, including both its genuine capabilities and its limitations. But the credibility gap between a 15-person creative technology team and a 500-person manufacturing firm has not been bridged. The favourable conditions under which VisionClaw operates — high technical literacy, low process rigidity, flat power dynamics, intrinsic motivation, the system architect as the primary user — are the most favourable possible environment.

### Key Insight

VisionClaw is an existence proof under ideal conditions, not a proof of general applicability. The coordination collapse thesis requires bridging evidence that the architecture works in less favourable environments: regulated industries, non-technical workforces, unionised labour, legacy ERP systems, and middle managers who did not choose the system.

Four bridging strategies are available:

1. Run a pilot in a genuinely different context (regulated industry, non-technical workforce).
2. Partner with an established consulting framework (Deloitte, BCG) for independent validation.
3. Publish the methodology as open-source and invite external testing.
4. Position VisionClaw explicitly as “existence proof under favourable conditions” and build the evidence base incrementally.

The fourth strategy is the one this report adopts. The others represent the research agenda.

## 17.11 Additional Failure Modes

The report documents two failure modes in detail: the ant death spiral (agents responding to agents in infinite loops) and the vigilance degradation problem (humans falling asleep at the wheel as AI quality improves). But an agentic mesh at organisational scale introduces additional failure modes that this analysis does not adequately address:

- **Cascading errors:** a single incorrect agent output propagating through downstream agents before the judgment broker can intervene.
- **Single point of failure in the judgment broker:** if the broker is unavailable, incapacitated, or overwhelmed, the mesh has no fallback governance mechanism.
- **Ontology drift:** the OWL 2 ontology gradually diverging from the actual business domain as the organisation evolves, producing increasingly inaccurate agent routing.
- **Metric gaming:** agents optimising for the four KPIs in ways that satisfy the metrics while degrading actual performance — Goodhart’s law applied to agentic governance.
- **Cultural rejection:** the mesh may function technically while being socially rejected by the workforce, producing compliance without engagement.

**Research direction:** systematic failure mode analysis (FMEA) for agentic mesh architectures, with probability and severity estimates calibrated to organisational context.





# Chapter 18

## Conclusion: The Through Line

The organisation is becoming a compounding learning system — not a command structure. The critical question is where irreducible human judgment sits in that system.

### 18.1 Synthesis

For two thousand years, organisations have paid a coordination tax — layers of humans whose primary function is routing information. The Roman *contubernium*, the Prussian General Staff, McCallum’s organisational chart, Taylor’s functional pyramid: each was an engineering response to the same constraint. A human leader can effectively manage three to eight people. Beyond that threshold, you must add a layer. Each layer adds latency, distortion, and cost. Two millennia of organisational innovation have been attempts to work around this tradeoff without breaking coherence.

AI breaks the tradeoff.

Information routing — synthesising status across workstreams, aggregating context across functions, translating across departmental boundaries, maintaining alignment without weekly meetings — is now computationally cheap. The structural reason for the organisational chart, as it has existed since the Roman legions, is dissolving. This is the coordination collapse.

Three models have emerged for what replaces it.

Dorsey and Botha (2026) propose the **Intelligence Layer**: eliminate middle management, build a Company World Model, centralise coordination in AI. The diagnosis is precise. The prescription is reckless. Ninety-five per cent of AI-generated code at Block still requires human modification. Klarna replaced 700 customer service agents with AI and reversed course when quality collapsed (Klarna 2025). The vigilance problem (Dell’Acqua 2023) demonstrates that the better AI becomes, the harder it is for humans to maintain oversight — and Block’s model removes the humans who would maintain it. Spicer (2026) observes that every wave of of-

fice technology produces delaying that subsequently reverses as complexity reasserts itself. The Self-Automator mode of human–AI collaboration — the mode that Block’s architecture embodies — produces the weakest outcomes in the BCG experiment (Randazzo et al. 2024).

Shipper (2026) describes the **Agent Colony**: give every employee a persistent personal AI agent, let culture self-organise. The experiment is the most honest account available of what actually happens when personal agents are deployed across an organisation. The insight that “everyone is a manager now” — that management skill democratises rather than disappears — directly contradicts Block’s thesis and is supported by the evidence. But the model hits structural limits at scale. The ant death spiral, the 41–86.7 per cent failure rates documented across naive multi-agent frameworks (Cemri et al. 2025), and the memory gaps that cause context loss between sessions are failure modes that cultural norms cannot resolve at organisational scale.

The **Dynamic Agentic Mesh** retains human judgment at the intersection points. Middle managers evolve from information routers to **judgment brokers** — federated guardians of a system where agentic orchestration handles the information routing that hierarchy was built for, while humans handle what AI cannot: navigating the jagged frontier, maintaining vigilance, adjudicating ethical dilemmas, and sustaining cultural coherence.

Since the first edition, the argument has acquired an economics and an evidence base it did not have in April. Liu (2026) formalised the coordination tax at the agent layer — contextual transaction cost — and, working from organisational-economics first principles rather than consulting practice, independently arrived at the interface organisation: the academic twin of what this report calls the judgment broker. Pereira, Graylin, and Brynjolfsson (2026) showed, across 51 production deployments, that escalation architectures outperform approval workflows by a wide margin: 71 per cent median productivity gains against 30 per cent. The Remote Labor Index (Mazeika 2026) quantified how fast the frontier itself is moving. None of this evidence existed when this report first argued that hierarchy is an information-routing protocol under algorithmic pressure. The thesis has moved from provocative framing towards corroborated theory.

## 18.2 Block Was Right About the Diagnosis

This report has been critical of Block’s Intelligence Layer model, and that criticism stands. But the criticism is of the *prescription*, not the *diagnosis*. The diagnosis is correct:

- Hierarchy is an information routing protocol, not a management philosophy.
- That protocol is constrained by human bandwidth.

- AI removes the bandwidth constraint.
- The structural logic of the organisational chart dissolves when information routing becomes computationally cheap.

Where Block errs is in the inference that follows. “The information routing function is automated, therefore eliminate the humans who performed it” ignores the jagged frontier (Chapter 4), the vigilance problem (Chapter 5), the Klarna evidence, and the historical pattern of delaying reversal. The correct inference is: “The information routing function is automated, therefore the humans who performed it must be redeployed to the functions that cannot be automated — judgment, legitimation, accountability, political negotiation, and cultural coherence.”

The judgment broker is not a consolation prize for displaced middle managers. It is the role that the evidence demands.

## **18.3 Everyone Is a Manager Now — And That Changes Everything**

Every’s experiment (Shipper 2026) reveals the deepest implication of the coordination collapse. When every employee has a personal AI agent, management skill — the ability to delegate, evaluate, coach, and coordinate — becomes the universal skill. The graduate with a well-configured agent can approximate the information routing that middle managers were uniquely paid to do.

This is genuinely threatening to the current middle management cohort. The report does not pretend otherwise. The 67 per cent role obsolescence fear documented by Capgemini Research Institute (2025) is rational. The skill floor has risen. What was the middle manager’s ceiling is now everyone’s floor.

But the conductor problem resolves the threat. Everyone in an orchestra can play an instrument. That does not make the conductor redundant — it makes the conductor essential in a different way. The musicians’ skill is playing. The conductor’s skill is ensuring that 80 individually excellent performances produce coherent music rather than 80 simultaneous solos.

When every employee manages their own agent, the organisation has 200 individual human-agent pairs each optimising for their own task. Without a mesh coordinator — without the judgment broker — this produces the ant death spiral documented by Every at 20 people, the systematic, taxonomisable failure documented in peer-reviewed multi-agent systems research (Cemri et al. 2025), and the quality collapse documented by Klarna in customer service. The evidence is consistent: uncoordinated agent deployment fails.

The middle manager’s new differentiating skill is not managing an agent. Everyone does that. It is managing the *coherence of the mesh* — ensuring that 200 individually optimised human-agent pairs produce organisational outcomes rather than organisational noise. This is the difference between individual agent management (Stage 2–3 on Klaassen’s compound engineering ladder) and mesh coherence management (Stage 4–5). The former is a personal productivity skill. The latter is systems thinking. It is genuinely hard. And it is the role that makes the coordination collapse survivable.

## 18.4 The Compound Flywheel



**Figure 18.1:** The compound organisational flywheel. Discovery feeds codification; codification feeds validation by the judgment broker; validated workflows propagate through the mesh; propagation increases organisational capability; increased capability accelerates the next cycle of discovery. Each revolution makes the next one faster.

The flywheel in Figure 18.1 is the core mechanism of the coordination collapse thesis. It is not a strategy diagram. It is an operational description of how the mesh produces compounding returns. Each shadow workflow surfaced by the Discovery Engine becomes institutional capability through the Insight Ingestion Loop. Each validated DAG makes the mesh smarter. Each cycle of the flywheel reduces the time required for the next cycle — the metric captured by Mesh Velocity.

The compound flywheel addresses Acemoglu’s scepticism about AI productivity gains (Acemoglu 2025). His 0.66 per cent TFP estimate applies to the economy as a whole. An individual organisation running a compound flywheel captures gains that are disproportionate to the macroeconomic average — precisely because the gains compound, and because the organisation’s competitors have not yet built the flywheel. The strategic advantage is not in using AI. It is in building a system that learns from its own use of AI.

## 18.5 What the Evidence Supports

The following propositions are supported by the evidence reviewed in this report:

1. Hierarchy is an information routing protocol. The coordination tax is real and measurable. AI can perform the information routing function at a fraction of the cost and latency.

2. The boundary between what AI can and cannot do is jagged, unpredictable, and dynamic (Dell’Acqua et al. 2026). Wholesale automation is empirically reckless.
3. Human oversight degrades as AI quality improves (Dell’Acqua 2023). “Human in the loop” without architectural support for maintaining vigilance is an unreliable governance mechanism.
4. Three collaboration modes exist — Centaur, Cyborg, and Self-Automator — and the Self-Automator mode produces the worst outcomes (Randazzo et al. 2024).
5. Organisations that revise their KPIs to account for AI are three times more likely to see financial benefit (Boston Consulting Group and MIT Sloan Management Review 2025).
6. The 1-9-90 adoption pattern is predictable, and investing in the 9 per cent yields the highest return (Boston Consulting Group 2025b).
7. Psychological safety is the strongest predictor of successful AI adoption (A. C. Edmondson and Lei 2025; Google 2025). Protected experimentation time produces 3.1 times higher adoption rates.
8. Providing sanctioned AI tools reduces unauthorised use dramatically. Ignoring shadow AI is not a neutral choice.
9. Human-imitation organisational forms underperform agent-native forms in agent collectives; collective efficiency collapses at high contextual transaction cost (Liu 2026).
10. Escalation-based oversight outperforms approval-based oversight in production deployments, with 71 versus 30 per cent median productivity gains (Pereira, Graylin, and Brynjolfsson 2026).
11. Multi-agent failure is systematic and taxonomisable, with 41–86.7 per cent failure rates documented in naive frameworks (Cemri et al. 2025).
12. Headcount reduction is the most common but not the majority outcome of successful AI deployment: 45 per cent, against 55 per cent for growth, redeployment, and avoided hiring combined (Kharazian, Simon, and Stevens 2026; Pereira, Graylin, and Brynjolfsson 2026).

## 18.6 What the Evidence Does Not Yet Support

The following propositions are advanced by this report but not yet validated by independent evidence:

1. The full organisational compounding flywheel — where each discovered insight makes the next discovery easier at the systems level — has not been demonstrated at scale.
2. The proposed KPIs (Mesh Velocity, Augmentation Ratio, Trust Variance, HITL Precision)

now have a theoretical anchor in Liu’s contextual-transaction-cost and collective-efficiency framework (Chapter 7), but they remain empirically unvalidated at scale.

3. The judgment broker role has not been tested with brokers who did not build the underlying system.
4. The Dynamic Agentic Mesh has not been proven beyond a 15-person team in a favourable context.
5. The governance architecture (declarative governance, ontological provenance, cascading trust) has not been stress-tested in regulated industries.
6. The mapping between Liu’s simulated organisation forms and production organisational architectures is interpretive, not demonstrated: blackboard memory and adaptive meta-organisation were validated against synthetic tasks and small real-LLM traces, not against functioning enterprise organisations.

Acknowledging these gaps is not a weakness. It is a statement of where the research frontier lies and an invitation for others to test the propositions in their own contexts.

## 18.7 Closing Messages

### 18.7.1 For Middle Managers

You are not being replaced. You are being promoted — from information router to judgment broker. The promotion is not in title or compensation (those are organisational decisions outside this report’s scope). The promotion is in *complexity, scope, and strategic importance*. The skill you need is not technical. It is the willingness to map the jagged frontier in your domain and the judgment to know when to trust the mesh and when to override it.

This is no longer only a projection. Ford rehired approximately 350 veteran “greybeard” engineers over three years specifically because AI tooling alone could not sustain quality, and used them to retrain the AI tools themselves — becoming the top mainstream brand in the J.D. Power Initial Quality Study in the process (Naughton 2026). The escalation-model evidence quantifies exactly what the judgment path is worth: production deployments built around human review of exceptions, rather than approval of every output, show 71 per cent median productivity gains against roughly 30 per cent for approval-on-everything models (Pereira, Graylin, and Brynjolfsson 2026). The judgment broker is not a hopeful reframing of a shrinking role. It is what the highest-performing deployments already look like.

The organisations that will outcompete are not the ones that eliminate middle management. They are the ones that give their middle managers experimentation time (3.1 times higher adop-

tion), psychological safety (2.6 times more likely to report successful AI integration), and a clear path from coordination to judgment.

Not every current middle manager will make this transition. The 1-9-90 data is honest about that. But the path is real, the evidence supports it, and the role that emerges on the other side is more valuable, more intellectually demanding, and more strategically important than the one it replaces.

### **18.7.2 For Change Managers**

Your playbook is not broken. It is incomplete. AI transformation has no end state. You are not managing a transition; you are building an organisation's capacity for continuous adaptation.

The frameworks exist: Continuous Transformation (Wade and Bonnet 2025), Directed Autonomy (Microsoft 2025a), Dynamic Capabilities (Teece 2025). The mesh provides the architecture. The 90-day roadmap in Chapter 15 is the bootstrap sequence. Phase 3 never ends. Your job is perpetual — and that is not a burden. It is a description of a role that will be needed for as long as the jagged frontier moves, which is to say permanently.

### **18.7.3 For AI Leaders**

The secret cyborgs are your greatest asset, not your greatest risk. Surface them. Sanction them. Learn from them. The shadow workflows they have already created are the raw material for your compounding loop.

Providing approved tools drops unauthorised use by 89 per cent. The alternative — 1,200 unsanctioned applications per enterprise (Gartner, Inc. 2025), \$4.63 million per breach (IBM Security 2025) — is untenable. The governance paradox has a resolution: embed governance into the architecture, and it becomes an accelerator rather than a brake.

But surfacing shadow workflows requires psychological safety. Workers who have automated 90 per cent of their role will not volunteer that information if they believe it invites headcount reduction. Mollick (2025) is direct on this point: organisations that delegate AI strategy to IT without CEO engagement guarantee failure.

The Stanford Digital Economy Lab's study of 51 successful deployments (Pereira, Graylin, and Brynjolfsson 2026) converges on the same imperatives from the opposite direction — not theory, but ground truth from organisations that got it right: start with the invisible work, because 77 per cent of the hardest challenges were change management and process redesign, not technology; define your KPIs before deployment, not after; and build multi-model, abstraction-layer architecture from day one, because the durable advantage sits in the orchestration layer, not in

any single foundation model. Chapter 16 demonstrated this report's own organisation applying exactly these prescriptions to its own systems — the mesh audited, not merely proposed.

#### **18.7.4 For All Three**

Block diagnosed the right problem. The question is whether you agree with their prescription. The evidence reviewed in this report suggests you should not — but you should take the diagnosis deadly seriously.

The coordination collapse is real. What you build in its place will determine whether your organisation compounds or collapses.

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# Research Methodology

This appendix documents the research methodology used to produce this report, including the multi-agent research architecture, source selection criteria, codebase audit procedures, and Wardley mapping approach.

## .1 Multi-Agent Research Architecture

The research was conducted using five parallel research agents, each assigned a distinct domain with defined source priorities and quality thresholds. The agents operated concurrently across a shared memory layer (RuVector, 1.17 million vector embeddings with HNSW indexing) that enabled cross-domain pattern detection and prevented duplication of effort.

**Table 1:** Research agent assignments and source priorities

| Agent   | Domain                          | Primary Sources                                                                                                                                                          |
|---------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agent 1 | Mollick / BCG / Jagged Frontier | Dell’Acqua et al. (2026), Randazzo et al. (2024), Dell’Acqua (2023), Mollick’s “One Useful Thing” corpus, Boston Consulting Group and MIT Sloan Management Review (2025) |
| Agent 2 | Block Critique                  | Dorsey and Botha (2026), Zamost (2026), Spicer (2026), Klarna (2025), Orgvue and Forrester Research (2025)                                                               |
| Agent 3 | Agentic Organisations           | McKinsey & Company (2026b), McKinsey & Company (2026a), Boston Consulting Group (2025b), Boston Consulting Group (2025a), Deloitte (2026)                                |
| Agent 4 | Change Management               | Wade and Bonnet (2025), Teece (2025), Microsoft (2025a), Prosci (2024), Capgemini Research Institute (2025)                                                              |
| Agent 5 | Every / Compound Engineering    | Shipper (2026), Klaassen (2025), CyberArk (2025), Towards Data Science (2026)                                                                                            |

Each agent maintained a source provenance log in the shared RuVector memory, recording the retrieval date, URL, and a confidence assessment for each source. Cross-referencing between agents was automated through semantic vector search: when Agent 2 (Block Critique) encountered a claim about AI productivity, the memory layer surfaced related findings from Agent 1 (Mollick / BCG) and Agent 5 (Every / Compound) for triangulation.

## **.2 Source Selection and Quality Assessment**

Sources were categorised into four tiers based on methodological rigour and evidentiary weight.

### **.2.1 Tier 1: Peer-Reviewed and Pre-Registered Research**

The highest evidentiary weight was assigned to peer-reviewed publications and pre-registered experiments. The primary load-bearing sources in this category are:

- Dell’Acqua et al. (2026): pre-registered field experiment with 758 BCG consultants, published in *Organization Science*. This is the evidential foundation for the jagged frontier concept.
- Randazzo et al. (2024): follow-up study with 244 BCG consultants identifying the Centaur, Cyborg, and Self-Automator collaboration modes.
- Dell’Acqua (2023): field experiment with HR recruiters demonstrating the vigilance degradation problem.
- Acemoglu and Restrepo (2019): the task-based automation framework that grounds the economic analysis of AI and labour.
- Acemoglu (2025): the sceptical macroeconomic analysis of AI productivity, arguing for at most 0.66 per cent TFP gain over a decade.

### **.2.2 Tier 2: Institutional Research with Disclosed Methodology**

Consultancy reports and institutional research with disclosed sample sizes and methodologies. These carry less evidentiary weight than peer-reviewed work but are treated as credible within stated limitations:

- Boston Consulting Group and MIT Sloan Management Review (2025): joint research with MIT Sloan,  $n = 3,000+$  across 25 industries.
- Microsoft and LinkedIn (2024): large-scale survey with disclosed methodology.
- World Economic Forum (2025): the World Economic Forum’s Future of Jobs Report.

- Capgemini Research Institute (2025): middle manager resistance patterns with percentage-level findings.
- A. C. Edmondson and Lei (2025): psychological safety and AI experimentation, grounded in the established literature on team effectiveness.

**Limitation:** Consultancy reports are produced by organisations with commercial interests in the phenomena they describe. Methodology sections are often absent or sparse. Sample composition and response biases are rarely disclosed. Statistics from these sources are used directionally — to establish trends and orders of magnitude — rather than as precise measurements.

#### .2.3 Tier 3: Expert Commentary and Primary Accounts

Expert commentary, practitioner accounts, and journalistic reporting. These provide context and narrative but are not treated as primary evidence:

- Dorsey and Botha (2026): primary source for the Block Intelligence Layer model.
- Shipper (2026) and Klaassen (2025): primary accounts of the Every agent colony experiment.
- Mollick (2025): expert synthesis from a researcher with deep domain expertise.
- Spicer (2026) and Zamost (2026): expert and insider critique of Block.

#### .2.4 Tier 4: Community and Vendor Sources

Sources with limited editorial oversight or commercial incentives. Used cautiously and flagged where they appear:

- Towards Data Science (2026): the original  $17\times$  error multiplication statistic. Superseded in this edition by Cemri et al. (2025) (Chapter 17); retained here in the bibliography for the correction record, not as an active source.
- IBM Security (2025) and Gartner, Inc. (2025): vendor-funded research with commercial incentives. Treated as directionally correct, per the methodological note in Chapter 17.

### .3 Web Search and Academic Database Approach

Research was conducted through a combination of five source channels:

1. **Academic databases:** Google Scholar, SSRN, NBER Working Papers, HBS Working Papers. Search terms included “agentic organisation,” “AI coordination,” “human-AI collaboration,” “middle management AI,” “organisational design automation,” and “jagged

technological frontier.”

2. **Consultancy publication portals:** McKinsey Insights, BCG Publications, Deloitte Insights, Gartner Research. Filtered to 2024–2026 publications on organisational AI.
3. **Practitioner platforms:** Substack (particularly Mollick’s “One Useful Thing” and Every’s “Chain of Thought”), podcasts, and conference proceedings.
4. **News and analysis:** Financial Times, New York Times, Fortune, CNBC, VentureBeat. Used for event-driven reporting (Block restructuring, Klarna reversal) with cross-referencing against multiple outlets.
5. **Regulatory sources:** EUR-Lex for EU AI Act text, UK Legislation website for the Employment Rights Act, ICO guidance on workplace monitoring.

## .4 Codebase Audit Methodology

The VisionClaw codebase was audited in April 2026 to reconcile claims made in earlier versions of this document against the actual code. The audit procedure:

1. **Claim extraction:** all factual claims about VisionClaw’s capabilities were extracted from the document and catalogued.
2. **Code search:** each claim was tested against the codebase using structural code analysis (call graphs, dependency trees) and text search. The codebase-memory MCP tool (48,159 nodes, 95,766 edges) was used for structural queries.
3. **Verification categories:** each claim was classified as (a) verified and accurate, (b) partially accurate with caveats, or (c) not supported by the code.
4. **Correction:** claims in categories (b) and (c) were corrected in Chapter 14. The corrections are presented transparently, with both the original claim and the corrected version visible.

### Conflict of Interest

The codebase audit was conducted by the system architect (Dr. John O’Hare), who is also the report author and the founder of DreamLab AI Consulting. This introduces a conflict of interest: the auditor has a commercial interest in the system being described favourably. The audit found and corrected seven overclaims (documented in Chapter 14), which provides some evidence of good faith. Independent code review by a third party would strengthen the credibility of the technical claims but was not available at the time of writing.

## .5 Wardley Mapping Methodology

Five Wardley maps were produced for this report, each following Simon Wardley’s standard mapping methodology:

1. **Value chain identification:** for each map, the user need was identified and the value chain of components required to serve that need was enumerated.
2. **Evolution assessment:** each component was placed on the evolution axis (Genesis → Custom → Product → Commodity/Utility) based on its current maturity and market availability.
3. **Movement indication:** where components are expected to evolve (e.g., information routing moving from custom toward commodity), movement arrows indicate the direction and anticipated pace.
4. **Strategic interpretation:** each map was interpreted for strategic implications relevant to the report’s argument.

The maps are interpretive tools, not empirical measurements. The placement of components on the evolution axis reflects the author’s assessment informed by the evidence assembled in this report. Different analysts may position components differently, particularly for novel concepts (judgment brokerage, mesh velocity) whose evolutionary stage is inherently uncertain.

## .6 The July 2026 Revision

This edition was produced through a second, distinct research and drafting cycle three months after the first. The process:

1. **Verification.** Four research agents, each with live web search, verified every transcript-derived claim in the April edition against primary sources — the CAIS blog, arXiv, Bloomberg, the *New York Times*, Ramp’s own research publication, and court-reporting coverage of the Hangzhou ruling — correcting names, dates, and figures before any revision text was written. Corrections of substance are flagged explicitly at the point the original claim appears in the main text.
2. **Ecosystem exploration.** Three further agents mapped the current state of the VisionClaw, AgentBox, and federation codebases to ground Chapter 16, which applies this report’s own findings to DreamLab’s operational systems.
3. **Drafting.** Chapter revisions were produced by section-scoped writing agents working against a single shared evidence brief with fixed citation keys, house style, and verified

figures, so that no two chapters could independently introduce inconsistent numbers for the same claim. The drafts were then editorially reviewed for voice, argument continuity, and consistency with chapters outside each agent’s assigned scope.

The new sources introduced in this revision were tier-assigned using the same rubric as above:

- **Tier 1:** (Cemri et al. 2025) — peer-reviewed, NeurIPS 2025, Datasets and Benchmarks Track.
- **Tier 2:** (Liu 2026) (preprint with disclosed methodology, single author); (Pereira, Graylin, and Brynjolfsson 2026) (disclosed methodology, acknowledged selection bias toward successful deployments); (Mazeika, Gatti, Menghini, et al. 2025), (Mazeika 2026), and (Kwa, West, Becker, et al. 2026) (benchmark reports with disclosed methodology); (Kharazian, Simon, and Stevens 2026) (corporate research, disclosed methodology, correlational with matched controls).
- **Tier 3:** news reporting — (Prakash and Boesler 2026), (C. Edmondson 2026), (Naughton 2026), (Chatterji 2026).
- **Tier 4:** (Levie 2026) — a single social-media source with no published methodology, flagged wherever it is used.

## .7 Limitations of the Research Process

1. **Single-author synthesis:** while the research was conducted using multiple agents, the synthesis and editorial judgment are those of a single author. The report reflects one analytical perspective on a multi-faceted phenomenon.
2. **Temporal constraints:** this edition incorporates sources through early July 2026; sources published after that point are not included. The jagged frontier, in particular, continues to shift — the Remote Labor Index update alone recorded a 6.4-fold increase in client-quality automation in the eight months preceding this revision (Mazeika 2026), and there is no reason to expect the pace to slow before the next edition.
3. **English-language bias:** the research draws almost exclusively on English-language sources. Organisational innovations in non-Anglophone contexts — particularly Haier’s micro-enterprise model in China — are discussed secondhand rather than from primary sources.
4. **Analyst-vendor conflict:** the author is the founder of DreamLab AI Consulting and the architect of VisionClaw. This conflict is acknowledged throughout the report and addressed directly in Chapter 17.
5. **Sample bias in practitioner evidence:** the Every experiment (~20 people) and VisionClaw deployment (~15–50 people) are both small-scale, technically sophisticated organi-

sations. The generalisability of findings from these contexts to larger, less technical organisations is an open question.

## **.8 Disclosure**

The author of this report, Dr. John O'Hare, is the founder and principal of DreamLab AI Consulting Ltd and the architect of the VisionClaw system described in Chapter 14. The report simultaneously analyses the problem space, proposes a solution framework, and describes a commercial offering from the author's firm. This positioning conflict is acknowledged in Chapter 17 and is offered as context for the reader's own evaluation of the argument.





# Competitor Comparison Matrix

This appendix positions the Dynamic Agentic Mesh against six frameworks that the report’s intended audiences are likely to be evaluating. The main body of the report positions the mesh against Block’s Intelligence Layer and Every’s Agent Colony — neither of which are consulting frameworks a COO or change manager would commission. This appendix addresses the gap identified in Chapter 17: differentiation from the frameworks the audience is actually considering.

## .9 Comparison Dimensions

Seven frameworks are compared across six dimensions:

1. **Coordination Model:** how the framework proposes to handle the information routing function that hierarchy currently performs.
2. **Management Role:** what happens to middle management in the framework’s model.
3. **Governance Approach:** how the framework addresses oversight, accountability, and compliance.
4. **AI Integration:** the depth and nature of AI’s role in the framework.
5. **Scale Proven:** the largest organisational context in which the framework has been deployed or validated.
6. **Evidence Base:** the type and strength of evidence supporting the framework’s claims.

## .10 Comparison Tables

The comparison is split across three tables for readability, each covering two or three frameworks alongside the Dynamic Agentic Mesh as the reference column.

**Table 2:** Competitor comparison: DreamLab Agentic Mesh, McKinsey Agentic Org, and SAFe 6.0

| Dimension       | DreamLab Agentic Mesh                                                                       | McKinsey Agentic Org                                                                          | SAFe 6.0                                                                      |
|-----------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Coordination    | Declarative governance + DAG orchestration with OWL 2 ontology                              | Five pillars: business model, operating model, governance, workforce/culture, technology/data | Agile Release Trains with AI-specific practices added in 2024                 |
| Management role | Judgment broker at jagged frontier intersections                                            | “Above the loop”: employees shift from performing tasks to orchestrating outcomes             | Existing agile roles (RTE, Product Owner) extended with AI practices          |
| Governance      | Embedded in architecture; bias detection, security, value-alignment in every DAG transition | Strategic governance pillar; framework-level guidance, not architectural specification        | Process-based: PI cadences, inspect-and-adapt cycles                          |
| AI integration  | Deep: OWL 2 ontology, agent orchestration, GPU visualisation, Nostr identity                | Advisory: describes what organisations should do, not how the AI system works                 | Additive: AI practices layered onto existing SAFe structure                   |
| Scale proven    | 15–50 people (VisionClaw)                                                                   | Consulting engagements across Fortune 500; no single reference implementation                 | Enterprise-wide (thousands of developers) across multiple industries          |
| Evidence base   | Codebase-audited system + third-party research synthesis                                    | Proprietary client data; “Six Shifts” framework from consulting practice                      | Broad adoption data; criticised for process overhead; AI additions are recent |

**Table 3:** Competitor comparison: Block Intelligence Layer, Every Agent Colony, and Holacracy 5.0

| Dimension       | Block Intelligence Layer                                                                          | Every Agent Colony                                                                        | Holacracy 5.0                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Coordination    | AI Company World Model + Customer World Model; centralised intelligence                           | Cultural norms; public visibility in shared channels; emergent self-organisation          | Distributed authority through roles and circles; constitutional governance            |
| Management role | Eliminated; replaced by ICs, DRIs, and Player-Coaches                                             | “Everyone is a manager”: each employee manages their personal agent                       | No managers; authority distributed to roles; Lead Links for operational coordination  |
| Governance      | AI-enforced through the Intelligence Layer; no separate governance function described             | Social accountability; agent trust derivative of human reputation                         | Constitutional governance; explicit policies and domains; integrative decision-making |
| AI integration  | Foundational: AI is the coordination mechanism itself                                             | Deep at individual level: personal agents with memory, specialisation, personality        | Minimal: Holacracy is a governance system; AI integration is an add-on                |
| Scale proven    | 10,000+ employees (intended, not fully validated)                                                 | ~20 people                                                                                | Hundreds (Zappos at 1,500 with significant attrition); version 5.0 addresses scale    |
| Evidence base   | 95% of AI code requires human modification; Klarna analogue reversed; stock surge on announcement | First-person practitioner account; ant death spiral documented; no independent validation | Decades of practice; academic critiques of Zappos; updated methodology                |

**Table 4:** Competitor comparison: Microsoft Copilot Studio

| Dimension       | Microsoft Copilot Studio                                                                        | DreamLab Agentic Mesh (reference)                                               |
|-----------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Coordination    | Copilot agents deployed across Microsoft 365; orchestration through Microsoft Graph             | Declarative governance + DAG orchestration with OWL 2 ontology                  |
| Management role | Unchanged; Copilot augments existing roles rather than transforming organisational structure    | Judgment broker at jagged frontier intersections                                |
| Governance      | Microsoft's responsible AI framework; Entra ID for agent identity; admin-controlled deployment  | Embedded in architecture; ontological provenance; Trust Variance monitoring     |
| AI integration  | Deep platform integration; agents operate within Microsoft ecosystem; limited cross-platform    | OWL 2 + Claude-Flow; cross-platform by design; Nostr for decentralised identity |
| Scale proven    | Enterprise-wide across Microsoft's customer base; largest deployment footprint of any framework | 15–50 people                                                                    |
| Evidence base   | Microsoft's internal productivity studies; third-party adoption data; vendor-published metrics  | Codebase-audited system + third-party research synthesis                        |

## .11 Strategic Positioning Analysis

### .11.1 The Architecture Gap

McKinsey describes what organisations should become (five pillars, six shifts) but does not specify how the coordination mechanism works at the system level. SAFe 6.0 adds AI practices to an existing process framework but does not redesign the coordination architecture. Microsoft Copilot Studio provides a platform but does not propose an organisational model. The Dynamic Agentic Mesh occupies the gap between strategic framework and operational platform — it specifies both the organisational model and the technical architecture. This is its primary differentiator and its primary vulnerability: it attempts more than the alternatives, and therefore has more to prove.

### .11.2 The Governance Gap

Holacracy provides the most rigorous pre-AI governance model (constitutional governance with explicit policies and domains) but was designed before AI was a factor in organisational coordination. Block provides no separate governance function. Every relies on cultural norms. The

Agentic Mesh embeds governance architecturally — in the DAG transitions, in the ontological provenance chains, in the Trust Variance monitoring — and provides a specific human role (the judgment broker) for edge cases that exceed automated governance. This architectural embedding of governance is the feature that most clearly distinguishes the mesh from all competitors.

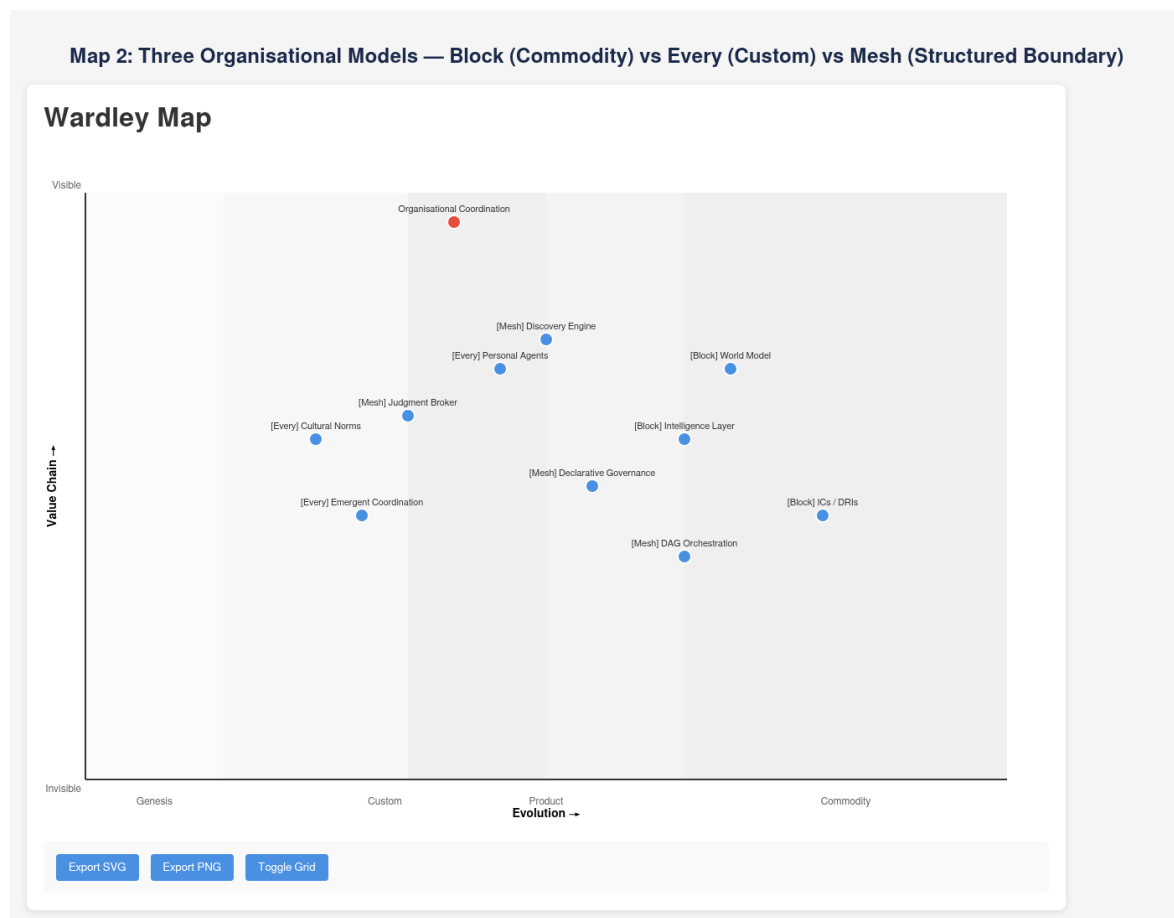
### **.11.3 The Scale Gap**

The Agentic Mesh has the smallest proven deployment footprint of any framework in the comparison. Microsoft Copilot Studio has the largest. SAFe 6.0 is proven at enterprise scale. McKinsey’s framework is backed by Fortune 500 consulting engagements. This is the most significant competitive disadvantage for the Agentic Mesh and the primary research priority identified in Chapter 17.

### **.11.4 The Evidence Gap**

The evidence bases differ in kind, not just in scale. The Agentic Mesh draws on third-party peer-reviewed research (Acemoglu 2025; Dell’Acqua et al. 2026; Randazzo et al. 2024) for its theoretical foundations but relies on a single codebase-audited deployment for its architectural claims. McKinsey and BCG draw on proprietary client data that is not independently verifiable. SAFe draws on broad adoption data but is criticised for process overhead. Holacracy draws on decades of practice but also decades of critique.

## .12 Wardley Map: Three Models Compared



**Figure 2:** Wardley map comparing the three primary models (Block, Every, DreamLab) on the evolution axis. Block’s Intelligence Layer positions AI coordination as a product/commodity; Every’s Agent Colony treats coordination as genesis/custom; the Agentic Mesh positions the judgment broker at the custom stage while using commodity AI for information routing.

## .13 Choosing the Right Framework

### Selection Guidance

- **Already in SAFe:** Consider integrating mesh concepts into existing PI planning — particularly the judgment broker role and the four KPIs. SAFe and the mesh are not incompatible; the mesh can operate within SAFe’s cadence structure.
- **Microsoft-native:** Start with Copilot Studio for individual augmentation; layer mesh governance concepts on top as agentic deployment matures. The two address different layers of the problem.
- **Greenfield or transforming:** The Agentic Mesh provides the most comprehensive model but requires the most organisational commitment and technical expertise. The 90-day roadmap in Chapter 15 is designed for this context.
- **Seeking cultural transformation:** Holacracy addresses power distribution directly; the mesh addresses information routing. They solve different problems and could in principle be combined.
- **C-Suite buy-in needed first:** McKinsey’s strategic framework provides the executive-level framing; the mesh can serve as the implementation architecture beneath it.
- **Enterprise scale with proven track record:** Microsoft Copilot Studio offers the lowest-risk deployment path. The mesh offers the most ambitious organisational re-design. The right choice depends on the organisation’s appetite for structural change versus incremental augmentation.

## .14 Interpretation

No single framework in this comparison is complete. McKinsey provides the strategic vision but not the operational architecture. SAFe provides the process but not the AI-native coordination model. Holacracy provides the governance but not the AI integration. Microsoft provides the platform but not the organisational transformation model. Block provides the boldest thesis but the weakest evidence for its prescription.

The Dynamic Agentic Mesh is differentiated by three features that no competitor offers in combination:

1. **Ontology-grounded orchestration:** formal OWL 2 semantics for agent routing, providing compositionality and inference that informal knowledge representations lack.

2. **Architecturally embedded governance:** governance as code in every DAG transition, not as a separate oversight function.
3. **The compound flywheel:** a specific, measurable mechanism (the Insight Ingestion Loop) by which the organisation's capability compounds over time, tracked by novel KPIs (Mesh Velocity, Augmentation Ratio, Trust Variance, HITL Precision).

The primary weakness is scale. The primary research priority is bridging the gap between the 15-person existence proof and the enterprise deployments that competitors can claim. This comparison clarifies that gap. It does not close it.



# Glossary of Key Terms

## 1-9-90 Pattern

The predictable adoption distribution identified by Boston Consulting Group (2025b) in AI deployment: 1 per cent are natural innovators who adopt independently, 9 per cent become champions with organisational support (the judgment broker candidates), and 90 per cent follow once social proof exists. Strategic investment in the 9 per cent yields the highest return. See Chapter 11.

## Agent Colony

An organisational model (associated with Every.to) in which every employee is given a persistent, personalised AI agent. Coordination emerges from cultural norms and public visibility rather than architectural design. Proven at approximately 20-person scale; subject to the ant death spiral at larger scales. See Chapter 6.

## Agent-Native Organisation Forms

Coordination structures designed for the properties of agent collectives rather than copied from human org charts — shared-state (blackboard) memory and adaptive, CTC-aware meta-organisation. In simulation, they outperform human-imitation forms (pipelines, hierarchies, committees) by wide margins (Liu 2026). See Chapter 7.

## Agentic Mesh

See *Dynamic Agentic Mesh*.

## Ant Death Spiral

A failure mode in multi-agent systems where agents respond to each other's outputs in an uncontrolled loop, consuming resources (tokens, compute, time) without producing useful outcomes. Named by analogy with army ant death spirals in entomology. Documented by Shipper (2026) in Every's agent colony experiments at 20-person scale. See Chapter 6.

## Augmentation Ratio (AR)

A proposed KPI:  $AR = \text{Cognitive Load Offloaded} \div \text{Total Cognitive Load}$ . Measures the proportion of decision-making volume successfully handled by agents without requiring human escalation. Target:  $> 65\%$ . Too low indicates the mesh is not yet effective; too high indicates potential vigilance degradation. See Chapter 9.

## BYOAI

Bring Your Own AI — the practice of employees using personal AI tools and accounts for

work purposes, bypassing corporate-approved systems. Microsoft and LinkedIn (2024) reports 78 per cent of knowledge workers engage in BYOAI.

**Centaur**

A human–AI collaboration mode identified by Randazzo et al. (2024) in which there is a clear division of labour between human and AI. The human performs tasks where human judgment is strongest; the AI handles tasks inside the frontier. Distinct handoff points separate the two. The Dynamic Agentic Mesh is designed to produce Centaur organisations. Contrast with *Cyborg* and *Self-Automator*. See Chapter 4.

**Compound Engineering**

A principle articulated by Klaassen (2025): “Each unit of engineering work should make subsequent units easier — not harder.” The 50/50 Rule prescribes spending half of one’s time building features and half improving the system that builds features. Extended in this report from the individual level to the organisational level as the basis for the compounding organisation. See Chapter 8.

**Compounding Organisation**

An organisation in which each unit of organisational learning makes the next one easier. The flywheel proceeds: workforce discovers a shortcut, agents detect and codify it, the judgment broker validates it, the pattern propagates across the mesh, and the organisation’s capability compounds. See Chapter 8.

**Contextual Transaction Cost (CTC)**

The cost of making task context usable across boundaries in an agent collective — token and latency burden, handoffs, compression loss, semantic drift, verification burden, and governance cost (Liu 2026). The agent-layer formalisation of this report’s coordination tax. See Chapter 7.

**Coordination Collapse**

The central thesis of this report. For two thousand years, organisations have paid a “coordination tax” — layers of humans whose primary function is routing information. AI collapses the cost of that coordination function to near-zero, dissolving the structural logic of hierarchical organisational design. Chapter 17 notes the qualification that “information routing collapse” may be a more precise term. See Chapter 3.

**Coordination Tax**

The cost — in headcount, latency, distortion, and overhead — of maintaining hierarchical layers whose primary function is routing information between organisational levels. See Chapter 2.

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## Cyborg

A human–AI collaboration mode identified by Randazzo et al. (2024) characterised by continuous interleaving: human and AI work on the same analysis, passing work back and forth across the frontier boundary without clean separation of responsibilities. One of the two high-performing modes alongside Centaur. Every’s Agent Colony model produces Cyborg organisations. Contrast with *Centaur* and *Self-Automator*. See Chapter 4.

## DAG (Directed Acyclic Graph)

A graph structure in which edges have direction and no cycles exist. In the Dynamic Agentic Mesh, DAGs represent formalised workflows: sequences of tasks with dependencies, ownership, governance constraints, and SLAs. Shadow workflows are codified as DAGs when promoted into the mesh via the Insight Ingestion Loop. See Chapter 6.

## Declarative Governance

A governance model in which policies are expressed as code and enforced automatically at every system transition, rather than being applied through periodic human review. Bias thresholds, security protocols, access controls, and value-alignment checks are embedded in each DAG transition. Analogous to declarative infrastructure (Kubernetes, Terraform) in software engineering. See Chapter 10.

## Discovery Engine

The bottom-up layer of the Dynamic Agentic Mesh that captures organisational insights. Passive agent monitoring surfaces shadow workflows; natural-language Insight Portals allow anyone to propose improvements; tacit knowledge is codified into agentic instructions. The Discovery Engine is the input mechanism for the Insight Ingestion Loop. See Chapter 6.

## Dynamic Agentic Mesh

The organisational architecture proposed in this report. Three layers (Discovery Engine, Orchestration Layer, Declarative Governance) retain human judgment at intersection points while agentic orchestration handles information routing. Middle managers evolve into judgment brokers who operate at the jagged frontier. See Chapter 6.

## Escalation Model

An operating model in which AI handles 80 per cent or more of a workflow autonomously and humans review only exceptions; associated with 71 per cent median productivity gains, versus roughly 30 per cent for approval-on-every-output models (Pereira, Graylin, and Brynjolfsson 2026). The deployment-scale expression of the judgment broker pattern. See Chapters 5 and 13.

**HITL (Human-in-the-Loop)**

The practice of maintaining human oversight or decision-making authority within an automated system. This report argues that naïve HITL implementations degrade as AI quality improves (the vigilance problem) and that HITL must be architecturally supported through deliberate friction and active engagement. See Chapter 5.

**HITL Precision (HP)**

A proposed KPI:  $HP = \text{Correct Escalations} \div \text{Total Escalations}$ . Measures whether the edge cases the mesh flags for human intervention genuinely require human judgment. Target:  $> 90\%$ . Low precision produces alert fatigue; high precision indicates the mesh has learned where the jagged frontier falls. See Chapter 9.

**Information Routing Protocol**

This report's characterisation of hierarchy: a system designed to filter, aggregate, and relay information across organisational scales, constrained by the human cognitive limit of managing three to eight direct reports. See Chapters 2 and 3.

**Insight Ingestion Loop**

The five-stage process by which the Dynamic Agentic Mesh converts individual discoveries into organisational capability: (1) Discovery — an employee finds a shortcut; (2) Codification — the system maps it as a proposed DAG; (3) Validation — the judgment broker reviews for strategic fit and compliance; (4) Integration — the validated workflow enters the live mesh with SLAs; (5) Amplification — the pattern propagates to other teams. Mesh Velocity measures how fast this loop runs. See Chapters 6 and 8.

**Intelligence Layer**

The organisational model proposed by Dorsey and Botha (2026) in which middle management is eliminated and coordination is handled by AI world models (Company World Model and Customer World Model) composed by an Intelligence Layer. Three roles remain: Individual Contributors, Directly Responsible Individuals, and Player-Coaches. See Chapter 6.

**Interface Organisation**

The designed arrangement connecting human accountability with agentic context coordination — which outputs enter human workflows, what evidence and uncertainty travel with them, what permissions agents hold, which traces persist, and where human judgment is mandatory (Liu 2026). The academic articulation of judgment brokerage plus declarative governance. See Chapter 7.

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## **Jagged Frontier**

The unpredictable boundary between tasks that AI performs well and tasks that AI performs poorly. Identified by Dell’Acqua et al. (2026) based on a pre-registered experiment with 758 BCG consultants. The frontier is “jagged” because adjacent, similar-looking tasks can fall on opposite sides. This concept is the intellectual load-bearing wall of the report: it is the reason wholesale automation is reckless, change playbooks cannot be written once, and judgment brokers have a genuine future. See Chapter 4.

## **Judgment Broker**

The evolved middle management role proposed in this report. The judgment broker operates at the jagged frontier, providing strategic direction, ethical adjudication, and relational intelligence that agentic systems cannot perform. Distinguished from the traditional information-routing role by its focus on edge cases, cross-functional coherence, and frontier navigation. See Chapter 13.

## **MAST (Multi-Agent System Failure Taxonomy)**

A peer-reviewed taxonomy of 14 multi-agent failure modes in three categories — specification issues, inter-agent misalignment, task verification — derived from 1,600+ annotated traces; documents 41–86.7 per cent failure rates across seven state-of-the-art frameworks (Cemri et al. 2025). Supersedes the informal  $17\times$  error-multiplication figure cited in the first edition.

## **Mesh Velocity (MV)**

A proposed KPI:  $MV = \Delta t(\text{insight} \rightarrow \text{codified workflow})$ . Measures the elapsed time from insight discovery to codified, sanctioned workflow in the mesh. Target:  $< 48$  hours. The primary indicator of organisational learning speed. See Chapter 9.

## **Ontological Provenance**

The property of an agentic system in which every decision traces back through a formal knowledge graph (OWL 2 ontology), providing auditable reasoning chains from output to input, policy, and human decision. See Chapter 10.

## **OWL 2**

Web Ontology Language 2 — a W3C standard for formal knowledge representation. In VisionClaw, OWL 2 provides shared semantics for agent skill routing through ontological subsumption (understanding that “risk assessment” is a sub-task of “governance review” and routing accordingly). The EL++ profile, supported by the Whelk-rs reasoner, enables tractable inference while maintaining semantic precision. See Chapter 14.

**Pseudo-Diversity**

The appearance of deliberation in multi-agent systems whose agents share the same model, prompts, and evidence, producing correlated paraphrase rather than independent judgment. The remedy is engineered diversity across models, tools, and retrieval paths (Liu 2026). See Chapter 7.

**Remote Labor Index (RLI)**

A benchmark measuring whether AI can complete real freelance projects at paying-client quality, judged against professional gold-standard deliverables (Mazeika, Gatti, Menghini, et al. 2025). The automation rate rose from 2.5 per cent (October 2025) to 16.1 per cent (July 2026) (Mazeika 2026). See Chapter 4.

**Secret Cyborg**

A term from Mollick (2023) describing workers who use AI tools without disclosing this use to their employer. The secrecy is driven by rational calculation: disclosure risks head-count reduction if the worker demonstrates they have automated significant portions of their role. Over half of generative AI users report using the technology covertly at least some of the time. See Chapter 3.

**Self-Automator**

A human–AI collaboration mode identified by Randazzo et al. (2024) in which the human delegates wholesale to AI with minimal oversight. The lowest-performing mode of the three. Block’s organisational model is characterised in this report as creating Self-Automator organisations. Contrast with *Centaur* and *Cyborg*. See Chapter 4.

**Shadow AI**

AI tools and applications used by employees without organisational awareness, approval, or governance. Shadow AI usage increased 156 per cent between 2023 and 2025; the average enterprise hosts approximately 1,200 unauthorised AI applications (Gartner, Inc. 2025). See Chapter 3.

**Shadow Workflow**

An informal work process created by an employee using AI tools, which bypasses or replaces the officially documented workflow without management awareness. Shadow workflows represent both a risk (ungoverned, undocumented) and an asset (raw material for the Insight Ingestion Loop and the compounding flywheel). See Chapter 3.

**Trust Variance (TV)**

A proposed KPI:  $TV = \sigma(\text{Agent Decision Quality})$  over a rolling 30-day window. Target:  $< 0.12\sigma$ . Provides real-time monitoring of drift or bias in the automated task layer. The

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early warning system for vigilance degradation — catching drift before it compounds into failure. See Chapter 9.

### **Vigilance Problem**

The empirically demonstrated tendency of human oversight quality to degrade as AI quality improves. Identified by Dell’Acqua (2023): HR recruiters given high-quality AI made worse decisions than those given low-quality AI because they stopped scrutinising outputs. The central challenge of human–AI governance. See Chapter 5.

### **VisionClaw**

The operational agentic mesh system built by DreamLab AI Consulting. Architecture: OWL 2 ontology engine (`horned-owl`, `Whelk-rs` EL++ inference), Claude-Flow orchestration (83 agent skills, RAFT consensus), WebXR 3D visualisation (Rust/CUDA, Babylon.js), Nostr agent identity (NIP-98), and RuVector memory (1.17M+ vector embeddings, HNSW indexing). Proven at 15-person scale in a creative technology context. See Chapter 14.

### **World Model**

In Block’s Intelligence Layer thesis (Dorsey and Botha 2026), the Company World Model is an AI system continuously tracking what is being built, what is blocked, and where resources are allocated. The Customer World Model is constructed from transaction data. Together they replace the context-holding function of middle management. The concept is analytically sound but operationally unproven beyond Block’s specific context of massive transaction data and remote-first culture. See Chapter 6.